Pokhara University Faculty of Science and Technology	
Course Code: CMP 215 (3 Credits)	Full Marks: 100
Course Title: Object Oriented Programming using Java (3-0-3)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: III	Program: Bachelor of Computer Application

1. Course Description:

This course equips students with comprehensive knowledge of various Object-Oriented aspects of Java, guiding them through fundamental OOP concepts such as Classes, Objects, Inheritance, Interfaces, Polymorphism, and Packages. It also covers the basics of graphical user interfaces. The skills acquired in this course will enable students to develop standalone applications for real-world use. By the conclusion of this course, students will have the ability to write efficient, error-free Java programs, establishing a strong foundation for future endeavours.

2. General Objectives:

The general objectives of this course are as follows:

a) To acquaint the students with Object Oriented Programming knowledge of java environment.
b) To acquaint the students to develop simple applications with OOPS concepts
c) To introduce interfaces, packages, JavaFX and Socket programming basic concepts.
d) To enhance programming skills through a series of practical exercises and assignments, encouraging the application of learned concepts in real-world scenarios.

3. Methods of Instructions:

- Lecture and discussion
- Practical
- Demonstration
- Presentation
- Case study

4. Course Contents:

Specific Objectives	Contents
Unit 1: Introduction to Java Programming	**4 hours**
• Describe the basic concepts of Object-Oriented Programming • Explain the user friendly in JAVA environment with basic structure of the program	1.1 History and Features of Java 1.2 C++ vs Java 1.3 Introduction 1.3.1 JDK 1.3.2 JVM 1.3.3 Bytecode 1.3.4 Java Environment. 1.4 Procedure Oriented Vs Object Oriented 1.5 Basic Concepts of Object-Oriented Programming 1.6 Structure of Java Programming, Naming conversions 1.7 Compiling and executing simple program 1.8 Scanner class
Unit 2: Java Fundamental Concepts	**7 hours**
	2.1 Java Tokens, Keywords, Statements

• Explore the Fundamental of JAVA language • Describe the concept of Arrays and Strings	2.2 Constants and Variable 2.3 Data Types 2.4 Operators 2.5 Control Statements 2.5.1 Branching 2.5.2 Looping Statements 2.5.3 Jump Statements 2.6 Arrays in Java 2.6.1 One dimensional 2.6.2 Two dimensional 2.7 Java Strings 2.7.1 String Length 2.7.2 Concatenation 2.7.3 String Comparison 2.7.4 String Buffer 2.8 Methods in Java 2.8.1 Predefined methods 2.8.2 User defined methods 2.8.3 Static methods
Unit 3: Java Object Oriented Programming	**8 hours**
• Familiarize the concept of Class and Object, methods in real life Application • Explore java. Lang package	3.1 Fundamental of Class 3.2 Object fundamental 3.3 new Keyword 3.4 Member methods of a class 3.5 Instance method 3.6 Constructor and its types 3.7 Finalize method 3.8 Abstraction 3.9 Encapsulation 3.10 Using this keyword 3.11 Method overloading 3.12 Wrapper classes 3.13 Autoboxing and unboxing 3.14 Visibility Modes
Unit 4: Inheritance, Polymorphism, Interface and Packages	**8 hours**
• Implement the Inheritance and interface concept in application	4.1 Inheritance 4.1.1 Using Extends 4.1.2 Member access 4.1.3 Super classes and sub classes 4.1.4 Single Inheritance 4.1.5 Multi-level Inheritance 4.1.6 Hierarchy Inheritance 4.2 Constructor and inheritance 4.3 Polymorphism-Method overriding 4.4 Abstract class 4.5 Using final keyword 4.6 Interface

	4.6.1 Creating and implementing an interface 4.6.2 Extending interface 4.7 Packages 4.7.1 Creating Packages 4.7.2 Accessing a Package
Unit 5: Multithreading, Java Collections	**6 hours**
• Explain how threads work in the real time applications • Implement the java. Utility package	5.1 Introduction Multithreading 5.1.1 Thread class and Runnable Interface 5.1.2 Creating Threads 5.1.3 Stopping and Blocking a Thread 5.1.4 Life Cycle of a Thread 5.1.5 Using Thread Methods 5.1.6 Thread Priority 5.2 Collection Framework in Java 5.2.1 Overview of java collection framework. 5.2.2 Commonly used collection classes- ArrayList, Vector
Unit 6: Exception Handling, Java IO and Streams	**7 hours**
• Familiarize the students with the exception handling process in real life applications • Explain the role of Stream and file handling	6.1 Managing Errors and Exceptions 6.2 Syntax of Exception Handling Code 6.3 Types of Exceptions 6.4 Using try and catch keywords 6.5 Multiple catches 6.6 Using Finally Statement 6.7 Throwing Exceptions 6.8 Streams: Byte Streams and Character Streams 6.9 Reading / Writing Console Input / Output 6.10 Reading and Writing files
Unit 7: Java JDBC	**4 hours**
• Describe and gain knowledge of database connectivity in java	7.1 JDBC introduction 7.2 JDBC drivers 7.3 Connecting to Database 7.4 Querying a database and processing the results
Unit 8: JavaFX	**4 hours**
• Explore the GUI based programming.	8.1 Introduction to JavaFX 8.2 JavaFX Architecture 8.3 JavaFX Application Basics-Line, rectangle and text

6. Practical Works:

The laboratory works should cover all the features of Object Oriented Systems using Java so that students would be motivated to solve the real-life problems.

1. Program to define a structure of a basic JAVA program.
2. Program to define the data types, variable, operators.
3. Program to define the arrays, string and control structures.
4. Program to define class and constructors.

5. Program to define class, methods and objects and method overloading.
6. Program to define inheritance and show method overriding.
7. Program to demonstrate Packages.
8. Program to demonstrate Exception Handling.
9. Program to demonstrate Multithreading.
10. Program to demonstrate I/O operations, demonstrate JDBC and JavaFX.

By incorporating all the lab activities, a mini project should be developed and presented individually. Course instructor must include the mini project weightage in the practical evaluation.

7. Evaluation System and Students' Responsibilities:

7.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50				

7.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

8. Prescribed Books and References:

Text Books

1. Herbert Schildt, Java the Complete Reference, 11th Edition, Tata McGraw-Hill Publishing Co. Ltd.
2. E. Balagurusamy, *"Programming with Java: A Primer",* Edition, Tata McGraw Hill Publication.

Reference Books

1. Y. Daniel Liang, Introduction to Java Programming Brief Version, 8th Edition Kindle Edition.
2. Cay S. Horstmann, Core Java Volume I–Fundamentals, 11th Edition, Pearson 2019.

Pokhara University Faculty of Science and Technology	
Course Code: CMP 227 (3 Credits)	Full Marks: 100
Course Title: Data Structure and Algorithms (3-0-2)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: III	Program: Bachelor of Computer Application

1. Course Description:

This course introduces students to fundamental data structures and algorithms. It covers core concepts, analysis techniques, and practical implementation. Students will learn to design efficient algorithms, analyze their performance, and apply them to problem-solving. The course emphasizes hands-on practice through coding assignments and projects.

2. General Objectives:

The general objectives of this course are as follows:

1. To familiarize students with fundamental data structures (stacks, queues, linked lists, trees, graphs) and algorithms, including sorting and searching techniques.
2. To equip students to analyze algorithm efficiency using time complexity metrics (Best, Worst, and Average case).
3. To enhance problem-solving and critical thinking skills through practical assignments, projects, and the comparison of recursive and iterative methods.
4. To prepare students for industry challenges by integrating modern tools, techniques, and the application of fundamental algorithms to modern technological fields.

3. Methods of Instructions:

- Lecture and discussion
- Practical
- Demonstration
- Presentation

4. Course Contents:

Specific Objectives	**Contents**
Unit 1: Introduction to Data Structures and Algorithms	**7 hours**
• Describe how data structures and algorithms (DSA) contribute to effective problem-solving. • Compare time complexity and asymptotic notations (Big O, Omega, Theta) to analyze algorithms. • Explore about the impact of dynamic memory allocation on performance.	1.1. Introduction to Data Structures 1.1.1 Data Types and Data Structures 1.1.2 Classification of Data Structures 1.1.3 ADTs (Abstract Data Types) 1.2 Introduction to Algorithms 1.2.1 Definition and Characteristic 1.2.2 Need/Importance of DSA 1.3. Algorithm design Divided and Conquer, Greedy Algorithm, Backtracking, Dynamic Programming 1.4 Algorithm Analysis 1.3.1 Time Complexity 1.3.2 Best, Worst and Average Case 1.3.3 Rate of Growth 1.3.3 Asymptotic Notation (Big O, Omega and Theta)

Unit 2: Stack, Queues and Recursion	**8 hours**
• Explain the concept of stacks as an ADT. • Explore the role queue types (linear, circular, priority). • Implement the recursive problems.	2.1. Concept of stacks and Stack ADT 2.1.1 Stack operations 2.1.2 Stacks implementation using arrays 2.1.3 Applications of stacks: 2.1.3.1 Conversion from infix to postfix 2.1.3.2 Evaluation of postfix 2.1.4 Recursion 2.1.4.1 Concept of recursion 2.1.4.2 Recursive vs. iterative solutions 2.1.4.3 Recursive problems: Factorial, Fibonacci sequence, GCD, Tower of Hanoi 2.1.5 Recursion and stack 2.2. Concept of queues and Queue ADT 2.2.1 Queue operations 2.2.2 Queue Implementation using arrays 2.2.3 Types of Queues 2.2.4. Linear and circular 2.2.5 Introduction to priority queues and double ended queue.
Unit 3: Linked List	**8 hours**
• Implementation of about different types of linked lists: Singly Linked List, Doubly Linked List, and Circular Linked List. • Describe basic operations in linked lists such as node creation, insertion, and deletion at various positions. • Implement stacks and queues using linked lists, with a focus on doubly and circular linked lists.	3.1 List –Definition and List ADT, array Implementation 3.2 Linked List – Definition and its operations 3.3 Types of Linked Lists: 3.3.1 Singly Linked List 3.3.2 Doubly Linked List 3.3.3 Circular Linked List 3.4 Implementation of Singly Linked Lists: 3.4.1 Node Creation 3.4.2 Node Insertion (Beginning, End, Specified Position) 3.4.3 Node Deletion (Beginning, End, Specified Position) 3.6 Linked List implementation of Stack and Queue
Unit 4: Tree	**8 hours**
• Implement and analyze various tree traversal algorithms for binary trees. • Implement binary search trees, including insertion, deletion, and searching operations. • Explain the need for balanced trees, search trees, and their applications.	4.1 Tree concepts and terminology 4.2 Binary trees: 4.2.1 Definition and types 4.2.2 Tree representations (array, linked-based) 4.2.3 Tree Traversal (Pre, Post and In Orders) 4.2.4 Construction of binary tree from traversal 4.2.5 Application of Tree: Huffman Algorithm 4.3 Binary search trees: 4.3.1 Construction of BST 4.3.2 Basic Operation on BST (Insert, delete, search) Node

	4.4 Introduction to balanced trees 4.4.1 Need for balanced trees 4.4.2 Definition of AVL (Adelson-Velskii and Landis) trees, insertion, deletion and Rotation operations 4.5 B-tree Definition and application
Unit 5: Sorting	**6 hours**
• Analyze and implement various sorting algorithms, including simple and advanced techniques. • Compare the efficiency of different sorting algorithms using time complexity analysis. • Implement the workings of distribution-based sorting algorithms.	5.1 Definition and Classification of Sorting algorithm (internal, external, stable, unstable) 5.2 Bubble Sort 5.3 Selection Sort 5.4 Insertion Sort 5.5 Merge Sort 5.6 Quick Sort 5.7 Radix Sort 5.8 Heap Sort as priority queues
Unit 6: Searching Algorithm and Hashing	**4 hours**
• Implement and analyze Binary Search for efficient data retrieval. • Apply the fundamentals of Hashing, including Hash Functions and Hash Tables. • Explore Collision Resolution Techniques: Separate Chaining, Linear Probing, Quadratic Probing	6.1Concept of Searching algorithms 6.2 Linear and Binary Search 6.3 Hashing: 6.3.1Concept of Hashing, Hash functions, Hash Table 6.4 Collision resolution techniques: 6.4.1. Open Hashing: Separate Chaining 6.4.2. Closed addressing : Linear Probing , Quadratic Probing, Double Hashing 6.5. Load Factor and Rehashing
Unit 7: Graphs	**4 hours**
• Describe graph representations through its types. • Master graph traversal techniques (DFS, BFS, topological sort). • Apply graph algorithms for finding minimum spanning trees (Kruskal's, Prim's) and shortest paths (Dijkstra's). • Apply graph concepts to real-world problems and applications.	7.1. Graph concepts and representations (adjacency matrix, adjacency list) and types 7.2. Graph traversals: 7.2.1. Depth-first search (DFS) 7.2.2. Breadth-first search (BFS) 7.2.3. Topological Sort 7.3. Minimum Spanning Tree- Kruskal's Algorithm and Prim's Algorithm 7.4. Shortest path problems- Dijkstra's algorithm

5. Practical Work:

Students should focus on manually implementing these data structures from scratch. Students are encouraged to implement these ADTs without use of Java's built-in libraries. At the end of practical works, student can identify the performance among the implementation.

1. Implement stack, linear and circular queue using arrays.
2. Implement Infix to postfix conversion using stack
3. Implement recursive algorithms- Factorial, Fibonacci series and Tower of Hanoi
4. Implement singly link list, binary search trees and implement traversal algorithms.
5. Implement graph traversal algorithms (DFS, BFS).

6. Implement sorting algorithms (quick sort, merge sort, heap sort).
7. Implement searching algorithms (sequential, binary search).
8. Implement hash system (separate chaining, liner and quadratic).
9. Implement Dijkstra's algorithm for shortest path.
10. Implement Kruskal's or Prim's algorithm for minimum spanning tree.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

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Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

Text Books

1. Langsam, Y., Augustin, M.J. and Tanenbaum, A.M: Data Structure Using C and C++, Prentice Hall of India
2. Michael T. Goodrich and Roberto Tamassia *Data Structures and Algorithms in Java, 0-471-73884-0, Fourth Edition, John Wiley & Sons, Inc.*

Reference Books

1. Rowe, G.W.: Introduction to Data Structure and Algorithms with C and C++, Prentice Hall of India
2. Kruse, R. L., & Ryba, A. J. (1998). *Data structures and program design in C++*. Prentice Hall, India..
3. Brassard, G., & Bratley, P. (1996). *Fundamentals of algorithmic*. Prentice-Hall, India.

Pokhara University Faculty of Science and Technology	
Course Code: CMP 221 (3 Credits)	Full Marks: 100
Course Title: System Analysis and Project Management (3-0-1)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: III	Program: Bachelor of Computer Application

1. Course Description:

This course provides a comprehensive overview of the core elements of system analysis, design, implementation, and project management. Students will explore system concepts, the System Development Life Cycle (SDLC), and the critical role of system analysts. Key topics include requirements analysis techniques, system design principles, input/output and database design, as well as system architectures. Additionally, the course covers system implementation processes such as testing, documentation, and user training, while also delving into project management tools and techniques for effective planning, monitoring, and control. Through practical case studies, students will gain the skills necessary to apply these concepts in real-world scenarios.

2. General Objectives:

The general objectives of this course are to equip students with fundamental knowledge and skills in System Analysis, Design, and Project Management. By the end of the course, students will be able to:

1. To understand the fundamental concepts of systems, their characteristics, and the phases of the System Development Life Cycle (SDLC).
2. To apply various techniques for system requirements analysis, including fact-finding, modeling with Data Flow Diagrams (DFDs), Entity-Relationship Diagrams (ERDs), and use case modeling.
3. To understand the processes and challenges of system implementation, including testing, documentation, and providing user support.
4. To manage project effectively through scheduling, monitoring, controlling, and risk management to ensure successful project outcomes.

3. Methods of Instructions:

- Lecture and discussion
- Practical
- Demonstration
- Presentation
- Case study

4. Course Contents:

Specific Objectives	Contents
Unit 1: Introduction to System Analysis and Design	**6 hours**
• Explain the fundamental concepts of systems analysis and design, and their importance in software systems. • Analyze the phases of the Systems Development Life Cycle (SDLC) and perform feasibility analysis.	1.1 **Overview of System Analysis and Design** • Introduction to system analysis and design concepts. 1.2 **Systems Development Life Cycle (SDLC)** • Phases of SDLC: Planning, Analysis, Design, Implementation, and Maintenance. • Introduction to system requests and feasibility analysis (technical, economic, legal, and organizational). 1.3 **Role and Skills of the Systems Analyst** • Introduction of programmer and system analyst • Skills and responsibilities of the systems analyst. • Characteristics of a system analyst • Interaction with stakeholders in system development projects. • The systems analyst as a bridge between technical teams and business stakeholders.

Unit 2. Project Management Fundamentals	**12 hours**
• Apply project selection and planning techniques to initiate and plan a system project. • Utilize project management tools like Gantt charts, CPM, PERT, and EVM to monitor and control project activities.	2.1 **Project Selection and Planning** • Selection methodologies and criteria for project approval. • Project planning: time, schedule and cost. 2.2 **Project Methodologies** • Traditional verses Agile methodologies. • Choosing the appropriate project management methodology. 2.3 **Managing and Controlling Projects** • Managing scope, time, and resources. • Staffing and coordinating project activities. • Risk management strategies. 2.4 **Project Management Tools** • Use of work breakdown structure (WBS), Gantt charts, Critical Path Method (CPM), and Program evaluation review technique (PERT). • Earned value management (EVM) and Key performance indicators (KPIs) for project monitoring and control.
Unit 3: Requirements Determination and Use Case Modeling	**8 hours**
• Implement requirements gathering techniques such as JAD, interviews, and observation to define system requirements. • Develop use case diagrams and descriptions to capture functional system requirements.	3.1 **Requirements Determination** • Techniques for gathering requirements: Interviews, Joint Application Development (JAD), document analysis, and observation. • Developing the requirements definition statement. 3.2 **Use Case Analysis** • Identifying use cases and creating use case diagrams. • Use cases as tools for capturing functional requirements, testing, and design
Unit 4: Process and Data Modeling	**8 hours**
• Create Data Flow Diagrams (DFDs) and Entity-Relationship Diagrams (ERDs) to model business processes and data structures. • Validate and refine process and data models for accuracy and completeness.	4.1 **Process Modeling** • Creating Data Flow Diagrams (DFDs) to define business processes. • Levels of DFDs: Context diagrams, Level 0, and Level 1 DFDs. • Process description and validation of models. 4.2 **Data Modeling** • Introduction to Entity-Relationship Diagrams (ERDs) for database modeling. • Elements of ERDs: Entities, attributes, relationships. • Creating and normalizing ERDs, and balancing ERDs with DFDs. • Data dictionary
Unit 5: System Design and Architecture	**8 hours**
• Design system architectures, including client-server and cloud-based systems, based on business requirements. • Apply user interface design principles to create and evaluate effective system interfaces.	5.1 **Transition from Requirements to Design** • Custom development vs packaged software vs outsourcing. • Developing system acquisition strategies based on business needs. 5.2 **System Architecture Design** • Designing client-server architectures and multi-tiered systems. • Introduction to cloud architecture and its relevance to modern system design. • Operational, performance, and security considerations in architecture design. 5.3 **User Interface Design** • Principles for effective UI design: Layout, aesthetics, consistency, and usability. • Prototyping and evaluating user interfaces.
Unit 6: Project Implementation and Post-Implementation Activities	**6 hours**
• Coordinate programming, testing, and system migration	6.1 **Managing the Programming Process** • Coordinating programming tasks and managing schedules.

strategies to ensure successful system implementation. • Conduct post-implementation reviews and manage ongoing system maintenance to meet organizational needs.	• Testing strategies: Unit tests, integration tests, system tests, and acceptance tests. 6.2 **System Transition and Change Management** • Migration strategies: Direct, parallel, phased, and pilot conversions. • Managing user resistance, training, and change management. 6.3 **Post-Implementation Reviews and Maintenance** • Conducting post-implementation reviews and ensuring system support. • Ongoing system maintenance and updates.

5. Practical Works:

The laboratory works should involve hands-on implementation and evaluation of concepts through unit-wise case studies, utilizing appropriate software tools. Students should work in groups to visualize and present their findings on systems analysis and project management.

1. Real-world application of systems analysis and project management concepts.
2. Hands-on experience with project management tools (Gantt charts, CPM, PERT) and modeling tools (DFDs, ERDs).
3. A comprehensive project involving system analysis, design, and management tasks.
4. Explore prototyping and methodologies such as RUP and CASE tools.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

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Total Internal Marks		**50**		
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

Text Books

1. Dennis, A., Wixom, B. H., & Roth, R. M. (2012). *System analysis and design*. John Wiley & Sons, Inc. http://www.wiley.com/college/dennis

Reference Books

1. Jeffery Whitten, Lonnie Bently, System Analysis Design Methods 7/E McGraw-Hill
2. J.F. George and J. S. Valacich, *Modern System Analysis and Design*, 8/E. Pearson Education Inc., 2017

3. J. Satzinger, R. Jackson and S. Burd, System Analysis and Design in a Changing World, 7/E, Cengage Learning, 2016s

Pokhara University **Faculty of Science and Technology**	
Course Code.: CMP 380 (3 Credits)	Full Marks: 100
Course Title: Web Technology I (3-0-3)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: III	Program: Bachelor of Computer Application

1. Course Description:

This course is designed to make solid foundation in web development. Students start by exploring the evolution of the web and understanding key concepts like client-server architecture and web protocols. The course covers HTML, CSS, JavaScript and advanced features to make web pages interactive including version control, UI/UX, wireframing, and SEO.

2. General Objectives:

The general objectives of this course are as follows:

- To provide how to create web pages using HTML, including working with elements, attributes, and various components like lists, links, images, and forms.
- To enable students to style and layout web pages using CSS, covering basics, the CSS box model, responsive design, and advanced layout techniques like Flexbox and Grid.
- To build students' JavaScript skills, including how to write scripts, handle events.
- To provide an overview of JavaScript frameworks and libraries, including jQuery, React and introduce popular frameworks like Angular, and Vue.js.
- To offer a foundation in web design principles, including UI/UX design, wireframing, typography, color theory, usability, accessibility, and SEO.

3. Methods of Instruction:

- Lecture and discussion
- Practical
- Demonstration
- Presentation
- Project

4. Course Contents:

Specific Objectives	**Contents**
Unit 1: Introduction to Web Technologies	**4 hours**
• Describe the client-server architecture and the roles of web browsers and servers. • Explain basic web protocols, DNS, and the differences between client-side and server-side scripting languages	1.1 Evolution of the Web 1.1.1 History of web 1.1.2 WWW 1.1.3 URL 1.1.4 Version of web 1.1.5 Search engine 1.1.6 Website and Web application 1.2 Client-Server Architecture 1.3 Overview of Web Browsers and Web Servers 1.3.1 Web browse, cross browser compabilities 1.3.2 Web server 1.4 Basic Protocols 1.4.1 Short overview of basic protocols

	1.4.2 HTTP/HTTPS 1.4.3 HTTP request and resoponse 1.5 DNS and its hierarchy 1.6 Client side and server side scripting language
Unit 2: HTML	**8 hours**
• Apply the HTML structure, tags and attributes in web design. • Implement lists, links, images, tables, and forms, and use semantic HTML for better web organization.	2.1 Introduction 2.1.1 History and evolution of HTML 2.1.2 Concept of tag 2.1.3 HTML document structure 2.1.4 HTML elements and attribbutes 2.1.5 Basic text formatting tags 2.2 Lists, Links and Images 2.2.1 Unordered, ordered and description lists, nested list 2.2.2 Creating hyperlinks, types of link 2.3 Adding images 2.4 Table and Forms 2.4.1 html table elements and attributes, nested table, mearging row and columns, grouping section of table 2.4.2 Common form elements like text fields, text areas, selects, buttons, radio button. 2.4.3 input date, time 2.4.4 form methods 2.5 Semantic HTML 2.5.1 Header, footer, nav, article, section, time, progress 2.6 Audio and video elements 2.7 div, span, aside, iframe, canvas
Unit 3: Cascading Style Sheets	**8 hours**
• Implement CSS including its syntax, box model, and layout techniques	3.1 Need of CSS 3.2 CSS Basics 3.2.1 CSS syntax, selectors, declarations, types, groping selectors, pseudo, comments, 3.2.2 CSS Specificity 3.2.3 Internal vs external CSS 3.2.4 Style text fonts, color, links 3.3 CSS Box Model 3.3.1 Box model properties for padding, border, width 3.3.2 Centering content with margins 3.3.3 Element backgrounds 3.4 Flexbox and Grid Layouts 3.5 Responsive Design Principles 3.5.1 Working with different screen size 3.5.2 Media queries in CSS 3.6 CSS Preprocessors (SASS, LESS)
Unit 4: JavaScript Basics	**8 hours**

• Explain the core concepts of JavaScript through variables, data types, and control structures, and to show how to use functions, manipulate the DOM, and handle events to make web pages interactive.	4.1 Introduction to JavaScript 4.2 Embeeding JavaScript to a webpage 4.2.1 External Script 4.2.2 Internal Script 4.2.3 Using onClick(), onSubmit(), onLoad() 4.3 Variables, Data Types, Operators, Expressions 4.4 Java Script Programming Construct: 4.4.1 Assignment, data declaration, 4.4.2 if, if else, switch 4.4.3 while, for, do while 4.4.4 label, break, continue 4.4.5 Arrays 4.5 Functions and Scope 4.6 Dialog boxes 4.7 Document Object Model 4.8 Event handling.
Unit 5: Advanced JavaScript Concepts	**6 hours**
• Analyses into advanced JavaScript features such as ES6+ syntax, promises, and async/await, and to understand how to handle errors and work with JSON and AJAX for more sophisticated web interactions.	5.1 ES6+ Features: 5.1.1 Let, Const, Arrow Functions, Promises, etc. 5.1.2 Objects 5.2 Regular Expressions 5.3 Asynchronous Programming (Async/Await) 5.3.1 Promises 5.3.2 Async/Await 5.3.3 Fetch 5.4 Error Handling 5.5 Introduction to JSON 5.5.1 JSON Syntax and Structure 5.5.2 Simple Example 5.6 Introduction to AJAX
Unit 6: Introduction to JS Frameworks and Libraries	**5 hours**
• Familiarize with JavaScript libraries like jQuery for easier DOM manipulation and to explore popular frameworks/library such as React, Angular, and Vue.js.	6.1 Jquery() 6.1.1 Why jQuery 6.1.2 Jquery fundamentals 6.1.3 Manipulating DOM elements. 6.2 Overview of Popular Frameworks/Library (React, Angular, Vue.js) 6.3 Choosing the Right Framework for Projects
Unit 7: Web Design Concepts	**5 hours**
• Explore the principles of UI/UX design, and apply concepts of typography, color theory, usability, accessibility, and SEO to create well-designed, user-friendly websites	7.1 Principles of User Interface Design 7.2 Introduction to UI/UX Design 7.2.1 Wireframing with its types 7.2.2 User Story 7.2.3 Responsive Design 7.3 Typography and Color Theory

	7.4 Usability and Accessibility Guidelines 7.5 SEO
5. Practical Works:	

Students should perform following HTML lab tasks:

- In the head section of a web page, make sure to:
  - Add a title tag to set the page's title, which appears in the browser tab.
  - Link to external stylesheets using relative paths to apply the desired styles.
  - Include meta tags to specify important information such as the character encoding, description, keywords, author, and viewport settings for responsive design.
- In the body of a web page, include:
  - A table with headers, rows, and cells, adjusting attributes for spanning and sizing.
  - Text, images, sound clips, and videos with proper controls, sizing, and alternate text.
  - The `<div>` tag for styling and layout.
  - Text elements using tags like `<h1>`, `<p>`, and `<li>`.
  - Classes and styles for formatting, including lists.
- Create hyperlinks from text and images to:
  - bookmarks on the same page
  - other locally stored web pages
  - a website using the URL
  - send mail to a specified email address
  - to open in a specified location (the same window, a new window)
- Create form for:
  - Login and signup purpose
  - Design registration form including Textbox, TextArea, Select, Checkbox, Radio button, Button elements and group them using fieldset and legend.
- Use elements to break down in different blocks.

CSS Labs

- Set background properties including colour and images, font properties, style table including size, background colour, horizontal and vertical alignment, spacing, padding, borders.
- Create external styles to be tagged in a web page including h1, h2, h3, p, li.
- Attach comments to an external stylesheet.
- Center a div with its elements.
- Design navigation menu for web page with proper stylling, when user put mouse over the menu the menu should be hilighted.
- Apply css to image, links, list
- Create product grid including product name, price, image, add to cart button and add effects when user put crouser on the product. Also make it responsive.

JavaScript:

- Link an external JavaScript file and manipulate DOM elements.
- Add and remove textbox from web page with add and delete button.
- Write a script to input information like Name, Address, Contact, Age abd display them in proper format when user clicks the submit button.
- Validate form inputs with JavaScrips
  - Validate login and signup form. the password should contain atleast letters, symbole and number and should be minimum 10 charactors, username field cannot be blank, to sign up user must be audult.
- Load another page inside a page when user click on load button without reloadind a page.

- Manupulate DOM emelents using jQuery.

VCS:

- Start a Git repository with **git init**, stage files with **git add**, and commit with **git commit** make some changes and push with **git push**

UI/UX:

- Prepare Low-Fidelity wireframe of products and product details page.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

1. Felke-Morris, T. (2020). *Web development and design foundations with HTML5*. Pearson.
2. Simpson, K. (2021). *You don't know JS yet: ES6 & beyond*. Independently published
3. Chacon, S., & Straub, B. (2014). *Pro Git*. Apress.

Pokhara University Faculty of Science and Technology	
Course Code.: CMP 230 (3 Credits)	Full Marks: 100
Course Title: Operating Systems (3-0-1)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: III	Program: Bachelor of Computer Application

1. Course Description:

This course is designed to provide the concepts of operating system to various system environments. It incites system software, internal structures, functions and security features of operating system to run the application software and perform various tasks with optimized throughput. This course also introduces the emerging new trended operating system for distributed environment like cloud and mobile systems. After completion of this course, students can select, apply and implement the operating system features and functions for the best utilization of the device specification.

2. General Objectives:

The general objectives of this course are as follows:

- To acquaint the students with structure of operating systems and their functionality.
- To acquaint the students with concepts of resource allocation and management with schedules.
- To acquaint the students with the knowledge of process and thread, I/Os, Memory, CPU, disk management, network security systems.
- To acquaint the students with basic concepts of operating systems new trends such as real-time, distributed, cloud systems and mobile.

3. Methods of Instructions:

- Lecture and discussion
- Practical
- Demonstration
- Case studies

4. Course Contents:

Specific Objectives	Contents
Unit 1: Basics of Operating System 6 hours	
• Familiarize with basic concepts of Operating systems, and its structures. • Explain the features of generation and types of operating systems.	1.1 Introduction, goals of operating system 1.2 Operating-System Structures and functions 1.3 Types of Operating Systems *Batch Processing, Multiprogramming, Multiprocessing, Networking, Real Time (RTOS), Distributed, Embedded system* 1.4 System Interface, System Calls 1.5 Virtual Machine
Unit 2: Process, Threads and Scheduling 13 hours	
• Familiarize with task, process and threads • Implement of resource allocation techniques • Describe the mutual exclusion for resource utilization	**2.1 Process** 2.1.1 The Process Model 2.1.2 Process Creation and Process Termination 2.1.3 Process Hierarchies 2.1.4 Process States 2.1.5 Implementation of Processes **2.2 Threads** 2.2.1 Thread Usage 2.2.2 Threads Models (Many-to-one model, One-to-One Model, Many-to many model)

	2.2.3 User Space and Kernel Space Threads 2.2.4 Hybrid Implementations 2.2.5 Difference between Threads and Processes **2.3 Inter-process Communication** 2.3.1 Race Conditions 2.3.2 Critical Regions 2.3.3 Mutual Exclusion with Busy Waiting 2.3.4 Sleep and Wakeup, Semaphores, Mutexes, Monitors, Message Passing 2.3.9 Avoiding Locks: Read-Copy-Update **2.4 Process Scheduling** 2.4.1 Basic Concept 2.4.2 Type of scheduling (Preemptive scheduling, Non-preemptive scheduling, batch, Interactive, real time scheduling), 2.4.3 Scheduling Criteria or performance analysis, Scheduling Algorithm (Round-robin, First come first served, Shortest-job- first, Shortest process next, Shortest remaining Time next, real time, priority fair share, guaranteed, Lottery scheduling) **2.5 Deadlocks** 2.5.1 System Resources: Preemptable and Non-preempable 2.5.2 Method of handling Deadlocks, 2.5.3 Deadlock prevention 2.5.4 Deadlock avoidance: Banker's Algorithm, *2.5.5* Protection- *System Protection, Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix, Implementation of Access Matrix, access Control list* **2.6** Research on Processes and Threads
Unit 3: Storage Management 13 hours	
• Explain the role and working procedure of memory • Familiarizing with virtual memory management • Implement the page replacement algorithms • Explain the mechanism file and filing	**3.1 Memory Management** 3.1.1 Logical & physical Address Space 3.1.2 Swapping 3.1.3 Contiguous Allocation 3.1.4 Paging, Structure of Page Table 3.1.5 Segmentation, Segmentation with Paging **3.2 Virtual Memory** 3.2.1 Background 3.2.2 Demand Paging, 3.2.3 Performance of Demanding Paging, 3.2.4 Page Replacement, Page Replacement Algorithms, Allocation of Frames, 3.2.5 Thrashing. **3.3. File System Interface and Implementation** 3.3.1 File System Interface- *The Concept of a File, Access methods, Directory Structure, File System Mounting, File Sharing, Protection* 3.3.2 File System Implementation- *File System Structure, File System Implementation, and Allocation methods, Free-space Management, Directory Implementation, Efficiency and Performance*

Unit 4: Input/output Management 7 hours	
• Describe the role of input/output devices • Explain the different approaches for optimal output	4.1 Principles of I/O Hardware- *I/O Device, Device Controller, Memory Mapped I/O, Direct Memory Access* 4.2 Principles of I/O Software- *Goals of I/O Software, Polled I/O verses Interrupt Driven I/O, Character User Interface and Graphical User Interface, Device Driver, Device Independent I/O Software, User-space I/O Software* 4.3 Mass Storage Structure - Overview of Mass Storage Structure, Disk Structure, Disk Attachment, Disk Scheduling, Disk Management, Swap space Management Redundant Array of Inexpensive Disks , RAM Disks, Optical Disk
Unit 5: New Trend in Operating System 4 hours	
• Familiarize with current trends of operating systems	5.1 Concept, character and role of Distributed, Cloud, Mobile and Multimedia operating systems 5.2 Memory wall and bottleneck for operating system
Unit 6: Case Study	**5 hours**
• Compare operating system structure and functionality	**6.1 UNIX/Linux -** Design principles, Inter-process communication, Kernel modules, Network structure **6.2 Windows -** Design principles, Programmer interface, System components **6.3** Security level in Linux/Windows

5. List of Practices:

Laboratory work should cover the operating system structure and functions of any two popular operating system. It also insist the students to design model of operating system with the reference of open source guideline. Students should complete the following tasks in laboratory:

1. Understanding and running all the internal command and external commands in Microsoft Disk Operating Systems.
2. Installation and user, application management in Windows (current version)
3. Simulation of Process Scheduling Algorithms
4. Simulation of Page Replacement Algorithms
4. Simulation of Disk Arm Scheduling Algorithms
5. System Administration (user, disk, role, etc.) in any open source operating system.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**		

Attendance / Class Participation	10%		**Semester End examination**	**50**
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

Books

1. Andrew S. Tanenbaum, Herbert Bos, *"Modern Operating Systems"*, Pearson, 4th Edition, ISBN-10: 0-13-359162-X. 2011

References

1. A. Silberschatz, P.B. Galvin, G. Gagne *"Applied Operating System Concepts*", Wiley, 8th Ed.
2. D. M. Dhamdhere , "*System Programming and Operating System*" - Tata McGraw-Hill, 20
3. Andrew S. Tanenbaum, *"Distributed Operating System*", Pearson
4. Naresh Chauhan, Principles of Operating Systems, Oxford University Press 2014, Oxford University Press, ISBN-13: 978-0-19-808287-3

Pokhara University Faculty of Science and Technology		
Course Code: ENG 121 (3 Credits)		Full Marks: 100
Course Title: English for IT Professionals (3-0-1)		Pass Mark: 45
Nature of the Course: Theory + Practical		Total Lectures: 48 hours
Level: Bachelor	Year: I/ Semester: I	Program: BCA

1. Course Description:

This course is designed to cover grammar, vocabulary, reading and writing components. Grammar incorporates the essential aspects of technical English usage in context. Vocabulary covers words from different academic fields. The reading component deals with a wide variety of carefully selected materials. They include, among other things, informative passages on contemporary and critical issues. The writing part includes materials geared to developing various writing skills required for effective communication on matters of general and academic interests.

2. General Objectives:

The general objectives of this course are as follows:

a) To introduce students to various skill sets for technical writers and involve them in collaborative writing processes
b) To assist students in refining their language usage, emphasizing grammatical accuracy and proficiency in writing.
c) To support students in enhancing their professional letter-writing skills, with particular emphasis on presentation and organization.
d) To foster students' reading proficiency, the course will offer essays and accompanying questions, encouraging active engagement to enhance their comprehension abilities.

3. Methods of Instructions:

- Lecture and discussion
- Demonstration
- Presentation
- Project work
- Self-study, etc.

It is expected that students are fully engaged with the teacher in subject matters and lessons to excel their interactive and presentation skills.

4. Course Contents

Specific Objectives	Course Content
Unit 1: Foundation of English Language	**8 hrs**
• Explain the basic concept of structures of English and academic language, focus on grammatical structures and developing vocabularies	1.1 Introduction to English Language 1.2 Academic English Basics 1.3 Parts of Speech 1.4 Word formation: prefixes and suffixes 1.5 Synonyms and antonyms 1.6 Phrasal verbs 1.7 Sentence structures: Simple, compound and complex 1.8 Voice- Active and passive voice 1.9 Verities of English

Unit 2: English Language Terminology	**6 hrs**
• Describe the role of appropriate tenses, prepositions and articles • Explain the use of supply the appropriate words	2.1 Tense 2.2 Prepositions 2.3 Article 2.4 Word order 2.5 Countable and uncountable nouns 2.6 Conjunctions, relative clauses
Unit 3: Review and Critical Thinking	**5 hrs**
• Explain the role of critical analysis, reviews orally	Reviews and critical thinking
Unit 4: General Communication Skills	**5 hrs**
• Describe the objectives and characteristics of Communication • Explain the ways of develop effective presentation strategies	4.1. Listening and speaking 4.2. Active listening 4.3. Effective speaking 4.4. Effective presentation strategy 4.5. Group communication
Unit 5: Reading and Writing	**5 hrs**
• Compare the ideas for reading texts and extract general ideas	5.1 Reading and its techniques 5.2 Technical writing 5.3 Paragraphs and essays
Unit 6: Leading and Misleading of Technical Communication	**5 hrs**
• Explain for the workplace; Style in technical prose	6.1 Ethics at work 6.1.1 Ethics at framework 6.1.2 Ethics for students 6.1.3 How is ethics related to technical communication? 6.2 Writing ethically
Unit 7: The Essentials of Writing	**8 hrs**
• Describe the various technical essays • Explain the characteristics of writing	7.1 The Structure of an Essay 7.1.1 The Introduction 7.1.2 Body Paragraphs 7.1.3 The conclusion 7.2 The writing processes
Unit 8: Reading Comprehension and Summary Writing	**6 hrs**
• Explore on language construction and find the theme	8.1 Note taking, summary and precis writing. 8.2 Computer makes the world smarter and smaller (Nancy Stern & Robert A. Stern) 8.3 How to become a programming expert? (Erich H. & John Ewan) 8.4 The Future of Information Technology (Erich H. & John Ewan).

5. List of Practical

Classwork should cover all the concepts of English language, highlighting grammatical structures. Also students should develop the presentation, group discussion, effective speaking and along with writing skills to solve the following problems as classwork.

- Conversation between the teacher and student, employer and employees
- Find the leading and misleading technical communication
- Implement the four elements of English language- speaking, listening, writing and reading
- Technical essay writing
- Writing summary and note taking

Students should submit a final classwork report that uses concepts of IT and academic language in this course. The marks for the practical will be based on the classwork, assignment and the project work.

6. List of Tutorials

The various tutorials activities that suit your course should cover all the content of the course to provide students a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments or classwork to the instructor for evaluation. The following tutorial activities should be conducted to cover the content of this course.

A. Discussion-based tutorial

1. Develop a technical writing and its skills. (class discussion)
2. Oral presentation in a group (topic will be assigned by subject teacher)
3. Purpose, structure and benefits of technical writing

B. Problem solving-based tutorials

1. Focus on text, an improvement in vocabulary and develop writing skills
2. Language competence through daily exercises
3. Visual Communication Exercise – Any two examples of visual communication (Technology, Communication, Employment, etc. Business Result *Elementary Student's Book by David Grant, John Hughes & Rebecca Turner*).

C. Review and Question/Answer based tutorials

1. Case study on general communication skills followed by oral presentation in class

7. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**40**	**Semester End**	**50**
Attendance and Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical/Assignment/Classwork		**10**		
Attendance/Class Participation	10%			
Tutorials	30%			
Assignment/Classwork	40%			
Viva	20%			
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Book:

1. Oxford English for Information Technology (Erich H. & John Ewan)

Reference Books:

1. Swan, M. (2005). *Practical English Usage* (Vol. 688). Oxford: Oxford university press.
2. Gardner, P., 2005. *New directions: Reading, writing, and critical thinking*. Cambridge University Press.
3. Graves, H. and Graves, R., 2012. *A Strategic Guide to Technical Communication-(US)*. Broadview Press.

Pokhara University Faculty of Science and Technology		
Course Code: MTH 131 (3 Credits)		Full Marks: 100
Course Title: **Mathematics (3-0-0)**		Pass Mark: 45
Nature of the Course: Theory		Total Lectures: 48 hours
Level: Bachelor	Year: I/ Semester: I	Program: BCA

1. Course Description:
This course includes set theory and real number system, functions and graphs, differential calculus and its application, partial derivative and its application, symbolic logics, matrices and determinants which are essential as mathematical foundation for computing.
2. General Objectives:
The general objective of this course is to provide the students with basic mathematical skills required to understand Computer Application Courses.
3. Methods of Instructions:
Lecture, Tutorial, Discussion, Assignments and Practical works.

4. Contents in Detail

Specific Objectives	Contents
• Solve problems related to sets, Absolute value and intervals.	**Unit 1: Set Theory and Real Number (6 Hrs.)** 1.1 Introduction 1.2 Types of sets 1.3 Operations on Sets (Union, Intersection, Difference, Complement) 1.4 Venn diagram 1.5 Cardinal Number of Set and Problems Related to Sets 1.6 Real Number 1.6.1 Types of real numbers 1.7 Absolute value of real numbers 1..8 Open and closed intervals
• Solve the problem related to functions and graph • Identify and plot the curves using any software tools (MATLAB/ Octave/ Mathematica, etc.) • Solve inequality and draw in graph.	**Unit 2: Functions and Graphs (5 Hrs)** 2.1 Definition of function 2.2 Domain and Range of a function 2.3 Inverse function 2.4 Type of functions: Constant, Algebraic (linear, Quadratic, Cubic), Trigonometric, exponential and Logarithmic and their graphs. Composite function. 2.5 Linear inequality and their graph (up to two variables)
• Solve the problems related to limit, continuity, discontinuity, demand and profit function.	**Unit 3: Limits & Continuity of Functions (5 Hrs.)** 3.1 Introduction 3.2 Limit of functions 3.3 Continuity & discontinuity of functions 3.4 Demand & Profit function

• Solve the problems related to derivative and its applications. • For visualization of curve use any software tools (MATLAB/ Octave/ Mathematica, etc.)	**Unit 4: Derivative and its application (10 Hrs.)** 4.1 Introduction 4.2 Techniques of differentiation 4.3 Derivative of algebraic, exponential, logarithmic & simple trigonometric functions. 4.4 Higher order derivative (up to 3rd order) 4.5 Application of derivative 4.5.1 Technique of finding limits (using L-hospital rule) 4.5.2 Increasing & decreasing function 4.5.3 Maxima & minima of function of one variable 4.5.4 Concavity of the function 4.5.5 Inflection point 4.6 Average cost & Marginal cost 4.6.1 Average revenue & marginal revenue 4.6.2 Profit maximization under perfect competition 4.6.3 Profit maximization under monopoly
Find partial derivatives and its applications.	**Unit 5: Multivariable Functions and Partial Derivatives (8 Hrs.)** 5.1 Introduction 5.2 Partial derivatives 5.3 Homogeneous function 5.4 Euler's theorem 5.5 Extreme values and saddle points for multivariable functions. 5.6 Lagrange multipliers
Solve the problems related to Logics.	**Unit 6: Symbolic Logics (6 Hrs.)** 6.1 Introduction 6.2 Statements 6.3 Logical connectives 6.4 Conjunction, Disjunction, Negation, conditional or Implication, Bi-conditional 6.5 Logical equivalence 6.6 Negation of compound events 6.7 Tautology & contradiction 6.8 Mathematical induction
• Solve the problems related to matrices and determinants. • Operation of matrices and determinants by using any software tools (MATLAB/ Octave/ Mathematical, etc.)	**Unit 7: Matrices and Determinant (8 Hrs.)** 7.1 Introductions of Matrices 7.2 Types of Matrices 7.3 Equality of Matrices 7.4 Algebra of Matrices 7.5 Transpose, Minors and Cofactors of Matrix 7.6 Determinant 7.7 Properties of determinants 7.8 Inverse of matrices 7.9 Singular and non-singular matrix 7.10 Solution of Linear equation by matrix inversion method, Cramer's rule, Gauss elimination method

5. List of Tutorials	
1.	Problem of sets, absolute value and intervals.
2.	Function and graph, domain and range, linear inequality and their graph.
3.	Limits, continuity, demand and profit function.

4.	Derivative of algebraic, exponential, logarithmic & simple trigonometric functions. Higher order derivative, Increasing & decreasing function, Maxima & minima of function of one variable, Average cost & Marginal cost, Profit maximization under perfect competition, Profit maximization under monopoly.
5.	Partial derivative, Verification of Euler's theorem, Extreme values and saddle points for multivariable functions, Lagrange multipliers.
6.	Exercise for each topic, Mathematical induction.
7.	Algebra of Matrices, Transpose, Minors and Cofactors of Matrices, Evaluation of determinant by using properties, Inverse of matrices, Solution of Linear equations.

6. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**50**	**Semester End**	**50**
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

Text Books:

1. Budnick F.S. ***Applied Mathematics for Business Economics and the Social Sciences*** McGraw-Hill Ryerson Limited
2. Thomas and Finney ***Calculus*** Pearson

References:

1. D.R. Bajracharya, R.M. Shrestha &et al**,** ***Basic Mathematics I, II*****,** Sukunda Pustak Bhawan, Nepal
2. Rudra Pratap ***Getting Started with MATLAB***, Oxford University Press 2010
3. Yamane,Taro: ***Mathematics for Economist***, Prentice Hall of India

Pokhara University Faculty of Science and Technology		
Course Code: ELX 111 (3 Credits)		Full Marks: 100
Course Title: **Digital Logic Systems (3-0-1)**		Pass Mark: 45
Nature of the Course: Theory and Practical		Total Lectures: 48 hours
Level: Bachelor	Year: I/ Semester: I	Program: BCA

1. **Course Description**

This course is designed to covers the concepts of digital logic systems. This course includes the role of information representation, fundamental concept of computing, principles and properties of Boolean algebra and its application in simplification, circuit analysis and gate implementation. It also covers the use of flip-flops in the design of synchronous and asynchronous sequential logic circuits. Finally, students will be capable to study functional units of simple computer ALU. Group discussion, lab works, simulation through software tools and group project are major instructional approaches.

2. **General Objectives**

The course is designed with the following general objective:

- To provide basic knowledge of logic systems.
- To familiarize student with the basic tools to study various digital circuits.
- To develop the skill among the student about to analyze the problem and develop concept for software development.

3. **Methods of Instruction**

- Lecture
- Tutorial
- Laboratory work
- Group discussion
- Grouped project work

4. **Contents in Detail**

Specific Objectives	Contents
• Explain the number representation in different number systems and their conversions • Identify the method of computing and role of complements in subtraction • Compare among binary codes, alphabets and other characters in secure computing.	**Unit 1 : Number system and Coding (8 hours)** 1.1 Analog and digital system- introduction and advantages 1.2 Digital computer and numerical representation 1.3 Number system 1.3.1 Decimal 1.3.2 Binary 1.3.3 Octal 1.3.4 Hexadecimal 1.4 Conversion among binary, octal, decimal and Hexadecimal number systems 1.5 Complements (Radix and diminished radix for binary and decimal only) 1.6 Subtraction using complements 1.7 Codes 1.7.1 Weighted codes (BCD, 84-2-1, 2421)

	1.7.2 Non weighted codes (Excess 3 , Gray) 1.8 Alphanumeric and instruction codes
• Interpret facts, theorems and principles based on Boolean algebra. • Emphasize on logic gates, Boolean function and truth tables and gate implementation.	**Unit 2 : Boolean algebra and Logic gates (5 hours)** 2.1 Basic definition and properties of Boolean algebra 2.2 Postulates and theorems of Boolean algebra 2.3 Logic gates, truth tables and Boolean function 2.4 Duality principle and complements 2.5 Implementation of NAND and NOR gates
• Explain the Boolean function simplification methods • Identify role of don't care conditions while simplifying Boolean functions.	**Unit 3 : Simplification of Boolean functions (6 hours)** 3.1 Venn diagram 3.2 Canonical forms and standard forms 3.3 Karnaugh map up to four variables 3.4 Don't care conditions 4.5 Simplification in SOP and POS using K-map
• Describe how to design various combinational logic circuits • Describe internal logic diagram, operation and circuit implementation	**Unit 4 : Combinational logic (10 hours)** 4.1 Adders and Subtractors 4.3 Code conversion *(BCD, 84-2-1, 2421, Ex-3 and Gray)* 4.4 Analysis procedure 4.5 NAND and NOR implementation 4.6 Parity generation and checking 4.7 Binary Parallel adder and subtractor 4.8 BCD adder 4.9 Decoder and encoder 4.10 Multiplexer and demultiplexer 4.11 ROM and PLA
• Explain the type of flip flops with their role in sequential circuits • Identify the key characters of each flip-flop for Implementation	**Unit 5 : Sequential logic (6 hours)** 5.1 Block diagram of sequential circuit 5.2 Flip flop (RS, D, JK, T, Master-slave), truth table, excitation table and characteristic equation 5.3 Triggering of flip flops 5.4 State diagram and state table
• Describe about registers, shift registers and types with timing sequences • Design synchronous, asynchronous and Mod counters	**Unit 6 : Registers and Counters (5 hours)** 6.1 Register, Register with parallel load 6.2 Shift register and types of shift register 6.3 Ripple counters (Binary and BCD counter) 6.4 Up-down counter, Mod-6 counter, Johnson
• Describe about read and write operation in RAM • Describe about the processor unit and its diagram	**Unit 7 : Functional Units of Memory and ALU (8 hours)** 7.1 Random access memory 7.2 Arithmetic logic unit 7.3 Status register 7.4 Processor unit

5. **Laboratory work:**
   - Familiarization with logic gates.
   - Familiarization with boolean functions.
   - Design of simple combinational circuits.
   - Adder and subtractor
   - Encoder and decoder
   - Multiplexer and demultiplexer
   - Demonstration of working principles of flip flops.
     Experiments are demonstrative either in trainer Kit or Simulation using appropriate software.

6. **Evaluation system and Students' Responsibilities**

Internal Evaluation

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows. The components may differ according to the nature of the subject.

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End**	**50**
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		**20**		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

Student Responsibilities:

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUILIFIED (NQ) and the student will not be eligible to appear the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for re-exam.

7. **Prescribed Books and References**

Text Book

1. M. Morris Mano, *Digital Logic and Computer Design* Pearson India, 2017.

Reference Book

1. M. Morris Mano, *Digital Design* California State University, Los Angeles.
2. A. Anand Kumar, *Fundamental of Digital Circuits,* Fourth Edition, PHI Learning Private Limited, 2016.

Pokhara University Faculty of Science and Technology		
Course Code: CMP 116 (3 Credits)		Full Marks: 100
Course Title: **Computer Fundamentals and Application (3-0-2)**		Pass Mark: 45
Nature of the Course: Theory and Practical		Total Lectures: 48 hours
Level : Bachelor	Year : I/ Semester: I	Program: BCA

1. Course Description

This course is designed to conceptualize the computer fundamental knowledge and its application to solve the real-life problems. This course covers the concept of computer, computer organization, software, hardware, functional behavior of peripherals and important accessories, data communication, computer networks and burning technologies. In this syllabus unit I and II covers the fundamental concept of computer; unit III and VII covers the software, unit IV and V covers the hardware where unit VI and VIII covers the data sharing tools and trends.

2. Course Objectives

The general course objectives of this course are outlined as:

1. To familiarize the student with fundamental knowledge about computer system
2. To make the student knowledgeable about software and its working procedure
3. To enhance the knowledge about various components as input, storage, processing and output devices
4. To equip the student about assembling and dissembling, general maintenance and networking the computers

3. Methods of Instruction

The medium of instruction is English and the faculty members can apply the various tools and techniques for teaching methodologies. As per the student's ability subject teachers faculties have their rights to use the teaching methodologies like Class Room/Lecture-based, Discussion-based, Project-based learning(PBL), Active Learning, Cooperative/Experimental Learning, Peer Teaching, etc. by motivating the students. After completion of each unit, faculty members can evaluate the students theoretically and practically as well as whenever possible.

4. Course Contents

Specific Unit wise Objective	**Course Contents**
Unit 1: Introduction to Computer	**5 hours**
1) Explain the conceptual knowledge of computer with historical and generation background 2) Describe the types of computer with characteristics and applications	1.1 Definition 1.2 History of computer 1.3 Generation of computer 1.4 Types of Computer 1.5 Characteristics of computer 1.6 Applications of computer
Unit 2: Basic Organization of Computer	**4 hours**

1) Explain the organization of computer with their functional components 2) Describe the working mechanism of computer system	2.1 Basic function of computer 2.2 Basic functional organization of computer 2.2.1 Input unit 2.2.2 Output unit 2.2.3 Storage unit 2.2.4 Arithmetic and logic unit 2.2.5 Control unit 2.2.6 Central Processing Unit (CPU) 2.3 The system concept
Unit 3: Computer Software	**6 hours**
1) Describe the concept of computer software with their roles 2) Explain the software architectures their application	3.1 Introduction to software 3.2 Logical system architecture 3.3 Types of software 3.3.1 System Software 3.3.2 Application Software 3.3.3 Utility Software 3.4 Firmware 3.6 Middleware
Unit 4: Computer Accessories	**9 hours**
1) Familiarized with input and output accessories with their application 2) Identify the role of agronomical design in related devices	4.1 The Input Accessories 4.1.1 Keyboard Devices 4.1.2 Point and Draw Devices 4.1.3 Yoke 4.1.4 Data Scanning Devices 4.1.5 Digitizer 4.1.6 Microphone 4.1.7 Electronic Cards Based Devices 4.1.8 Speech Recognition Devices 4.1.9 Vision Based Devices 4.2 The Output Accessories 4.2.1 Monitor 4.2.2 Printer 4.2.3 Plotter 4.2.4 GPS 4.2.5 Projectors 4.2.6 Headphones 4.2.7 Soundcard/video Card 4.2.8 Speaker 4.2.9 Voice Response System 4.2.10 Computer Output Microfilm (COM) 4.2.11 SGD Speech Generation Device 4.3 Ergonomically Designed Devices
Unit 5: Storage Devices	**7 hours**
1) Describe the characters of primary and secondary storage devices 2) Explain the function and behavior of different types of memories .	5.1 Introduction 5.2 Types of Storage Devices 5.3 Cache memory 5.4 Registers 5.5 Primary Memory 5.5.1 Introduction 5.5.2 Characteristics of main memory 5.5.3 Types of primary memory 5.5.4 Difference between RAM and ROM, DRAM and SRAM, primary and secondary memory 5.6 Secondary Memory

	5.6.1 Introduction 5.6.2 Characteristics of secondary memory 5.6.3 Types of Secondary memory 5.6.4 Difference between Hard disk and SSD, magnetic and optical disk, direct and sequential access magnetic memory
Unit 6: Data Communication and Computer Network	**5 hours**
1) Interpret the method of data communication including transmission modes and media 2) Explain the relationship of computer network and topologies with their advantages and disadvantages 3) Compare the role of email and internet in modern life	6.1 Data Communication 6.1.1 Introduction 6.1.2 Basic Elements of a communication system 6.1.3 Data Transmission Modes (Simplex, Half Duplex, Full Duplex) 6.1.4 Data transmission Speed (Narrowband, Voice band, Broadband) 6.1.5 Data Transmission Media (Twisted-pair wire, coaxial cable, Optical fibers, Microwave system Communication satellite) 6.2 Computer Network 6.2.1 Definition 6.2.2 Types of Computer Network (PAN, LAN, CAN, MAN and WAN), Differences, advantages disadvantages 6.2.3 Network Topologies, advantages, disadvantages 6.2.4 Intranet, Extranet, Internet 6.2.5 E-mail
Unit 7: Operating System	**6 hours**
1) Explain the features of operating system and its historical background 2) Describe the functional and architecture behaviors of various type of OS. 3) Explain the different installation methods for operating system	7.1 Introduction 7.2 History and evolution of OS 7.3 Objective of OS 7.4 Generation of OS 7.5 Functions of OS 7.6 Types of OS 7.7 System architecture of OS 7.8 Different between GUI and TUI/CUI 7.9 The booting system 7.10 Windows/ Linux operating system, settings, properties and installation guides
Unit 8: AI and Emerging Technologies	**6 hours**
1. Familiarize with Artificial Intelligence and its applications. 2. Explain about emerging trends of technologies.	8.1 Introduction to AI 8.2 AI and its Applications 8.2.1 Natural Language Processing 8.2.2 Machine vision 8.2.3 Expert system 8.3 Machine Learning 8.4 Neural Networks 8.5 Blockchain technology 8.6 IoT 8.7 Cloud Computing 8.8 Cyber Security

5. Laboratory Works

A. Office Automation

a) **Word-processing**
   1. Basic options of word-processing for typing, editing, formatting, margin setting, viewing, designing, printing a document.
   2. Creating, insetting, editing, formulating table for word-processing
   3. Document preparation with table, figure, page number, margin setting etc. and printing document as a report submission.

b) **Spreadsheet**
   1. Preparing sheet for data processing like, arithmetic, logical and other types of functional operation
   2. Preparing data table for calculation, analysis and creating various charts for presentation.
   3. Inserting picture, table, graphs into word-processing
   4. Printing documents after proper setting into the required format.

c) **Presentation**
   1. Creating various types of slides with master slide for presentation
   2. Setting the slide into the required format.

B. **Email and Internet**
   1. Setting various kinds of email account and using them for personal and group purpose
   2. Uploading and downloading the information form internet.

5. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End**	**50**
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		**20**		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. Prescribed Books and References

Text Books

1. P.K. and Priti Sinha, *Foundations of Computing*, BPB Publications, Third Edition

References Books

1. A Text Book of Computer Fundamental and Application, OCEM Publication, (Second Edition), by Hari Bhandari
2. Computer Fundamentals, BPB Publications by V.K Jain
3. Fundamental of Computers, by Balagurusamy E., New Delhi: Tata McGraw Hill.

Pokhara University **Faculty of Science and Technology**		
Course Code: CMP 117 (3 Credits)		Full Marks: 100
Course Title: **Programming Logic and Technique (3-0-0)**		Pass Mark: 45
Nature of the Course: Theory		Total Lectures: 48 hours
Level: Bachelor	Year : I / Semester : I	Program: BCA

6. Course Description

This course is most popular conceptual course to bachelor level of students for develops their career in programming filed. The main theme of this course is that the students without concept of programming, the concept of programming is only possible with logical creation of the system and the concept only to be implicated with proper technique and tools. Students are able to learn programming concept with languages in the first unit and concept can be applied in the program development. The third unit provides the logical and technical ideas, knowledge and tools for better programming where the fourth unit focuses to develop an error free program or software with maintaining budget, time and environment. Finally, the fifth unit familiarizes students about standalone, client-server programming, web programming and mobile computing.

7. Course Objectives

This course is designed with the following general objectives:

- To familiarize the students with the concept of fundamental problem analysis, modeling and coding techniques.
- To develop the skill about program, programing and various programming languages.
- To promote the knowledge about the phases of program development and its methodologies.
- To acquaint the student with create and innovate the ideas, skills using specific tools and techniques.
- To make the student able to develop a plan and procedure for a standardized program with minimizing errors and budget.
- To provide the concept about programming in client server, web server and mobile, cloud.

8. Methods of Instruction

The medium of instruction is English and the faculty members can apply the various tools and techniques for teaching methodologies. As per the student's ability faculties have their rights to use the teaching methodologies like Class Room/Lecture-based, Discussion-based, Project-based learning(PBL), Problem-based Learning (PBL), Flipped Classroom-based, Active Learning, Socratic Method, Cooperative Learning, Experimental Learning, Gamification, Inquiry-based Learning, Constructivist Approach, Collaborative Learning, Direct Instruction, Differentiated Instruction, Montessori Method, Reggio Emilia Approach, Waldorf Education, Peer Teaching etc. are called alternative method of teaching to motivate the students for

learning. After completion of each unit, faculty members can evaluate the students theoretically and practically if exists. They can conduct VIVA, Supersized test, Questionnaire test, Assignment test, Project work and Practical work, even Terminal examination as per the requirement.

9. Course Contents

Specific Objective	Course Contents
Unit 1: Programming Languages	**14 hours**
• Describe the computer languages, generation and their types with their features • Explain the program translation processes from a source program to object/executable file. • Interrupt the loading, linking and relocating the program into the memory while compilation.	1.1 Introduction 1.2 Analogy with Natural Language 1.3 Classification of Computer Language (Low Level Language and High Level Language) 1.3.1 Low Level Languages, (Machine and Assembly Language) 1.3.2 Advantages and Disadvantages of Machine Level Language 1.3.3 Advantages of Assembly Language over machine Level Language 1.3.4 Disadvantages of Assembly Language 1.4 High Level Language 1.4.1 Advantages and Limitations of High Level Language 1.4.2 Difference between Low Level and High Level Language 1.4.3 Program Language Translator 1.4.3.1 Types of Translator (Compiler, Assembler and Interpreter) 1.4.3.2 Different among Translator 1.5 Linker and Loader 1.5.1 Types of Linder and Loader 1.6 Generation of Computer Languages 1.6.1 Introduction to 3GL and 4GL 1.6.2 Advantages of 4GL over 3GL 1.6.3 Feature of 4GL (Object Oriented Language) 1.7 Machines Independent and Portability of programs 1.8 Important types of High-Level Language
Unit 2: Program/ Software Development	**10 hours**
• Describe the program development step and role of life cycle. • Explain the role of budget in a program development • Identify the role of programming tools for the good programming • Explain the role of creative, logical and analytical thoughts for requirement study and efficient program development	2.1 Introduction to program development/software development 2.2 Program Development Life Cycle. 2.2.1 Problem identification 2.2.2 Problem Analysis/Requirement Analysis 2.2.3 Program Design Phases (Data design, Input, Output, Procedure, File design) 2.2.4 Program coding 2.2.5 Program Testing and Debugging 2.2.6 Program Delivery/Implementation and Evaluation 2.2.7 Program Maintenance 2.2.8 Documentation 2.3 Basic Programming Tools 2.3.1 Algorithm 2.3.2 Flowchart 2.3.3 Decision Table 2.3.4 Pseudocode 2.4 Comparison of Programming Tools 2.5 Advantages and Disadvantage of Programming Tools 2.6 Example of Programming Tools
Unit 3: Programming Techniques	**8 hours**

• Describe programming techniques and models • Compare programming techniques and model for easily identify the advantages and disadvantages • Compare cohesion and coupling with their types • Explain the internal relationship of modules for communication in programming.	3.1 Introduction 3.2 Type of Programming Technique (Approaches) 3.2.1 Modular, Top down, bottom up, Structure and Object Oriented approach 3.2.2 Advantages and Disadvantages 3.3 Program Development Methods (Models) 3.3.1 Types of Program Development Models (Water fall or Traditional model, Prototype or Transformation model, Spiral model, Iterative model, V-model, RAD model, Bog Bang model, Evolutionary models, Agile) 3.3.2 Advantage and Disadvantages of Waterfall, Prototype and Spiral Model (Simple Introduction to other Models) 3.4 Cohesion and coupling 3.4.1 Types of Cohesion and Coupling 3.4.2 Use of Cohesion and Coupling in Program Development 3.5 Logics o Program Development 3.5.1 Types of Logics (Sequential, Selection Iteration and Recursion) 3.5.2 Differentiation 3.6 Communication between modules
Unit 4: Program Maintenance	**8 hours**
• Describe the software maintenance, its cost issues and time factors • Compare the ratio of the budget for development and maintenance of the programme. • Describe the role of program documentation and its standard • Explain the role of system flow and DFD in program design • Describe the requirement specification of the program and development procedures	4.1 Introduction 4.2 Types of Program maintenance (Corrective maintenance, Adoptive maintenance, Perfective maintenance, Emergency maintenance) 4.3 Problem Areas in Program Maintenance 4.4 Cost issues in software maintenance 4.5 Impact of software Errors 4.6 Program documentation and its standards 4.6.1 Requirements of Documentation 4.6.2 Importance of Documentation 4.6.3 Types of Documentation 4.7 Program Specification 4.8 System Flow Chart 4.8.1 Elements of System Flowcharts 4.8.2 Examples of System Flowchart 4.9 Data Flow Diagram 4.9.1 Element of DFD 4.9.2 Leveling the data flow diagram 4.9.3 Idea for Drawing DFD 4.9.4 Examples of DFD
Unit 5: Client Server and Web-based Programming Technology	**8 hours**

• Explain computer and communication, stand-alone and client server computing, two and three-tier architecture of client server system, web-based programming • Describe the role of platform independent feature of software	5.1 Standalone Programming concept 5.1.1 Introduction 5.1.1 Advantage and limitations of Standalone Programming 5.2 Client Server Programming (CSP) 5.2.1 Introduction to Client server programming 5.2.2 Advantages and Disadvantages of CSP 5.2.3 Distinguish between Standalone and Client Server Programming 5.2.4 Client Server Programming Architecture 5.2.5 Some of the client server programming and their applications 5.3 Web-based Programming 5.3.1 Introduction to web, versions of web 5.3.2 Advantages and Disadvantages of web-based programs 5.3.3 Architecture of web-based programming 5.3.4 Web-based System and Programming 5.3.5 Some important web-based programing and their applications 5.4 Introduction to mobile computing languages and their application 5.5 Requirement of Platform-Independents system

5. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**50**	**Semester End**	**50**
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the

specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. Prescribed Books and References

Text Books

1. P.K. and Priti Sinha, *Foundations of Computing*, BPB Publications, Third Edition

References Books

1. Er. Hari Bhandari, *A Text Book of Programming Logics and techniques*, OCEM Publication, Second Edition
2. V.K Jain, *Computer Fundamentals*, BPB Publications

Pokhara University **Faculty of Science and Technology**		
Course Code: CMP 111 (1 Credit)		Full Marks: 100
Course Title: **Computer Application Workshop (1-0-3)**		Pass Mark: 45
Nature of the Course: Practical		Total Lectures: 30 hours
Level: Bachelor	Year : I / Semester : I	Program: BCA

1. Course Description

This course provides the knowledge and hands-on skills of computer hardware, software, computer networking and enables students to identify and rectify the onboard computer hardware, software and network related problems. Students will be able to understand the hardware specifications for the operating system and various application programs.

2. General Objectives

The general objectives of this course are:

- To familiarize the students with the computer systems hardware, basic practical works and computer networking concepts
- To make the students competent to install/update operating system and various application software, manage data backup and restore operations on a computer system

3. Methods of Instruction

Specific instructional techniques: Lab works, Project works

4. Contents in Detail

Specific Objectives	**Contents**
- Recognize the computer hardware and accessories.	**Unit 1: Introduction to Desktop/Workstation Assembly (2 Hrs.)**

	1.1 Demonstration of computer hardware and peripheral devices: Processors, Motherboard, Memory, Storage devices, I/O Devices. 1.2 Demonstration of Power Supply-SMPS, Internal cablings and Ports.
- Assemble and Dissemble PC and laptop with power supply, internal cabling, Motherboard and PCI Device	**Unit 2: Assembling Desktop and Laptop Computer (4 Hrs.)** 2.1 Assembling steps and precautions. 2.2 Setting of motherboards, memory, hard disk, processors, CDROM, SMPS and other devices.
- Install OS (Linux and Windows) in Desktop/Laptop	**Unit 3: Laptop/Desktop OS Installation (6 Hrs.)** 3.1 BIOs setting, firmware types, BOOT configuration. 3.2 Hard disc partitioning. 3.3 OS installation (Linux, Windows). 3.4 User account management 3.5 Printer, Scanner installation
- Install and update application software and utility software. - Scan and remove viruses from computer system.	**Unit 4: Software Installation (5 Hrs.)** 4.1 Installation of application and utility software. 4.2 Update of firmware patches, 4.3 Anti-virus installation and scanning.
- Recognize computer networks-wired PAN, CAN, LAN and Wireless LAN configuration.	**Unit 5: Networking and Internet Setup (5 Hrs.)** 5.1 Introduction to Computer Network, Network topologies, wired and wireless networking media. 5.2 Network cabling, cable types and connectors. 5.3 IP Address setting, Sharing of Printer/ Network Devices 5.4 Basic wireless AP configuration.
- Study of server of Mail/Data/Domain/FTP. - Apply FTP/SCP Client tool to upload/download files to FTP server.	**Unit 6: Server Installation and Backup (3 Hrs.)** 6.1 Introduction to server. 6.2 Study of different server (Email, Data, Domain, FTP).
- Use the basic troubleshooting tools and utilities.	**Unit 7: Hardware and Software Troubleshooting (5 Hrs.)** 7.1 Basic Repair and Maintenance of Desktop/Laptop. 7.2 Replacement of Passive Components. 7.3 Uses of basic software troubleshooting commands. 7.4 Uses of trouble shooting tools and utilities.

5. Evaluation system and Students' Responsibilities

Evaluation System

The evaluation of a student may consist of assignments, attendance, test-exams, term-exams, lab reports and projects etc. The tabular presentation of the evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	0	Attendance & Class Participation	5%	100
		Assignments/Lab Report	10%	
		Quizzes/MCQ	10%	
		Project Work/Lab Test	10%	
		Viva	15%	
		Internal Exam	50%	
	0	**Total Internal**		**100**
Full Marks: 0 + 100 = 100				

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. Prescribed Books and References

References

1. Rosch, W. R. *The Winn L. Rosch Hardware Bible*. USA: Sams.

2 Bose, A. K. *Hardware and Software of Personal Computers*. New Delhi: New Age International.
3. Rajaraman, V. and Adabala, N. *Fundamental of Computers*. New Delhi: PHI Learning.

Pokhara University **Faculty of Science and Technology**		
Course Code: ENG 122 (3 Credits)		Full Marks: 100
Course Title: **Business and Technical Communication (3-0-1)**		Pass Mark: 45
Nature of the Course: Theory and Practical		Total Lectures: 48 hours
Level: Bachelor	Year: I / Semester: II	Program: BCA

1. Course Description:

This course is designed for the foundations of communication through writing letters, memos, emails, preparing reports and proposals, seminar papers, structuring of essay and verbal and nonverbal communications. The course includes a practicum component in that students towards the end of the semester to submit a portfolio that includes a range of writing assignments such as technical correspondence and job application process. This course follows various teaching and learning methodologies specially, the seminar model of classroom teaching with discussions and presentations using modern tools and techniques. And it offers conceptual insight into the field, principles of communication design, style, grammar and various writing skills of general and academic interests.

2. General Objectives:

The general objectives of this course are as follows:

a. To be able to comprehend and take notes after listening.
b. To develop the ability to deliver technical knowledge orally in English.
c. To fasten reading skills in technical and non-technical reading materials
d. To be able to write proposals, reports, letters, description on technical talks.

3. Methods of Instructions:

- Lecture and discussion
- Demonstration
- Presentation
- Explanation and illustration
- Group and individual work
- Project work
- Self-study etc.

It is expected that students are fully engaged with the teacher in subject matters and lessons to excel their interactive and presentation skills.

4. Course Content in details

Specific Objectives	**Course Content**
• Explain the communication and its processes • Describe the principles of communication	**Unit 1: Introduction to Communication 3 hrs.** 1.1 Definition of communication 1.2 Process of Communication 1.3 7 C's Principles of communication
	Unit 2: Technical communication Process 8 hrs

• Define technical communication with its forms and types • Explain the role competency and contrast with its core competencies along with attitudes, values and personality. • Describe the skill sets for technical writers	2.1 Definition 2.2 Evolution of technical communication 2.3 Forms of technical communication 2.4 Types of technical communication 2.5 Importance of technical communication 2.6 Definition of Competency 2.7 Attitudes, values and personality 2.8 Core competencies in technical communication 2.9 Skill sets for technical writers
• Describing the technical communication process at two different levels- document and collaboration processes.	**Unit 3: Technical communication process 5 hrs** 3.1 The Document Process: - ❖ Planning ❖ Writing the draft, ❖ Revising ❖ Delivery 3.2 The Collaboration Process ❖ The pros and cons of collaboration ❖ Collaborative writing process ❖ Strategies for effective collaborative writing
• Comparing the techniques of writing letters to apply for job • Describe the role of correspondence in the form of memoranda and letters	**Unit 4: Correspondence 8 hrs** 4.1 Memo writing ❖ Meaning and definition of memo writing ❖ Purpose of memos ❖ Format of memo 4.2 Letter Writing ❖ Introduction ❖ Features of letter ❖ Types of letters ❖ Job application- CV ❖ Letter of inquiry ❖ Quotation letter ❖ Order Letter ❖ Claim Letter ❖ Adjustment Letter
5 Highlight the nature and methods of effective job interviews, meeting and minutes	**Unit 5: Interviews, Meetings and Minutes 7 hrs** **Interview:** 5.1 Introduction 5.2 Effective techniques for interview 5.3 Interview and body language **Meeting:** 5.4 Introduction 5.5 Purposes of Meeting 5.6 Notice (format of Meeting) 5.7 Minutes of Meeting 5.8 Purpose of Minutes 5.9 Format of Minutes
	Unit 6: Technical Talk and Non-verbal Communication 3 hrs

• Explain the methods to develop the presentation skill and strategies • Describe the non-verbal communication and its forms	6.1 Oral communication 6.2 Oral presentation 6.3 Types of non-verbal communication 6.4 Importance of non-verbal communication 6.5 Characteristics of effective presentation
• Explain effective and efficient methods for writing the reports and proposals	**Unit 7: Writing Proposals and Reports 8 hrs** **Proposal writing** 7.1 Introduction 7.2 Components of proposals: Title, Introduction, Statement of Problem, Literature Review, Methodology, Budgeting, Output 7.3 Types of proposal **Report Writing** 7.4 Introduction 7.5 Different types of Report 7.6 Elements of Report :Title, Acknowledgement, Abstract, Introduction, Methodology, Results/Findings, Conclusion and recommendation, References
• Enhance the quality of life job interviews • Explain how to resolve IT related conflicts involving group discussions	**Unit 8: Real-Life Scenarios and Role-Playing 3 hrs** 8.1 Mock job interviews 8.2 Negotiating contracts and agreements 8.3 Resolving IT-related conflicts 8.4 Group discussions on IT trends and innovations
• Describe grammatical items and transform the sentences as required. • Explain the varieties of English with focus on usage	**Unit 9: Grammar and Usage 3 hrs** 9.1 Phrases, clauses and sentences 9.2 Functional and structural division of sentences 9.3 Transformation of sentences 9.4 Voice- Active and Passive 9.5 Varieties of English

5. List of Practical/Assignment

Classwork should cover the concepts of technical communication from all chapters studied in the lectures. Students should develop the skills on presentation, letter writing, technical talk and non-verbal communication to solve following problems in classwork.

- Comprehensive coverage of topics essential for enhancing the employability skills and performance at workplaces
- Finding the forms of technical communication
- Explaining the document process and its terms
- Pros and Cons of collaboration
- Keeping records of books, items and access them
- Solving verbal and non-verbal communication problems
- Implementing the presentation skills presenting power point slides
- Pronunciation, accent, grammar and vocabulary basics to effective technical communication
- Develop the oral and written media in academic tasks every day to face the challenges communicating effectively

Students should submit a final project that uses the concept of communication studied in this course. The marks for the practical will be based on the project work, class performance and assignments.

6. List of Tutorials

The various tutorials that suit the course should cover all the content of the course to provide a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments to the instructor for evaluation and supervision.

A. Discussion based Tutorials

1. Evolution of basics of communication and the use of technology.
2. Structures of effective presentation strategies, interview and group discussion
3. The pros and cons of environmental pollution and its solutions, technical education and its necessities, role of internet (oral presentation).
4. Fundamentals of effective writing

B. Problem Solving based tutorials

1. Write technical documents namely letters, memos, proposals, reports based on a given scenarios.
2. Develop the constituents of effective writing through relevant and effective ways of the reading.

C. Review and Question/Answer-based Tutorials

1. Case study on development of communication with process, correspondence, verbal and non-verbal communication followed by oral presentation in the class.
2. Visual Communication Exercise – Any two examples of visual communication (Training, Performance Career, etc. from Business Result *Pre-intermediate Student's Book by David Grant, Jane Hudson*).

7. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**40**	**Semester End**	**50**
Attendance and Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical/Assignment/Classwork		**10**		
Attendance/Class Participation	10%			
Tutorials	30%			
Assignment/Classwork	40%			
Viva	20%			
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End

Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Book:

1. Business Communication for Success, UNIVERSITY OF MINNESOTA LIBRARIES PUBLISHING EDITION, 2015, ISBN -13

References:

1. Adhikari, Dharma, and Upadhyaya, Phanindra, *Technical Communication*, Buddha Publication, 2012
2. Adhikari, Dharma, and Upadhyaya, Phanindra, *Technical Communication-II*, Buddha Publication

Pokhara University **Faculty of Science and Technology**		
Course Code: MTH 132 (3 Credits)		Full Marks: 100
Course Title: **Mathematics II (3-0-0)**		Pass Mark: 45
Nature of the Course: Theory		Total Lectures: 48 hours
Level: Bachelor	Year: I / Semester: II	Program: BCA

1. Course Description:

This course covers fundamental of integrals, application of integration, differential equations, vector Space, complex numbers and function of complex variables, sequence and series and Fourier series which are essential as mathematical foundation for computing.

2. General Objectives:

The general objective of this course, is to provide the students with basic mathematical skills required to understand Computer Application Courses

3. Methods of Instructions:

Lecture, Tutorial, Discussion, Assignments and Practical works.

5. Contents in Detail

Specific Objectives	**Contents**
Explain • Indefinite • Definite • Improper and • Double integration • Symbolic calculation of integration using any software tools	**Unit 1: Fundamental of integrals** **10 Hrs** 1.1 Introduction 1.2 Indefinite integrals 1.3 Techniques of integration 1.3.1 Integration by substitution 1.3.2 Integration by parts 1.3.3 Integration by partial fractions 1.4 Definite integrals

(MATLAB/Mathematica/Octave etc.)	1.5 Improper integrals 1.6 Beta and Gamma functions 1.7 Double integral (concept only)
• Evaluate area and volume by integration	**Unit 2: Application of integration 7Hrs** 2.1 Introduction 2.2 Application in economics (Determination of total cost and total revenue function) 2.3 Area between the curves 2.4 Arc length of curves 2.5 Volume of solid of revolution (Disks and Washers) 2.6 Area of surface of revolution, 2.7 Consumer's surplus and producer's surplus
• Solve first and second order differential equations.	**Unit 3: Differential equations 7 Hrs** 3.1 Introduction 3.2 Order and degree of ordinary differential equations. 3.3 Solution of differential equations of first order by 3.3.1 Separation of variables 3.3.2 Homogeneous 3.3.3 Linear 3.3.4 Equation reducible to linear form (Bernoulli's equation) 3.3.5 Linear and exact differential equations 3.4 Second order homogenous ODE with constant coefficients. 3.5 Second order Non homogenous ODE (Concept only)
• Solve the problem related to Vector spaces, subspaces, linear dependent and independent, and Eigen values and Eigen vectors	**Unit 4: Vector Space 6 Hrs** a. Introduction b. Vector spaces and subspaces with example c. Linear combination of vectors d. Linear dependence and independence of vectors e. Basis and dimension of vector space f. Eigen values and Eigen vectors.
• Solve and analyze complex number related problems	**Unit 5: Complex numbers and Function of complex variables 7 Hrs** 5.1 Introduction 5.2 Algebra of the complex numbers 5.3 Geometric representation of complex numbers

	5.4 Conjugate and absolute values of complex numbers 5.5 Polar form of complex numbers 5.6 Complex variables and function of complex variables 5.7 Analytic functions 5.8 Necessary and sufficient conditions for f(z) to be analytic (without proof) 5.9 Harmonic functions 5.10 Conformal mappings
• Find Sum of series • Expand function in series	**Unit 6: Sequence and series** **6 Hrs** 6.1 Introduction 6.2 Arithmetic and Geometric series 6.3 Sum of finite natural numbers 6.4 Sum of square of first 'n' natural numbers 6.5 Sum of cubes of first 'n' natural numbers. 6.6 Convergence of geometric series 6.7 Taylor series (one and two variables) 6.8 Maclaurin series.
• Find Fourier series, Fourier sine and cosine series	**Unit 7: Fourier series** **5 Hrs** 7.1 Introduction 7.2 Periodic functions and trignomteric series 7.3 Fourier series 7.4 Fourier sine and cosine series

5. List of Tutorials.

1. Problems of indefinite, definite and improper integration.
2. Area, volume, Consumer's surplus and producer's surplus.
3. Solution of first and second order differential equations.
4. Problem related to Vector spaces, subspaces, linear dependent and independent, and Eigen values and Eigen vectors.
5. Solve complex numbers & complex function related problems
6. Sum of series and expansion of function in series
7. Fourier series, Fourier sine and cosine series

6. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**50**	**Semester End**	**50**
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

Text Books:

1. Erwin Kreyszig ***Advance engineering Mathematics,***
2. Thomas and Finney ***Calculus*** Pearson

References:

3. D.R. Bajracharya, R.M. Shrestha &et al**,** ***Basic mathematics I, II*****,** Sukunda Pustak Bhawan, Nepal
4. Budnick F.S. ***Applied Mathematics for Business Economics and the Social sciences*** MCGraw-Hill Ryerson Limited
5. K.K. Shrestha & R. K. Thagurathi, Applied Mathematics
6. Rudra Pratap ***Getting Started with MATLAB***, Oxford University Press 2010

Pokhara University **Faculty of Science and Technology**		
Course Code: ACC 131 (3 Credits)		Full Marks: 100
Course Title: **Financial Accounting (3-0-1)**		Pass Mark: 45
Nature of the Course: Theory and Practice		Total Lectures: 48 hours
Level: Bachelor	Year: I / Semester: II	Program: BCA

1. Course Description:

This course aims to explore the fundamental principles and practices of Financial Accounting. This comprehensive course covers the entire accounting cycle, including inventory management and bank reconciliation statements. In this course, students will learn in-depth about journals, ledgers, and trial balances; the accounting equation; income statements, statements of retained earnings, and balance sheets; cash flow statements; inventory management; and bank reconciliation statements. By the end of this course, students will have a strong grasp of financial accounting concepts and will be well-equipped to analyze financial statements, make informed business decisions, and pursue further studies in accounting or related fields. Whether students aspire to become accountants, entrepreneurs, or simply have an interest in financial literacy and software development, this course will provide them with valuable skills for the business world. In summary, this course provides students with a solid foundation in the aspects of financial reporting and analysis.

2. General Objectives:

Studying Financial Accounting typically involves several general objectives that help students gain a comprehensive understanding of financial reporting and analysis. This course is designed with the following general objectives:

- To make students able to understand financial transactions.
- To enable the students to prepare and interpret key financial statement.
- To make students familiar with accounting standards such as GAAP, IFRS, NAS and so on.

- To acquaint the students with the various types of activities conducted in an organization.
- To familiarize the students with the conceptual foundation of accounting.
- To acquaint the students with basics of corporate reporting.
- To enable students to process and record business transactions.
- To make students understand the use of computer in the field of financial accounting.

3. Methods of Instructions:

3.1. General Instructions:

- Lecture
- Group discussion
- Question-answers
- Home Assignment and Self Study
- Classroom Presentation
- Problem Solving

3.2. Specific Instruction:

After completing each unit, students must present their views and opinions related to the taught subject matters**.**

4. Course Contents in Details

Specific Objectives	**Contents**
Explain the accounting principles and theories.	**Unit 1: The Conceptual Foundation of Accounting (7 Hrs)**
	1.1 Accounting as a Language of Business 1.2 Forms of Business Organizations 1.3 Types of Activities Performed by Business Organizations 1.4 Users of Accounting Information: Internal and External 1.5 Qualitative Characteristics of Accounting Information 1.6 The Accounting Profession – Role and Activities of an Accountant 1.7 The Accounting Framework – GAAP 1.8 Accounting Terminology 1.9 Accounting Information System in Modern Business Organizations 1.10 Use of Computers in Accounting Process
Describe the financial information clearly to stakeholders and ensure adherence to accounting standards and regulations.	**Unit 2: Basics of Corporate Reporting (5 Hrs)**
	2.1 Legal Requirement of Accounting: Provisions of Company Act Relating to Accounting 2.2 Introduction to Accounting Standards: IFRS and NAS 2.3 Annual Report 2.3.1 Definition 2.3.2 Major Components/Elements of Annual Report 2.4 Financial Statements 2.4.1 Definition 2.4.2 Objectives of Financial Statement (Primary & Secondary) 2.4.3 Components of Financial Statement 2.4.3.1 Income Statement 2.4,3.2 Statement of Retained Earnings 2.4.3.3 Balance Sheet 2.4.3.4 Statement of Cash Flow 2.4.3.5 Statement of Changes in Stockholders' Equity 2.5 Accounting Policies and Notes 2.6 Introduction to Audit

	2.7 Legal Provisions Regarding Audit of Accounts in Nepal
Explain in precise systematic documentation of all financial activities within an organization.	**Unit 3: Processing and Recording Business Transactions (6 Hrs)**
	3.1 The Basic for Recording Transactions 3.1.1 Sources of Accounting Information 3.1.1.1 Types of Source Document 3.1.1.2 Role/ Importance of Source Documents 3.1.2 Recording of Transaction and Evens 3.1.2.1. External Transactions (Events) 3.1.2.2 Internal Transactions (Events) 3.1.3 The Accounting Equation and Analysis of Transactions 3.1.3.1 Steps Involved in Developing Accounting Equation 3.2 The Double Entry System 3.2.1 The Account and its Analysis 3.2.2 Debits and Credits and its Rules 3.2.3 The Journals (General Journals and Compound Journals0 3.2.4 General Ledger (Standard Format and T-Account) 3.2.5 Normal Account Balances 3.2.6 Trial Balance (Meaning/ Objectives/ and Preparation) 3.2.7 Use of Excel in Processing Business Transactions
Describe an organization's profitability, health, financial performance, sources, and utilization of fund over a specific period through the income statement, balance sheet, and cash flow statement.	**Unit 4: Preparation of Financial Statement (15 Hrs)**
	4.1 Income Statement 4.1.1 Concept of Income Statement 4.1.2 Basic Concepts, Principles, and Conventions used in Income Statement 4.1.3 Major Components of Income Statement 4.1.4 Preparation of Income Statement (vertical multi-step format) 4.1.5 Statement of Retained Earnings 4.2 Balance Sheet 4.2.1 Concept of Balance Sheet 4.2.2 Basic Concept, Principles, and Conventions used in Balance Sheet 4.2.3 Major Components of Balance Sheet 4.2.4 Preparation of Balance Sheet 4.2.5 Use of Computers in Preparation of Income Statement and Balance Sheet 4.3 Cash Flow Statement 4.3.1 Meaning of Cash Flow Statement 4.3.2 Purpose of the Cash Flow Statement 4.3.3 Importance of Cash Flow Statement 4.3.4 Classification of Cash Flow Activities 4.3.5 Preparation of Cash Flow Statement 4.3.6 Two Methods of Reporting Cash Flow from Operating Activities 4.3.7 Accruals and Cash Basis Accounting 4.3.8 Reconciling Cash Flow under Operating Activity using Indirect Method 4.3.9 Use of Computers in Preparation of Cash Flow Statement 4.4. Financial Ratios based on Financial Statements 4.4.1 Liquidity Ratios 4.4.2 Leverage Financial Ratios 4.4.3 Efficiency Ratios 4.4.4 Profitability Ratios
	Unit 5: Accounting for Cash and Cash Equivalent (6 Hrs)

Explain "cash and cash equivalents" in financial management, enabling effective liquidity management strategies and minimizing financial risk for the organization.	5.1. Components of Cash and Cash Equivalent 5.2 Need for Adjustment to Accounting Records 5.3 Preparation of the Bank Reconciliation Statement 5.4 Petty Cash 5.5 Balance Sheet Presentation of Cash and Cash Equivalent 5.6 Internal Control System 5.7 Cash Control (Receipt and Disbursement)
Compare the inventory management techniques and its impacts on profitability and operational efficiency of the organization.	**Unit 6: Inventories and Cost of Goods Sold (9 Hrs)** 6.1 Introduction to Inventory 6.2 Meaning and Nature of Inventory 6.3 Determining Inventory Quantities 6.4 Cost Included in Inventory 6.5 Cost of Goods Sold Models 6.6 Inventory System: Periodic and Perpetual 6.7 Inventory Costing Methods with Periodic System 6.8 Inventory Costing Methods with Perpetual System 6.9 The Choice of Inventory Method 6.10 Inventory Valuation and Income Measurement 6.11 Disclosure in the Financial Statement 6.12 Accounting Principles Related to Inventory 6.13 Methods of Inventory Estimation 6.14 Analysis of Inventory

5. List of Practical

1. Preparing income statement, balance sheet, and cash flow statement using accounting software.
2. Prepare and present a case study on accounting software used by any public or private organization.

6. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**40**	**Semester End**	**50**
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		**10**		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		**50**		

Full Marks: 50 + 50 = 100

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

Text Books:

1. Porter, G.A., & Norton, C.L. *Financial Accounting*: *The Impact on Decision Makers.* USA: The Dryden Press

References:

1. Hermanson, H.R. and Edwards, D.J. Financial Accounting: *A Business Perspective.* USA: Von Holffmann Press.
2. Kimmel, P.D., Weygandt, J.J., & Kieso, D.E. *Financial Accounting*. New Delhi: Wiley India Pvt. Ltd.
3. Narayanswamy, R. *Financial Accounting: A Managerial Perspective.* New Delhi: Prentice Hall of India.
4. Koirala, M.P., Acharya, C., Sharma, L.P.B., Sharma, N., & Gautam, C.M. *Financial Accounting*. Kathmandu: Buddha Academic Enterprises.
5. Nepal Accounting Standard (NASs).
6. International Accounting Standards (IASs) / International Financial Reporting Standards (IFRSs).

Pokhara University **Faculty of Science and Technology**		
Course Code: CMP 118 (3 Credits)		Full Marks: 100
Course Title: **Programming in C (3-0-3)**		Pass Mark: 45
Nature of the Course: Theory + Practical		Total Lectures: 48 hours
Level: Bachelor	Year: I / Semester: II	Program: BCA

10. Course Description

This course is designed to encompass the concepts of the C programming through a combination of theory and practical components. This course covers syntax, data types, control structures, functions, arrays, pointers, file handling, and dynamic memory allocation. The course aims to equip students with the skills necessary to write efficient and error-free C programs and apply their knowledge to real-world programming challenges.

By the end of this course, students will be able to write efficient and error-free C programs, understand the underlying principles of the language, and apply knowledge to solve real-world programming challenges.

This course will lay a solid foundation for students whether they aspire to pursue a career in software development, systems programming, or any other field that requires C programming skills.

11. General Objectives

- To provide the comprehensive understanding of the programming language.
- To introduce the develop proficiency in writing structured and error-free programs.
- To utilize different data types and control structures effectively in programming.
- To gain hands-on experience with arrays, pointers, and file handling in C.
- To enhance programming skills through a series of practical exercises and assignments, encouraging the application of learned concepts in real-world scenarios.
- To develop the ability to critically analyze and solve practical programming problems using the programming language, fostering a strong foundation for future programming endeavors.

12. Methods of Instruction

These are sample Strategies,which teachers can use to accelerate the attainment of the various course outcomes.

-
- Use of Video/Slides/Animations to explain functioning of various concepts.
- Encourage collaborative (Group Learning)Learning in the class.
- Adopt Problem Based Learning, which fosters student's Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.
- Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them.
- Discuss how every concept can be applied to the real world-and when that's possible, it helps to improve the students' understanding.

13. Course Contents in Details

Specific Objective	Course Contents
Unit-1 Introduction to C Programming	**3 hours**
▪ Review about algorithm and flowchart. ▪ Describe history and features of C ▪ Explain the basic construct of structured programming	1.1 Review of Flowchart and Algorithms 1.2 History of C programming language 1.3 Features of C programming language 1.4 Structure of a C program 1.5 Preprocessor directives
Unit-2 Variables, Operators, and Datatypes	**7 hours**
▪ Explain the basic constructs of C language	2.1 C character set, Tokens, 2.2 Keywords and reserved words 2.3 Identifiers with naming convention. 2.4 Constants and Variables 2.4.1 Introduction to Constants and Variables 2.4.2 Variable Declaration 2.4.3 Variable types 2.5 Datatypes 2.6 Operator 2.6.1 Types 2.6.2 Precedence and associativity of operators 2.7 Formatted and Unformatted I/O functions 2.8 Comments- Single line and Block

Unit-3 Control Statements **8 hours**	
▪ Describing the role of control statement with its types	3.1 Clean code practices 3.2 Sequential control Statement 3.3 Selection Control Statement 3.3.1 Decision-making with if-else statements 3.3.2 Nested if 3.3.3 Conditional Operator 3.3.4 Switch statement 3.4 Iteration control statement 3.4.1 Loop using for, while and do…while keyword 3.4.2 Nested loop 3.5 Jump Statements 3.6 Best practices for writing structured code
Unit-4 Arrays and Strings **10 hours**	
▪ Describe the arrays and their declaration, accessing and manipulation in C ▪ Introduce the strings and string manipulation functions ▪ Explain the role of multi-dimensional arrays and its implementation	13.1 Introduction to arrays and their declaration 13.2 Accessing array elements and array indexing 13.3 Manipulating array elements 13.4 Multi-dimensional arrays 13.5 Working with strings and string manipulation functions
Unit-5 Functions **7 hours**	
▪ Describing the basics of functions declaration, definition, calling and returning ▪ Compare the pass by value and pass by address ▪ Explain the role of recursion and pass arrays/strings to functions	5.1 Introduction to function 5.1.1 Types of function (Library and User defined) 5.2 Declaration, definition, and calling 5.3 Function arguments and return values 5.4 Function prototypes: 5.4.1 No Arguments and No return values 5.4.2 Arguments but No return Values 5.4.3 Arguments with return values 5.4.4 No arguments but return a value 5.5 Recursive function 5.6 Iteration versus recursion 5.7 Passing arrays and strings to functions
Unit-6 Structure and Union **4 hours**	
• Describe the user defined data types • Compare features and function of Structure and Unions • Introduce code modularity and maintainability, reusability and reduce complexity.	6.1 Introduction to Structure and Union 6.2 Declaration, accessing members, initialization 6.3 Nested structures 6.4 Arrays of structures 6.5 Passing structures to functions, returning structures. 6.6 Difference between structure and union
Unit-7 Pointers **3 hours**	
▪ Describe the role of pointers with its operator. ▪ Introduce dereferencing and access memory location ▪ Explain the role of pointer arithmetic ▪ Compare the features and functions for dynamic memory allocation	7.1 Introduction to pointers and pointer declaration 7.2 Dereferencing pointers and accessing memory locations. 7.3 Returning multiple values using pointers 7.4 Pointer arithmetics 7.5 Pointers and arrays. 7.6 Pointers to structures 7.7 Pointer as a function argument

	7.8 Dynamic Memory Allocation
Unit-8 File Handling 6 hours	
▪ Discuss the role of file and file handling in C with its modes and functions	8.1 Opening, reading, writing, and closing files using different techniques 8.2 Sequential and random access to files 8.3 Error handling and file input/output operations 8.4 File handling best practices

14. Laboratory and Assignments

The instructor should design laboratory and assignments by taking into consideration the necessary knowledge, technological aspects, practicality, and current trends related to the topic. The assignment guidelines should cater to the average students while also providing the opportunity to engage and challenge the more advanced ones. It is advisable for the instructor to create multiple sets of assignments and distribute them among different groups of students. It is also preferable that the assignments are based on real-world problems or applications. Students should be encouraged to utilize Hungarian notation, practice proper indentation, and include comments in their code.

Furthermore, the instructor can assign a mini-project that showcases the application of the learned concepts in a real-life scenario. Additionally, they may assign an assignment or mini-project that aligns with the respective branch of study, going beyond the syllabus.

6. Evaluation system and Students' Responsibilities

Student academic performance is assessed through internal and external methods. Internal evaluation is carried out by faculty members, monitoring day-to-day student progress. On the other hand, external evaluation is conducted by the university, involving semester-end examinations worth 100 marks. This evaluation system operates as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End examination**	**50**
Attendance and class Participation	10%			
Assignments	20%			
Project work/Presentations	10%			
Internal Examination	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab report	20%			
Practical Exam	40%			
Viva	30%			
Total Internal Marks		**50**		

Full marks=50+50 = 100

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Text Books and references

Text Books:

1. S.K. Srivastava and Deepali Srivastava: C in Depth

References:

1. Kely and Pohl: A book on C
2. Bhim Bhandari, A Complete Concept and Examples in Programming in C, 2024
3. Wait, Mitchell, Steven Prata and Donald Martin: C Primer Plus

Pokhara University **Faculty of Science and Technology**		
Course Code: ELX 112 (3 Credits)		Full Marks:
Course Title: : Microprocessor and Computer Architecture (3-0-1)		Pass Mark
Nature of the Course: Theory + Practical		Total Lectures: 48 ho
Level: Bachelor	Year: I / Semester: II	Program: B

1. Course Description

This course is designed to encompass the fundamental concepts of 8085 and 8086 with assembly level programming. It also provides major notions for the design of processing unit, control unit architectures. It presents the various computer arithmetic algorithms to solve the computer arithmetic problems. This course also introduces memory and input/output organization and the advance computer architecture. After completion of this course, students can design very simple CPU and Control Unit.

2. General Objectives

The major objectives of this course are to provide knowledge of:

- To acquaint the students with basic concepts of basic architecture details of microprocessor (8085/8086)
- To acquaint the students with concepts of assembly level programming using 8-bit microprocessor
- To acquaint the students with the fundamentals of computer systems.
- To acquaint the students with the knowledge of computer architecture and associated processing, control unit and ALU unit of very simple central processing unit.
- To apprise the students with the architectural and associated components of computer systems.

- To aware the students about the architecture of the computer systems available in the market.

3. Methods of instructions

Lecture, Project work and Practical

4. Content in details

Specific objectives	Contents
• Familiarize with fundamentals of microprocessor-based system along with their real-world applications.	**Unit 1: An overview of computer and Microprocessor** **3 Hrs** 1.1. Brief overview of microprocessor and microcontroller 1.2. Organization of computer systems 1.3. Architecture of computer 1.4. Applications of microprocessor and microcontroller
• Describe the architecture of 8085 microprocessor and basic assembly level programming. • Explain the machine cycles involved during the execution of instructions.	**UNIT 2 : Architectural and assembly language programming of 8085** **12 Hrs** 2.1 Pin Configuration, 2.2 Functional Block Diagram 2.3.1 Timing and Control Unit, 2.3.2 Registers, 2.3.3 Data and Address Bus, 2.3.4 ALU 2.3.5 I/O Interface 2.3 Intel 8085 instructions types 2.3.1 Data transfer instructions 2.3.2 Arithmetic instructions 2.3.3 Logic instructions 2.3.4 Decision and Branching instructions 2.3.5 Machine control instruction 2.4 Operation Code and Operands, 2.5 Addressing Modes, 2.6 Interrupts and Flags 2.7 Instructions Types and Data Flow inside 8085, 2.8 Timing Diagram (two examples of 4, 7, 10 and 13 T-states) 2.9 Basic Assembly Language Programming Using 8085 Instruction Sets
• Familiarize with the architecture of 16 bit microprocessor 8086. • concept of memory segmentation and pipelining in modern processor.	**UNIT 3 : Overview of 8086 microprocessor** **5 Hrs** 3.1 Features of 8086 microprocessor 3.2 Functional diagram of 8086 microprocessor 3.3 Registers and Flags 3.4 ALP Development Tools: Editor, Assembler and linker
• Familiarize with CPU Fundamental	**UNIT 4 : CPU Fundamental** **4 Hrs** 4.1 CPU organization/Structure 4.2 Register organization and data path 4.3 Arithmetic and Logic units 4.4 Design principal for modern system
• Examine the basic structure of a micro-sequencer and Hardwired Control Unit	**UNIT 5 Control Unit Design** **7 Hrs** 5.1 Control of the processor 5.2 Hardwired Control Unit(Control unit inputs and logic)

	5.3 Microprogramed control units(Micro instruction and its types) 5.4 Architecture of micro programmed control unit 5.5 Microinstruction sequencing and execution 5.6 Application of hardwired and micro programmed control units
• Understand the representation of binary numbers in signed and unsigned notation along with the algorithms used for the basic arithmetic operations.	**UNIT 6 : Computer Arithmetic** **7 Hrs** 6.1 Numeric format and representation of binary number in signed and unsigned notation 6.2 Addition and subtraction in signed and unsigned notation. 6.3 Shift and add multiplication algorithm, Booth's algorithm. (signed and unsigned)
• Review memory Hierarchy of computer system and study the concept of associative and cache memory in real world scenario.	**UNIT 7 : Memory Organization** **4 Hrs** 7.1 Memory hierarchy 7.2 Memory interfacing diagram (RAM and ROM with 8085) 7.3 Associative memory 7.4 Cache Memory and mapping techniques
• Familiarize with serial and parallel communication interfaces and introduce various methods for improving I/O performances.	**UNIT 8 : Input/Output Organization** **4 Hrs** 8.1 Serial and parallel communication interfaces 8.2 Programmed I/O 8.3 Interrupts, types of interrupts, Interrupt processing, Interrupt Hardware and priority 8.4 Direct Memory Access, I/O Processors
• Understand with the concept of instruction pipelining and multicore architecture in modern processor.	**UNIT 9 : Advance Architectures** **4 Hrs** 9.1 RISC and CISC Fundamentals 9.2 Instruction Pipeline, Register window 9.3 Flynn's Taxonomy, MIMD system topologies and architectures 9.4 Introduction to multicore architecture

5. Laboratory Works

The laboratory works should include followings:

1. Assembly language program using 8085 microprocessor trainer kit or any software simulator.
   1.1 Use of all types of instructions and addressing modes. Programs including basic arithmetical, logical, looping, bitwise and branching instructions.
2. Computer Arithmetic can be implemented with C/C++.

6. List of Tutorials:

The various tutorial activities that outfits this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of instructor. Students should submit tutorials as assignments or class works to the instructor for evaluation. The following tutorial activities of 15 hrs should be conducted to cover all the content of this course:

A. Discussion based Tutorials

1. Comparative analysis of Von-Neumann, Harvard and modified Harvard Architecture.
2. Applications of microprocessor and microcontroller based system in real world.
3. Hardware and programmable module of 8 bit microprocessor.
4. Interpret the concept of memory segmentation and pipelining in 16 bit microprocessor.
5. Review each entity memory hierarchy for modern processors.
6. Interfacing of DMA, interrupt controller and UART with 8085 microprocessor.
7. Comparative analysis of different aspects of computing system as defined in Flynn's Classification.

B. Assembly level programming tutorials

1. Assembly level programming illustrating data transfer instructions, arithmetic instructions, logical instructions and branching instructions in 8085 microprocessor.
2. Enlighten PUSH and POP instructions using appropriate assembly level programs.
3. Assembly level programming for simple sequencing program, multiplication, division, table processing and sorting.
4. Timing diagram for different types of instructions and RTL of each machine cycle involved during the execution of instructions.

C. Design tutorials

1. Design a CPU for any given registers set, instruction set and state diagram. Show the RTL code for each execution cycle.
2. Develop a control unit for any given state diagram.
3. Design a micro-sequencer control unit for any given specifications following design procedure.

D. Computer Arithmetic tutorials

1. Perform arithmetic addition and subtraction in signed and unsigned notation for any given numbers.
2. Perform Multiplication operation for any given numbers using shift-add multiplication algorithm and Booth's algorithm.
3. Perform Division operation for any given numbers using restoring and Non restoring Division algorithm.

E. Project work

Develop a case study report about any of the modern Advance superscalar processors. It should include the architecture of processor, control unit, memory as well as input output organization in detail. An oral presentation with the submission of report should be a part of work and must be included as a component for evaluation.

7. Evaluation system and Students' Responsibilities

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**		
Attendance and class Participation	10%			

Assignments	20%		**Semester End examination**	**50**
Project work/Presentations	10%			
Internal Examination	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab report	20%			
Practical Exam	40%			
Viva	30%			
Total Internal Marks		**50**		
Full marks=50+50 = 100				

Students Responsibility:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester End Examination. Students are advised to attend all the classes, formal exam, and test and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Text Books and references

Text Books:

2. Gaonkar, Ramesh S., Microprocessor Architecture, Programming, and Applications with 8085, Prentice Hall. New Delhi
3. Stallings, W., "*Computer Organization and Architecture*", Eighth Edition, 2011, Pearson.

References:

1. Hall, Douglas V. Microprocessor and Interfacing programming and Hardware, McGraw Hill, New Delhi
2. Carpineili, John D., Computer system Organization and Architecture, Addison Wesley. Pearson Education Asia (LPE),2001
3. Malvino: Digital Computer Electronics and Introduction to Microcomputers
4. Dougals V. Hall: Microprocessor and Interfacing programming and Hardware, McGraw Hill.
5. Mano, M.M., "*Computer Systems Architecture*", Third Edition, 2011, Pearson.
6. Tanenbaum, A.S., "*Structured Computer Organization*", Fourth Edition, 2003, Pearson Education.
7. Rajaraman, V. et all, "*Computer Organization and Architecture*", 2011, PHI.
8. Sima, D. et all, "*Advanced Computer Architecture*", 2000, Addison Wesley.

Pokhara University **Faculty of Science and Technology**	
Course Code: PRJ 151 (1 Credit)	Full Marks: 100
Course Title: Project I **(1-0-2)**	Pass Mark: 45
Nature of the Course: Theory and Practical	Total Lectures: 48 hours

Level: Bachelor	Year: I/ Semester: I	Program: BCA

1. Course Description:

This course is designed to blend the theoretical concept learned from Programming Logic and Techniques, Computer Fundamental and Application, Computer Application Workshop, English for IT Professionals and Programming in C with real-life problems. This course mainly implements the programming knowledge acquired from the C programing language.

2. Objective:

1. To develop practical skills for implementation of the C programming constructs.
2. To use and implement C language to develop real world applications.
3. To provide practical knowledge for planning and formulating project, design, development, documentation and presentation.

3. Procedure:

A group of maximum four students can be formed to work in the project. Students are themselves responsible to form the group and select project on any domain of common interest. A project supervisor is assigned for guidance, tutoring and mentoring of the overall project activities of a section. The project shall be evaluated by the evaluation panel that includes internal evaluator(s) and/or supervisor and external examiner(s) appointed by the respective department. Appointment of External Examiner is mandatory for final defense of the project. Each group is required to complete all project work phases and develop software product that is SMART (Specific, Measurable, Achievable, Relevant, and Time-Bound.

4. Project Phases:

The entire project work shall be divided in to 3 phases and evaluation shall be done accordingly:

Phase I: Proposal Submission and Defense

In Phase I, students are required to form a team comprising of maximum 4 team members and draft (conceptual framework) for their project work in the form of a proposal. The project team must submit and present the proposal in presence of examiner/supervisor in a formal presentation and be approved.

Phase II: Mid-Term Progress Defense

The project team needs to present the work progress to Examiner(s)/Supervisor and demonstrate as required. For the progress defense, students must have finished the design phase including the overall system/architectural design. The progress defense can be scheduled after 4 weeks of project proposal defense.

Phase III: Final Submission and Defense

Before the semester ends, students must submit and defend (oral presentation and demonstration) the project final work in presence of the evaluation panel inclusive of external examiner. The project work should be presented in complete format that include project requirement analysis, design, coding and testing. Students are required to submit the final report 1 week prior to the final defense. Students must submit the hardcopy/softcopy of the project in the standard format prescribed by respective department of the college.

5. Evaluation Criteria:

The overall project evaluation shall be cumulative of Phase I, Phase II and Phase III evaluations. Phase I and II shall be evaluated by internal examiner and Phase III evaluation shall be done by evaluation panel inclusive of external examiner. The evaluation shall be done based on following criteria.

Phase I (20%)

Tasks required to be accomplished in Phase I.

- Feasibility Study
  - Requirements Analysis and Specification
  - Project plan
  - Creativity, Innovativeness and Usefulness of the Idea

Phase II (30%)

Tasks required to be accomplished in Phase II.

- System/Architectural Design
- Progress
- Level of achievement
- Group/Team Effort
- Ability to propose solutions

Phase III (50%)

- Presentation Skills (10 Marks)
- Project Completeness and Final Output (10 Marks)
- Project Demonstration (10 Marks)
- Viva-Voce (10 Marks)
- Documentation (10 Marks)

6. Documentation Structure

The documentation structure defines the components that must be included in the final report after the completion of the Project I. It may include the following components:

Preliminaries

- Cover Page

- Acknowledgment
- Letter of Approval
- Abstract
- Table of Content
- List of Figures
- List of Tables
- List of Abbreviation

Chapter 1: Introduction

- Background
- Problem Statement
- Objective
- Project Schedule

Chapter 2: System Analysis and Design

- Feasibility study
- Algorithm
- Flowchart
- Context diagram
- Data flow diagram
- User interface

Chapter 3: Development and Testing

- Implementation (Development Environment and System Operation Procedure)
- Validation and Verification

Chapter 4: Results and Conclusion

Références: (Respective school/department may decide which citation and reference notation to choose in the report.)

Annex: (It may include implementation of main algorithm(s) of project, screenshots of user-interface and/or any other detail information that the project report requires to present.)

The documentation structure may slightly differ based on the nature of the project that the students have chosen. If so, students should take the approval from the respective department and/or their supervisor.

Pokhara University **Faculty of Science and Technology**	
Course Code: CMP 215 (3 Credits)	Full Marks: 100

Course Title: Object Oriented Programming using Java (3-0-3)	Pass Marks: 48
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: III	Program: Bachelor of Computer Application

1. Course Description:

This course equips students with comprehensive knowledge of various Object-Oriented aspects of Java, guiding them through fundamental OOP concepts such as Classes, Objects, Inheritance, Interfaces, Polymorphism, and Packages. It also covers the basics of graphical user interfaces. The skills acquired in this course will enable students to develop standalone applications for real-world use. By the conclusion of this course, students will have the ability to write efficient, error-free Java programs, establishing a strong foundation for future endeavours.

2. General Objectives:

The general objectives of this course are as follows:

a) To acquaint the students with Object Oriented Programming knowledge of java environment.
b) To acquaint the students to develop simple applications with OOPS concepts
c) To introduce interfaces, packages, JavaFX and Socket programming basic concepts.
d) To enhance programming skills through a series of practical exercises and assignments, encouraging the application of learned concepts in real-world scenarios.

3. Methods of Instructions:

1. Lecture and discussion
2. Practical
3. Demonstration
4. Presentation
5. Case study

4. Course Contents:

Specific Objectives	**Contents**
Unit 1: Introduction to Java Programming	**4 hours**
• Describe the basic concepts of Object-Oriented Programming • Explain the user friendly in JAVA environment with basic structure of the program	1.1 History and Features of Java 1.2 C++ vs Java 1.3 Introduction 1.3.1 JDK 1.3.2 JVM 1.3.3 Bytecode 1.3.4 Java Environment. 1.4 Procedure Oriented Vs Object Oriented 1.5 Basic Concepts of Object-Oriented Programming 1.6 Structure of Java Programming, Naming conversions 1.7 Compiling and executing simple program 1.8 Scanner class
Unit 2: Java Fundamental Concepts	**7 hours**
• Explore the Fundamental of JAVA language • Describe the concept of Arrays and Strings	2.1 Java Tokens, Keywords, Statements 2.2 Constants and Variable 2.3 Data Types 2.4 Operators 2.5 Control Statements 2.5.1 Branching

	2.5.2 Looping Statements 2.5.3 Jump Statements 2.6 Arrays in Java 2.6.1 One dimensional 2.6.2 Two dimensional 2.7 Java Strings 2.7.1 String Length 2.7.2 Concatenation 2.7.3 String Comparison 2.7.4 String Buffer 2.8 Methods in Java 2.8.1 Predefined methods 2.8.2 User defined methods 2.8.3 Static methods
Unit 3: Java Object Oriented Programming	**8 hours**
• Familiarize the concept of Class and Object, methods in real life Application • Explore java. Lang package	3.1 Fundamental of Object and Class 3.2 new Keyword 3.3 Member methods of a class 3.4 Instance method 3.5 Constructor and its types 3.6 Finalize method 3.7 Abstraction 3.8 Encapsulation 3.9 Using this keyword 3.10 Method overloading 3.11 Wrapper classes 3.12 Autoboxing and unboxing 3.13 Visibility Modes
Unit 4: Inheritance, Polymorphism, Interface and Packages	**8 hours**
• Implement the Inheritance and interface concept in application	4.1 Inheritance 4.1.1 Using Extends 4.1.2 Member access 4.1.3 Super classes and sub classes 4.1.4 Single Inheritance 4.1.5 Multi-level Inheritance 4.1.6 Hierarchy Inheritance 4.2 Constructor and inheritance 4.3 Polymorphism-Method overriding 4.4 Abstract class 4.5 Using final keyword 4.6 Interface 4.6.1 Creating and implementing an interface 4.6.2 Extending interface 4.7 Packages 4.7.1 Creating Packages 4.7.2 Accessing a Package
Unit 5: Multithreading, Java Collections	**6 hours**

• Explain how threads work in the real time applications • Implement the java. Utility package	5.1 Introduction Multithreading 5.1.1 Thread class and Runnable Interface 5.1.2 Creating Threads 5.1.3 Stopping and Blocking a Thread 5.1.4 Life Cycle of a Thread 5.1.5 Using Thread Methods 5.1.6 Thread Priority 5.2 Collection Framework in Java 5.2.1 Overview of java collection framework. 5.2.2 Commonly used collection classes- ArrayList, Vector
Unit 6: Exception Handling, Java IO and Streams	**7 hours**
• Familiarize the students with the exception handling process in real life applications • Explain the role of Stream and file handling	6.1 Managing Errors and Exceptions 6.2 Syntax of Exception Handling Code 6.3 Types of Exceptions 6.4 Using try and catch keywords 6.5 Multiple catches 6.6 Using Finally Statement 6.7 Throwing Exceptions 6.8 Streams: Byte Streams and Character Streams 6.9 Reading / Writing Console Input / Output 6.10 Reading and Writing files
Unit 7: Java JDBC	**4 hours**
• Describe and gain knowledge of database connectivity in java	7.1 JDBC introduction 7.2 JDBC drivers 7.3 Connecting to Database 7.4 Querying a database and processing the results
Unit 8: JavaFX	**4 hours**
• Explore the GUI based programming.	8.1 Introduction to JavaFX 8.2 JavaFX Architecture 8.3 JavaFX Application Basics-Line, rectangle and text

6. Practical Works:

The laboratory works should cover all the features of Object Oriented Systems using Java so that students would be motivated to solve the real-life problems.

1. Program to define a structure of a basic JAVA program.
2. Program to define the data types, variable, operators.
3. Program to define the arrays, string and control structures.
4. Program to define class and constructors.
5. Program to define class, methods and objects and method overloading.
6. Program to define inheritance and show method overriding.
7. Program to demonstrate Packages.
8. Program to demonstrate Exception Handling.
9. Program to demonstrate Multithreading.
10. Program to demonstrate I/O operations, demonstrate JDBC and JavaFX.

By incorporating all the lab activities, a mini project should be developed and presented individually. Course instructor must include the mini project weightage in the practical evaluation.

7. Evaluation System and Students' Responsibilities:

7.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50				

7.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

8. Prescribed Books and References:

Text Books

1. Herbert Schildt, Java the Complete Reference, 11th Edition, Tata McGraw-Hill Publishing Co. Ltd.
2. E. Balagurusamy, *"Programming with Java: A Primer",* Edition, Tata McGraw Hill Publication.

Reference Books

1. Y. Daniel Liang, Introduction to Java Programming Brief Version, 8th Edition Kindle Edition.
2. Cay S. Horstmann, Core Java Volume I–Fundamentals, 11th Edition, Pearson 2019.

Pokhara University Faculty of Science and Technology	
Course Code: CMP 227 (3 Credits)	Full Marks: 100
Course Title: Data Structure and Algorithms (3-0-2)	Pass Marks: 48
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: III	Program: Bachelor of Computer Application

1. Course Description:

This course introduces students to fundamental data structures and algorithms. It covers core concepts, analysis techniques, and practical implementation. Students will learn to design efficient algorithms, analyze their performance, and apply them to problem-solving. The course emphasizes hands-on practice through coding assignments and projects.

2. General Objectives:

The general objectives of this course are as follows:

1. To familiarize students with fundamental data structures (stacks, queues, linked lists, trees, graphs) and algorithms, including sorting and searching techniques.
2. To equip students to analyze algorithm efficiency using time complexity metrics (Best, Worst, and Average case).
3. To enhance problem-solving and critical thinking skills through practical assignments, projects, and the comparison of recursive and iterative methods.
4. To prepare students for industry challenges by integrating modern tools, techniques, and the application of fundamental algorithms to modern technological fields.

3. Methods of Instructions:

6. Lecture and discussion
7. Practical
8. Demonstration
9. Presentation

4. Course Contents:

Specific Objectives	Contents
Unit 1: Introduction to Data Structures and Algorithms	**7 hours**
• Describe how data structures and algorithms (DSA) contribute to effective problem-solving. • Compare time complexity and asymptotic notations (Big O, Omega, Theta) to analyze algorithms. • Explore about the impact of dynamic memory allocation on performance.	1.1. Introduction to Data Structures 1.1.1 Data Types and Data Structures 1.1.2 Classification of Data Structures 1.1.3 ADTs (Abstract Data Types) 1.2 Introduction to Algorithms 1.2.1 Definition and Characteristic 1.2.2 Need/Importance of DSA 1.3. Algorithm design Divided and Conquer, Greedy Algorithm, Backtracking, Dynamic Programming 1.4 Algorithm Analysis 1.3.1 Time Complexity 1.3.2 Best, Worst and Average Case 1.3.3 Rate of Growth 1.3.3 Asymptotic Notation (Big O, Omega and Theta)
Unit 2: Stack, Queues and Recursion	**8 hours**
• Explain the concept of stacks as an ADT. • Explore the role queue types (linear, circular, priority). • Implement the recursive problems.	2.1. Concept of stacks and Stack ADT 2.1.1 Stack operations 2.1.2 Stacks implementation using arrays 2.1.3 Applications of stacks: 2.1.3.1 Conversion from infix to postfix

	2.1.3.2 Evaluation of postfix 2.1.4 Recursion 2.1.4.1 Concept of recursion 2.1.4.2 Recursive vs. iterative solutions 2.1.4.3 Recursive problems: Factorial, Fibonacci sequence, GCD, Tower of Hanoi 2.1.5 Recursion and stack 2.2. Concept of queues and Queue ADT 2.2.1 Queue operations 2.2.2 Queue Implementation using arrays 2.2.3 Types of Queues 2.2.4. Linear and circular 2.2.5 Introduction to priority queues and double ended queue.
Unit 3: Linked List	**8 hours**
• Implementation of about different types of linked lists: Singly Linked List, Doubly Linked List, and Circular Linked List. • Describe basic operations in linked lists such as node creation, insertion, and deletion at various positions. • Implement stacks and queues using linked lists, with a focus on doubly and circular linked lists.	3.1 List –Definition and List ADT, array Implementation 3.2 Linked List – Definition and its operations 3.3 Types of Linked Lists: 3.3.1 Singly Linked List 3.3.2 Doubly Linked List 3.3.3 Circular Linked List 3.4 Implementation of Singly Linked Lists: 3.4.1 Node Creation 3.4.2 Node Insertion (Beginning, End, Specified Position) 3.4.3 Node Deletion (Beginning, End, Specified Position) 3.6 Linked List implementation of Stack and Queue
Unit 4: Tree	**8 hours**
• Implement and analyze various tree traversal algorithms for binary trees. • Implement binary search trees, including insertion, deletion, and searching operations. • Explain the need for balanced trees, search trees, and their applications.	4.1 Tree concepts and terminology 4.2 Binary trees: 4.2.1 Definition and types 4.2.2 Tree representations (array, linked-based) 4.2.3 Tree Traversal (Pre, Post and In Orders) 4.2.4 Construction of binary tree from traversal 4.2.5 Application of Tree: Huffman Algorithm 4.3 Binary search trees: 4.3.1 Construction of BST 4.3.2 Basic Operation on BST (Insert, delete, search) Node 4.4 Introduction to balanced trees 4.4.1 Need for balanced trees 4.4.2 Definition of AVL (Adelson-Velskii and Landis) trees, insertion, deletion and Rotation operations 4.5 B-tree Definition and application
Unit 5: Sorting	**6 hours**
• Analyze and implement various sorting algorithms, including simple and advanced techniques.	5.1 Definition and Classification of Sorting algorithm (internal, external, stable, unstable) 5.2 Bubble Sort

• Compare the efficiency of different sorting algorithms using time complexity analysis. • Implement the workings of distribution-based sorting algorithms.	5.3 Selection Sort 5.4 Insertion Sort 5.5 Merge Sort 5.6 Quick Sort 5.7 Radix Sort 5.8 Heap Sort as priority queues
Unit 6: Searching Algorithm and Hashing	**4 hours**
• Implement and analyze Binary Search for efficient data retrieval. • Apply the fundamentals of Hashing, including Hash Functions and Hash Tables. • Explore Collision Resolution Techniques: Separate Chaining, Linear Probing, Quadratic Probing	6.1Concept of Searching algorithms 6.2 Linear and Binary Search 6.3 Hashing: 6.3.1Concept of Hashing, Hash functions, Hash Table 6.4 Collision resolution techniques: 6.4.1. Open Hashing: Separate Chaining 6.4.2. Closed addressing : Linear Probing , Quadratic Probing, Double Hashing 6.5. Load Factor and Rehashing
Unit 7: Graphs	**4 hours**
• Describe graph representations through its types. • Master graph traversal techniques (DFS, BFS, topological sort). • Apply graph algorithms for finding minimum spanning trees (Kruskal's, Prim's) and shortest paths (Dijkstra's). • Apply graph concepts to real-world problems and applications.	7.1. Graph concepts and representations (adjacency matrix, adjacency list) and types 7.2. Graph traversals: 7.2.1. Depth-first search (DFS) 7.2.2. Breadth-first search (BFS) 7.2.3. Topological Sort 7.3. Minimum Spanning Tree- Kruskal's Algorithm and Prim's Algorithm 7.4. Shortest path problems- Dijkstra's algorithm

5. Practical Work:

Students should focus on manually implementing these data structures from scratch. Students are encouraged to implement these ADTs without use of Java's built-in libraries. At the end of practical works, student can identify the performance among the implementation.

1. Implement stack, linear and circular queue using arrays.
2. Implement Infix to postfix conversion using stack
3. Implement recursive algorithms- Factorial, Fibonacci series and Tower of Hanoi
4. Implement singly link list, binary search trees and implement traversal algorithms.
5. Implement graph traversal algorithms (DFS, BFS).
6. Implement sorting algorithms (quick sort, merge sort, heap sort).
7. Implement searching algorithms (sequential, binary search).
8. Implement hash system (separate chaining, liner and quadratic).
9. Implement Dijkstra's algorithm for shortest path.
10. Implement Kruskal's or Prim's algorithm for minimum spanning tree.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**		
Attendance / Class Participation	10%			

Assignments	20%		**Semester End examination**	**50**
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

Text Books

1. Langsam, Y., Augustin, M.J. and Tanenbaum, A.M: Data Structure Using C and C++, Prentice Hall of India
2. Michael T. Goodrich and Roberto Tamassia *Data Structures and Algorithms in Java, 0-471-73884-0, Fourth Edition, John Wiley & Sons, Inc.*

Reference Books

1. Rowe, G.W.: Introduction to Data Structure and Algorithms with C and C++, Prentice Hall of India
2. Kruse, R. L., & Ryba, A. J. (1998). *Data structures and program design in C++*. Prentice Hall, India..
3. Brassard, G., & Bratley, P. (1996). *Fundamentals of algorithmic*. Prentice-Hall, India.

Pokhara University Faculty of Science and Technology a	
Course Code: CMP 221 (3 Credits)	Full Marks: 100
Course Title: System Analysis and Project Management (3-0-1)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: III	Program: Bachelor of Computer Application

1. Course Description:

This course provides a comprehensive overview of the core elements of system analysis, design, implementation, and project management. Students will explore system concepts, the System Development Life Cycle (SDLC), and the critical role of system analysts. Key topics include requirements analysis techniques, system design principles, input/output and database design, as well as system architectures. Additionally, the course covers system implementation processes such as testing, documentation, and user training, while also delving into project management tools and techniques for effective planning, monitoring, and control. Through practical case studies, students will gain the skills necessary to apply these concepts in real-world scenarios.

2. General Objectives:

The general objectives of this course are to equip students with fundamental knowledge and skills in System Analysis, Design, and Project Management. By the end of the course, students will be able to:
To understand the fundamental concepts of systems, their characteristics, and the phases of the System Development Life Cycle (SDLC).
To apply various techniques for system requirements analysis, including fact-finding, modeling with Data Flow Diagrams (DFDs), Entity-Relationship Diagrams (ERDs), and use case modeling.
To understand the processes and challenges of system implementation, including testing, documentation, and providing user support.
To manage project effectively through scheduling, monitoring, controlling, and risk management to ensure successful project outcomes.

3. Methods of Instructions:

1. Lecture and discussion
2. Practical
3. Demonstration
4. Presentation
5. Case study

4. Course Contents:

Specific Objectives	Contents
Unit 1: Introduction to System Analysis and Design	**6 hours**
Explain the fundamental concepts of systems analysis and design, and their importance in software systems. Analyze the phases of the Systems Development Life Cycle (SDLC) and perform feasibility analysis.	1.1 **Overview of System Analysis and Design** • Introduction to system analysis and design concepts. 1.2 **Systems Development Life Cycle (SDLC)** • Phases of SDLC: Planning, Analysis, Design, Implementation, and Maintenance. • Introduction to system requests and feasibility analysis (technical, economic, legal, and organizational). 1.3 **Role and Skills of the Systems Analyst** • Introduction of programmer and system analyst • Skills and responsibilities of the systems analyst. • Characteristics of a system analyst • Interaction with stakeholders in system development projects. • The systems analyst as a bridge between technical teams and business stakeholders.

Unit 2. Project Management Fundamentals	**12 hours**
Apply project selection and planning techniques to initiate and plan a system project. Utilize project management tools like Gantt charts, CPM, PERT, and EVM to monitor and control project activities.	2.1 **Project Selection and Planning** • Selection methodologies and criteria for project approval. • Project planning: time, schedule and cost. 2.2 **Project Methodologies** • Traditional verses Agile methodologies. • Choosing the appropriate project management methodology. 2.3 **Managing and Controlling Projects** • Managing scope, time, and resources. • Staffing and coordinating project activities. • Risk management strategies. 2.4 **Project Management Tools** • Use of work breakdown structure (WBS), Gantt charts, Critical Path Method (CPM), and Program evaluation review technique (PERT). • Earned value management (EVM) and Key performance indicators (KPIs) for project monitoring and control.
Unit 3: Requirements Determination and Use Case Modeling	**8 hours**
Implement requirements gathering techniques such as JAD, interviews, and observation to define system requirements. Develop use case diagrams and descriptions to capture functional system requirements.	3.1 **Requirements Determination** • Techniques for gathering requirements: Interviews, Joint Application Development (JAD), document analysis, and observation. • Developing the requirements definition statement. 3.2 **Use Case Analysis** • Identifying use cases and creating use case diagrams. • Use cases as tools for capturing functional requirements, testing, and design
Unit 4: Process and Data Modeling	**8 hours**
Create Data Flow Diagrams (DFDs) and Entity-Relationship Diagrams (ERDs) to model business processes and data structures. Validate and refine process and data models for accuracy and completeness.	4.1 **Process Modeling** • Creating Data Flow Diagrams (DFDs) to define business processes. • Levels of DFDs: Context diagrams, Level 0, and Level 1 DFDs. • Process description and validation of models. 4.2 **Data Modeling** • Introduction to Entity-Relationship Diagrams (ERDs) for database modeling. • Elements of ERDs: Entities, attributes, relationships. • Creating and normalizing ERDs, and balancing ERDs with DFDs. • Data dictionary
Unit 5: System Design and Architecture	**8 hours**
Design system architectures, including client-server and cloud-based systems, based on business requirements. Apply user interface design principles to create and evaluate effective system interfaces.	5.1 **Transition from Requirements to Design** • Custom development vs packaged software vs outsourcing. • Developing system acquisition strategies based on business needs. 5.2 **System Architecture Design** • Designing client-server architectures and multi-tiered systems. • Introduction to cloud architecture and its relevance to modern system design. • Operational, performance, and security considerations in architecture design. 5.3 **User Interface Design** • Principles for effective UI design: Layout, aesthetics, consistency, and usability. • Prototyping and evaluating user interfaces.
Unit 6: Project Implementation and Post-Implementation Activities	**6 hours**

Coordinate programming, testing, and system migration strategies to ensure successful system implementation. Conduct post-implementation reviews and manage ongoing system maintenance to meet organizational needs.	**6.1 Managing the Programming Process** • Coordinating programming tasks and managing schedules. • Testing strategies: Unit tests, integration tests, system tests, and acceptance tests. **6.2 System Transition and Change Management** • Migration strategies: Direct, parallel, phased, and pilot conversions. • Managing user resistance, training, and change management. **6.3 Post-Implementation Reviews and Maintenance** • Conducting post-implementation reviews and ensuring system support. • Ongoing system maintenance and updates.

5. Practical Works:

The laboratory works should involve hands-on implementation and evaluation of concepts through unit-wise case studies, utilizing appropriate software tools. Students should work in groups to visualize and present their findings on systems analysis and project management.

1. Real-world application of systems analysis and project management concepts.
2. Hands-on experience with project management tools (Gantt charts, CPM, PERT) and modeling tools (DFDs, ERDs).
3. A comprehensive project involving system analysis, design, and management tasks.
4. Explore prototyping and methodologies such as RUP and CASE tools.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

Text Books

1. Dennis, A., Wixom, B. H., & Roth, R. M. (2012). *System analysis and design*. John Wiley & Sons, Inc. http://www.wiley.com/college/dennis

Reference Books

1. Jeffery Whitten, Lonnie Bently, System Analysis Design Methods 7/E McGraw-Hill
2. J.F. George and J. S. Valacich, *Modern System Analysis and Design*, 8/E. Pearson Education Inc., 2017
3. J. Satzinger, R. Jackson and S. Burd, System Analysis and Design in a Changing World, 7/E, Cengage Learning, 2016s

Pokhara University Faculty of Science and Technology	
Course Code.: CMP 380 (3 Credits)	Full Marks: 100
Course Title: Web Technology I (3-0-3)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: III	Program: Bachelor of Computer Application

1. Course Description:

This course is designed to make solid foundation in web development. Students start by exploring the evolution of the web and understanding key concepts like client-server architecture and web protocols. The course covers HTML, CSS, JavaScript and advanced features to make web pages interactive including version control, UI/UX, wireframing, and SEO.

2. General Objectives:

The general objectives of this course are as follows:

- To provide how to create web pages using HTML, including working with elements, attributes, and various components like lists, links, images, and forms.
- To enable students to style and layout web pages using CSS, covering basics, the CSS box model, responsive design, and advanced layout techniques like Flexbox and Grid.
- To build students' JavaScript skills, including how to write scripts, handle events.
- To provide an overview of JavaScript frameworks and libraries, including jQuery, React and introduce popular frameworks like Angular, and Vue.js.
- To offer a foundation in web design principles, including UI/UX design, wireframing, typography, color theory, usability, accessibility, and SEO.

3. Methods of Instruction:

10. Lecture and discussion
11. Practical
12. Demonstration
13. Presentation
14. Project

4. Course Contents:

Specific Objectives	**Contents**
Unit 1: Introduction to Web Technologies	**4 hours**
• Describe the client-server architecture and the roles of web browsers and servers. • Explain basic web protocols, DNS, and the differences between client-side and server-side scripting languages	1.1 Evolution of the Web 1.1.1 History of web 1.1.2 WWW 1.1.3 URL 1.1.4 Version of web 1.1.5 Search engine 1.1.6 Website and Web application 1.2 Client-Server Architecture 1.3 Overview of Web Browsers and Web Servers 1.3.1 Web browser, cross browser compabilities 1.3.2 Web server 1.4 Basic Protocols 1.4.1 Short overview of basic protocols 1.4.2 HTTP/HTTPS 1.4.3 HTTP request and response 1.5 DNS and its hierarchy 1.6 Client side and server side scripting language
Unit 2: HTML	**8 hours**
	2.1 Introduction 2.1.1 History and evolution of HTML

• Apply the HTML structure, tags and attributes in web design. • Implement lists, links, images, tables, and forms, and use semantic HTML for better web organization.	2.1.2 Concept of tag 2.1.3 HTML document structure 2.1.4 HTML elements and attributes 2.1.5 Basic text formatting tags 2.2 Lists, Links and Images 2.2.1 Unordered, ordered and description lists, nested list 2.2.2 Creating hyperlinks, types of link 2.3 Adding images 2.4 Table and Forms 2.4.1 html table elements and attributes, nested table, merging row and columns, grouping section of table 2.4.2 Ccommon form elements like text fields, text areas, selects, buttons, radio button. 2.4.3 input date, time 2.4.4 form methods 2.5 Semantic HTML 2.5.1 Header, footer, NAV, article, section, time, progress 2.6 Audio and video elements 2.7 div, span, aside, iframe, canvas
Unit 3: Cascading Style Sheets	**8 hours**
• Implement CSS including its syntax, box model, and layout techniques	3.1 Need of CSS 3.2 CSS Basics 3.2.1 CSS syntax, selectors, declarations, types, groping selectors, pseudo, comments, 3.2.2 CSS Specificity 3.2.3 Internal vs external CSS 3.2.4 Style text fonts, color, links 3.3 CSS Box Model 3.3.1 Box model properties for padding, border, width 3.3.2 Centering content with margins 3.3.3 Element backgrounds 3.4 Flexbox and Grid Layouts 3.5 Responsive Design Principles 3.5.1 Working with different screen size 3.5.2 Media queries in CSS 3.6 CSS Preprocessors (SASS, LESS)
Unit 4: JavaScript Basics	**8 hours**
• Explain the core concepts of JavaScript through variables, data types, and control structures, and to show how to use functions, manipulate the DOM, and handle events to make web pages interactive.	4.1 Introduction to JavaScript 4.2 Embedding JavaScript to a webpage 4.2.1 External Script 4.2.2 Internal Script 4.2.3 Using onClick(), onSubmit(), onLoad() 4.3 Variables, Data Types, Operators, Expressions 4.4 Java Script Programming Construct: 4.4.1 Assignment, data declaration, 4.4.2 if, if else, switch 4.4.3 while, for, do while 4.4.4 label, break, continue 4.4.5 Arrays 4.5 Functions and Scope 4.6 Dialog boxes

	4.7 Document Object Model 4.8 Event handling.
Unit 5: Advanced JavaScript Concepts	**6 hours**
• Analyses into advanced JavaScript features such as ES6+ syntax, promises, and async/await, and to understand how to handle errors and work with JSON and AJAX for more sophisticated web interactions.	5.1 ES6+ Features: 5.1.1 Let, Const, Arrow Functions, Promises, etc. 5.1.2 Objects 5.2 Regular Expressions 5.3 Asynchronous Programming (Async/Await) 5.3.1 Promises 5.3.2 Async/Await 5.3.3 Fetch 5.4 Error Handling 5.5 Introduction to JSON 5.5.1 JSON Syntax and Structure 5.5.2 Simple Example 5.6 Introduction to AJAX
Unit 6: Introduction to JS Frameworks and Libraries	**5 hours**
• Familiarize with JavaScript libraries like jQuery for easier DOM manipulation and to explore popular frameworks/library such as React, Angular, and Vue.js.	6.1 Jquery() 6.1.1 Why jQuery 6.1.2 Jquery fundamentals 6.1.3 Manipulating DOM elements. 6.2 Overview of Popular Frameworks/Library (React, Angular, Vue.js) 6.3 Choosing the Right Framework for Projects
Unit 7: Web Design Concepts	**5 hours**
• Explore the principles of UI/UX design, and apply concepts of typography, color theory, usability, accessibility, and SEO to create well-designed, user-friendly websites	7.1 Principles of User Interface Design 7.2 Introduction to UI/UX Design 7.2.1 Wireframing with its types 7.2.2 User Story 7.2.3 Responsive Design 7.3 Typography and Color Theory 7.4 Usability and Accessibility Guidelines 7.5 SEO
5. Practical Works:	

Students should perform following HTML lab tasks:

- In the head section of a web page, make sure to:
  - Add a title tag to set the page's title, which appears in the browser tab.
  - Link to external stylesheets using relative paths to apply the desired styles.
  - Include meta tags to specify important information such as the character encoding, description, keywords, author, and viewport settings for responsive design.
- In the body of a web page, include:
  - A table with headers, rows, and cells, adjusting attributes for spanning and sizing.
  - Text, images, sound clips, and videos with proper controls, sizing, and alternate text.
  - The tag for styling and layout.
  - Text elements using tags like <h1>, <p>, and <li>.
  - Classes and styles for formatting, including lists.
- Create hyperlinks from text and images to:
  - bookmarks on the same page
  - other locally stored web pages
  - a website using the URL
  - send mail to a specified email address
  - to open in a specified location (the same window, a new window)
- Create form for:
  - Login and signup purpose

- Design registration form including Textbox, TextArea, Select, Checkbox, Radio button, Button elements and group them using fieldset and legend.
- Use elements to break down in different blocks.

CSS Labs

- Set background properties including colour and images, font properties, style table including size, background colour, horizontal and vertical alignment, spacing, padding, borders.
- Create external styles to be tagged in a web page including h1, h2, h3, p, li.
- Attach comments to an external stylesheet.
- Center a div with its elements.
- Design navigation menu for web page with proper styling, when user put mouse over the menu the menu should be highlighted.
- Apply CSS to image, links, list
- Create product grid including product name, price, image, add to cart button and add effects when user put crouser on the product. Also make it responsive.

JavaScript:

- Link an external JavaScript file and manipulate DOM elements.
- Add and remove textbox from web page with add and delete button.
- Write a script to input information like Name, Address, Contact, Age abd display them in proper format when user clicks the submit button.
- Validate form inputs with JavaScrips
  - Validate login and signup form. the password should contain atleast letters, symbol and number and should be minimum 10 characters, username field cannot be blank, to sign up user must be adult.
- Load another page inside a page when user click on load button without reloading a page.
- Manipulate DOM elements using jQuery.

VCS:

- Start a Git repository with **git init**, stage files with **git add**, and commit with **git commit** make some changes and push with **git push**

UI/UX:

- Prepare Low-Fidelity wireframe of products and product details page.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

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Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such

score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

1. Felke-Morris, T. (2020). *Web development and design foundations with HTML5*. Pearson.
2. Simpson, K. (2021). *You don't know JS yet: ES6 & beyond*. Independently published
3. Chacon, S., & Straub, B. (2014). *Pro Git*. Apress.

Pokhara University Faculty of Science and Technology	
Course Code.: CMP 230 (3 Credits)	Full Marks: 100
Course Title: Operating Systems (3-0-1)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: III	Program: Bachelor of Computer Application

1. Course Description:

This course is designed to provide the concepts of operating system to various system environments. It incites system software, internal structures, functions and security features of operating system to run the application software and perform various tasks with optimized throughput. This course also introduces the emerging new trended operating system for distributed environment like cloud and mobile systems. After completion of this course, students can select, apply and implement the operating system features and functions for the best utilization of the device specification.

2. General Objectives:

The general objectives of this course are as follows:

- To acquaint the students with structure of operating systems and their functionality.
- To acquaint the students with concepts of resource allocation and management with schedules.
- To acquaint the students with the knowledge of process and thread, I/Os, Memory, CPU, disk management, network security systems.
- To acquaint the students with basic concepts of operating systems new trends such as real-time, distributed, cloud systems and mobile.

3. Methods of Instructions:

- Lecture and discussion
- Practical
- Demonstration
- Case studies

4. Course Contents:

Specific Objectives	Contents
Unit 1: Basics of Operating System	**6 hours**
• Familiarize with basic concepts of Operating systems, and its structures. • Explain the features of generation and types of operating systems.	1.1 Introduction, goals of operating system 1.2 Operating-System Structures and functions 1.3 Types of Operating Systems *Batch Processing, Multiprogramming, Multiprocessing, Networking, Real Time (RTOS), Distributed, Embedded system* 1.4 System Interface, System Calls 1.5 Virtual Machine
Unit 2: Process, Threads and Scheduling	**13 hours**
• Familiarize with task, process and threads • Implement of resource allocation techniques • Describe the mutual exclusion for resource utilization	**2.1 Process** 2.1.1 The Process Model 2.1.2 Process Creation and Process Termination 2.1.3 Process Hierarchies 2.1.4 Process States 2.1.5 Implementation of Processes **2.2 Threads** 2.2.1 Thread Usage

	2.2.2 Threads Models (Many-to-one model, One-to-One Model, Many-to many model) 2.2.3 User Space and Kernel Space Threads 2.2.4 Hybrid Implementations 2.2.5 Difference between Threads and Processes **2.3 Inter-process Communication** 2.3.1 Race Conditions 2.3.2 Critical Regions 2.3.3 Mutual Exclusion with Busy Waiting 2.3.4 Sleep and Wakeup, Semaphores, Mutexes, Monitors, Message Passing 2.3.9 Avoiding Locks: Read-Copy-Update **2.4 Process Scheduling** 2.4.1 Basic Concept 2.4.2 Type of scheduling (Preemptive scheduling, Non-preemptive scheduling, batch, Interactive, real time scheduling), 2.4.3 Scheduling Criteria or performance analysis, Scheduling Algorithm (Round-robin, First come first served, Shortest-job- first, Shortest process next, Shortest remaining Time next, real time, priority fair share, guaranteed, Lottery scheduling) **2.5 Deadlocks** 2.5.1 System Resources: Preemptable and Non-preempable 2.5.2 Method of handling Deadlocks, 2.5.3 Deadlock prevention 2.5.4 Deadlock avoidance: Banker's Algorithm, *2.5.5* Protection- *System Protection, Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix, Implementation of Access Matrix, access Control list* **2.6** Research on Processes and Threads
Unit 3: Storage Management	**13 hours**
• Explain the role and working procedure of memory • Familiarizing with virtual memory management • Implement the page replacement algorithms • Explain the mechanism file and filing	**3.1 Memory Management** 3.1.1 Logical and physical Address Space 3.1.2 Swapping 3.1.3 Contiguous Allocation 3.1.4 Paging, Structure of Page Table 3.1.5 Segmentation, Segmentation with Paging **3.2 Virtual Memory** 3.2.1 Background 3.2.2 Demand Paging, 3.2.3 Performance of Demanding Paging, 3.2.4 Page Replacement, Page Replacement Algorithms, Allocation of Frames, 3.2.5 Thrashing. **3.3. File System Interface and Implementation** 3.3.1 File System Interface- *The Concept of a File, Access methods, Directory Structure, File System Mounting, File Sharing, Protection*

	3.3.2 File System Implementation- *File System Structure, File System Implementation, and Allocation methods, Free-space Management, Directory Implementation, Efficiency and Performance*
Unit 4: Input/output Management	**7 hours**
• Describe the role of input/output devices • Explain the different approaches for optimal output	4.1 Principles of I/O Hardware- *I/O Device, Device Controller, Memory Mapped I/O, Direct Memory Access* 4.2 Principles of I/O Software- *Goals of I/O Software, Polled I/O verses Interrupt Driven I/O, Character User Interface and Graphical User Interface, Device Driver, Device Independent I/O Software, User-space I/O Software* 4.3 Mass Storage Structure - Overview of Mass Storage Structure, Disk Structure, Disk Attachment, Disk Scheduling, Disk Management, Swap space Management Redundant Array of Inexpensive Disks , RAM Disks, Optical Disk
Unit 5: New Trend in Operating System	**4 hours**
• Familiarize with current trends of operating systems	5.1 Concept, character and role of Distributed, Cloud, Mobile and Multimedia operating systems 5.2 Memory wall and bottleneck for operating system
Unit 6: Case Study	**5 hours**
• Compare operating system structure and functionality	**6.1 UNIX/Linux -** Design principles, Inter-process communication, Kernel modules, Network structure **6.2 Windows -** Design principles, Programmer interface, System components **6.3** Security level in Linux/Windows

5. List of Practices:

Laboratory work should cover the operating system structure and functions of any two popular operating system. It also insist the students to design model of operating system with the reference of open source guideline. Students should complete the following tasks in laboratory:

1. Understanding and running all the internal command and external commands in Microsoft Disk Operating Systems.
2. Installation and user, application management in Windows (current version)
3. Simulation of Process Scheduling Algorithms
4. Simulation of Page Replacement Algorithms
4. Simulation of Disk Arm Scheduling Algorithms
5. System Administration (user, disk, role, etc.) in any open source operating system.

6. 6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

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Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

Books

1. Andrew S. Tanenbaum, Herbert Bos, *"Modern Operating Systems"*, Pearson, 4th Edition, ISBN-10: 0-13-359162-X. 2011

References

1. A. Silberschatz, P.B. Galvin, G. Gagne *"Applied Operating System Concepts*", Wiley, 8th Ed.
2. D. M. Dhamdhere , "*System Programming and Operating System*" - Tata McGraw-Hill, 20
3. Andrew S. Tanenbaum, *"Distributed Operating System*", Pearson
4. Naresh Chauhan, Principles of Operating Systems, Oxford University Press 2014, Oxford University Press, ISBN-13: 978-0-19-808287-3

Pokhara University Faculty of Science and Technology	
Course Code: CMP 323 (3 Credits)	Full Marks: 100
Course Title: Software Engineering (3-0-2)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: IV	Program: Bachelor of Computer Application

1. Course Description:
This course introduces students to essential principles and methodologies of software engineering, focusing on Object-Oriented Analysis and Design (OOAD) using Unified Modeling Language (UML). It covers software development process models, software design, coding practices, software testing, maintenance, and quality assurance. Students will apply these concepts through case studies and project work, ensuring practical skills in software design and development.

2. General Objectives:
1) To provide students with a comprehensive understanding of the principles, methodologies, and best practices in software development.
2) To equip students with essential skills in Object-Oriented Analysis and Design (OOAD) using UML, software process models, coding standards, software testing, and software quality assurance.
3) To design, develop, and maintain software systems that meet high standards of functionality, quality, and performance in real-world scenarios.

3. Methods of Instructions:
15. Lecture and discussion
16. Practical
17. Demonstration
18. Presentation
19. Case study

4. Course Contents:

Specific Objectives	Contents
Unit 1: Introduction to Software Engineering	**5 hours**
• Describe concepts and challenges in software engineering. • Analyse the attributes of good software and ethical responsibilities in the software industry.	1.1 Overview of Software Engineering 1.2 Definition and characteristics of software. 1.3 Software engineering as a discipline and its challenges. 1.4 Professional Practices in Software Engineering 1.5 Attributes of good software. 1.6 Software development life cycle (SDLC) and ethical practices.
Unit 2: Software Development Process Models	**4 hours**
• Apply various software development process models to real-world software projects. • Analyse the role of CASE tools in the software development lifecycle. software process, software process model:	2.1 Software Process Models 2.2 The Waterfall Model, Evolutionary Development, Component Based Software Engineering(CBSE), Process Iteration, Incremental Delivery, Spiral Development, Rapid Software Development, Agile Methods, Extreme Programming, Rapid Application Development and Rational Unified Process (RUP). 2.3 Overview and classification of CASE tools. 2.4 Role of CASE tools in modern software engineering practices.
Unit 3: System Analysis in OOAD	**4 hours**

• Analyze and document system requirements in an object-oriented context. • Develop use case models to define functional requirements in OOAD.	1. Requirements in Object-Oriented Systems 2. Identifying functional and non-functional requirements in object-oriented systems. 3. Use of Use Case Diagrams to model system requirements. 4. Requirement Modeling in OOAD 5. Creating Use Case Descriptions and Use Case Diagrams in UML for capturing functional requirements.
Unit 4: Object-Oriented Design (OOD) with UML	**12 hours**
• Develop models of software systems using UML diagrams. • Apply object-oriented principles and design patterns in software system design	4.1 Introduction to OOD 4.2 Design Concept; Abstraction, Architecture, Patterns, Modularity Cohesion, Coupling Information Hiding, Functional Independence, Refinement 4.3 Object-oriented principles: abstraction, encapsulation, inheritance, polymorphism. 4.4 Benefits of OOAD in software engineering. 4.5 Class diagrams, object diagrams, and sequence diagrams, State machine diagrams, component diagrams, and deployment diagrams. 4.6 Introduction to common design patterns such as Singleton, Factory, Observer. **4.7** Procedural Design Using Structural Methods, User Interface Design, Human Computer Interaction, Information Presentation, Interface Evaluation, Design Notation
Unit 5: Software Testing and Quality Assurance	**9 hours**
• Compare different software testing techniques to ensure software quality. • Implement verification, validation, and quality assurance practices in software projects.	5.1 Testing Techniques 5.2 Unit testing, integration testing, and system testing. 5.3 Black-box and white-box testing techniques. 5.4 Verification and Validation 5.4 The importance of software testing in the development life cycle. 5.5 Software Quality Assurance (SQA) 5.6 Overview of software quality standards (ISO, CMM). 5.7 Importance of software quality assurance (SQA) processes. 5.8 Risk Categories, The RMM Plan
Unit 6: Coding Practices and Programming	**4 hours**
• Analyse the best coding practices to improve software quality and maintainability. • Compare appropriate programming languages and tools for developing software systems.	6.1 Programming Languages and Development Tools 6.2 Selecting programming languages and tools for software development. 6.3 Best Practices in Coding 6.4 Clean code practices, code readability, and refactoring. 6.5 Introduction to software development environments and integrated development environments (IDEs).
Unit 7: Software Maintenance and Evolution	**6 hours**
• Describe types of software maintenance and their significance in the software lifecycle.	7.1 Software Maintenance 7.2 Evolving Nature of Software, 7.3 Types of software maintenance: corrective, adaptive, perfective, preventive. 7.4 Re-engineering and Configuration Management

• Apply re-engineering and configuration management techniques to maintain and improve software systems.	7.5 Process of software re-engineering and version control. 7.6 Importance of configuration management in large-scale software systems.

5. List of Practical:

Lab

1. Requirements Gathering and Use Case Modeling: Conduct a requirements gathering session to identify functional requirements for a system. Create Use Case Diagrams and develop use case descriptions for the identified requirements.
2. Class Diagram Design: Design a Class Diagram for a given system based on the collected requirements. Model classes, attributes, methods, and relationships (inheritance, association, aggregation).
3. Sequence Diagram Design: Create a Sequence Diagram to represent the interaction between objects in a use case. Model the flow of messages and interactions between system components.

4: Activity Diagram for Workflow Modeling: Design an Activity Diagram to represent the flow of activities in a business process. Capture the flow of control, decision points, and concurrency in the system.

5: Coding Best Practices and Refactoring: Implement a software module using clean code practices. Perform code refactoring to improve readability, maintainability, and performance of an existing module.

6: Unit Testing with Junit: Write Unit Tests for a given software module using JUnit. Perform testing on individual components and analyze the test results.

7: Software Version Control with Git: Set up a project in Git and implement version control for the software development process. Use Git commands to track changes, commit, push, pull, and merge code.

8: Software Architecture Design: Design a multi-layered Software Architecture (e.g., client-server or layered architecture) for a system. Create Component Diagrams to visualize the system's architecture.

9: Software Quality Assurance (SQA) Techniques: Conduct a quality assurance review for a software project. Use static code analysis tools and apply software quality standards such as ISO or CMMI.

10: Black-Box and White-Box Testing: Perform Black-Box Testing and White-Box Testing on a software application. Design test cases, execute tests, and document results to ensure software quality.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

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Practical		**20**		

Attendance and Lab Participation	10%				
Lab Report	20%				
Lab Examination	40%				
Viva Examination	30%				
Total Internal Marks		**50**			
Full marks=50+50 =100					

6.2 Students' Responsibilities:

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7. Prescribed Books and References:

Text Books

1. Roger S. Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition, McGraw Hill International edition,2005

Reference Books

1. Ali Behforooz , and Frederick J. Hudson, "Fundamental of Software Engineering", OUP,1996
2. Ian Somerville, "Software Engineering", 9th Edition Addison-Wesley, 2010.
3. Rajib Mall," Fundamental of Software Engineering", 3rd Edition, 2010.

Pokhara University Faculty of Science and Technology	
Course Code.: CMP 226 (3 Credits)	Full Marks: 100
Course Title: Database Management System (3-0-1)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: IV	Program: Bachelor of Computer Application

1. Course Description:

This course is designed to encompass the fundamental concepts of database management system. These concepts include the concept of database design, database languages and the database system implementation. This course presents the introductory concepts of database security, query processing and optimization, transactions and their concurrency control and recovery policies. This course also introduces the emerging new trended databases such as NoSQL and Block-chain databases. After completion of this course, students can design and implement a database system to develop a software application.

2. General Objectives:

The general objectives of this course are as follows:

- To acquaint the students with the knowledge of relational database design using ER Model.
- To develop the skills in students to design normalized relational database required for a specified application.
- To acquaint the students with the knowledge of database security, query processing and optimization, files and record organizations, transaction, concurrency control, data recovery mechanisms.
- To acquaint the students with concepts of NoSQL databases and Block-chain Databases.

3. Methods of Instructions:

- Lecture and discussion
- Practical
- Project and presentation

4. Course Contents:

Specific Objectives	Contents
Unit 1: Introduction	**7 hours**
• Familiarize with basic concepts of database systems, and its architecture. • Explain the concept for database components and their roles	1. Database Management System purpose and applications, 2. Database Systems vs File Systems, Storage Management and Indexing 3. View of Data- Data Abstraction (Physical, logical and view level, Data Independence) 4. Instances and Schemas, 5. Database Languages (DDL, DML and DCL) 6. Database and Application Architecture- Database System Architecture and Database Application Architecture (two-tier and three-tier)
Unit 2: ER and Relational Model	**8 hours**
• Familiarize with ER Model and Relational Model and relational algebra. • Design the relational database using ER model.	1. Introduction to ER Model: 2. Entity sets, attributes and values, Relationship sets-participation, entity's role, descriptive attributes, degree of relationship set, Mapping Cardinalities, Attributes-simple, composite, single-valued, multi-valued, derived, Entity-Relationship (ER) Diagram, Specialization, Generalization, and Aggregation 3. Key and its types 4. Reducing ER diagrams to Relational Schema

	5. Structure of Relational Databases, Database Schema, Schema Diagrams **6.** Relational Algebra
Unit 3: Structured Query Language	**6 hours**
• Implement and write DDL and DML queries in the SQL.	1. Structured Query Language (SQL)- SQL DDL and DML 2. Basic Structure of SQL Queries, DDL queries, Basic Operations (Rename, String, Attribute Specification in the select clause, order by, where-clause), Set Operations, Null values, Aggregate Functions, 3. Nested Queries, 4. Join Expressions (Natural Join, Join Conditions, Outer Joins), Views, stored Procedures
Unit 4: Relational Database Design	**6 hours**
• Apply the integrity constraints to implement database securities. • Analyze the database to a defined normal form.	1. Integrity constraints- Domain Constraints, Entity Integrity Constrains, Referential Integrity Constraints, Assertions and Triggers 2. Features of Good Relational Designs 3. Functional dependencies and Armstrong's Axioms 4. Closure of a Set of Functional Dependencies and Closure of Attribute Sets 5. Database Normalization and Normal Forms- 1NF, 2NF, 3NF and BCNF 6. De-normalization for Performance
Unit 5: Query Processing and Optimization	**5 hours**
• Explore the mechanism of query processing and need of query optimization. • Familiarize with different file organization methods.	1. Introduction to Query Processing 2. Equivalence of Expressions 3. Query Cost Estimation 4. Query evaluation and execution plan 5. Query Optimization
Unit 6: Transactions and Concurrency Control	**5 hours**
• Implement ACID properties, isolation, serial schedule. • Familiarize with the need of Concurrency control.	1. Transaction Concepts 2. Transaction Model and State Diagram 3. ACID properties of transaction 4. Serializability- conflict and view serializability 5. SQL Standard Isolation Levels 6. Concurrency Control- Lock-Based Protocols
Unit 7: Security and Crash Recovery	**6 hours**
• Describe the recovery algorithms and techniques to protect and recover the data from various failures. • Explain the need of database security. • Compare the role of access control, authorization, views and encryption mechanism to provide database security.	1. Security and integrity violations 2. Access control and authorization, views 3. Failure classification 4. Recovery and Atomicity- log records, database modification, concurrency control and recovery, transaction commit, Redo and Undo Transactions using Log, Check Points, 5. Recovery Algorithm Using Log Records- Transaction Rollback, Recovery after a System Crash, Optimizing Commit Processing
Unit 8: Emerging Trend in Databases	**5 hours**

• Explain the concepts of NoSQL, ODBMS, Distributed Databases	1. NoSQL Databases- Characteristics, Categories, Advantages 2. Object Oriented Database and ORM 3. Parallel and Distributed Databases 4. Data Warehouse and Data Mining

5. Practical Works

Laboratory work should cover the database design, use of database languages and database system implementation using any relational database management system such as MS SQL or MySQL or Oracle etc. Students should complete the following tasks in laboratory:

SN	Tasks to Complete
1	Introduction to MS SQL (or any RDBMS), its datatypes and its installation.
2	All SQL DDL operations studied in Unit 2 such as creating database, creating tables, delete database, drop table, alter etc.
3	All SQL DML operations studied in Unit 2 such as database modification operation- insert into, delete, update etc.
4	Implementing Join Expressions (Natural Join, Join Conditions, Outer Joins)
5	Implementing Stored Procedures.
6	Illustration and implementation of Views.
7	Implementing Integrity constraints (Domain Constraints, Entity Integrity Constrains, Referential Integrity Constraints)
8	Implementing Assertions and Triggers.
9	Implementation of Database Security and Privileges: Grant and Revoke Commands, Commit and Rollback Commands.
10	Connecting database with connection string using any standard programming language and executing SQL queries.

Students should submit a project work that uses all the knowledge obtained from this course to design and implement a database system for any application that students chose. The students should design the database using ER model and present using ER diagram which are then reduced to relational schema. The students should apply the constraints studied in this course including triggers. The database should be in at least 3NF. The marks for the practical evaluation must be based on the project work submitted by students.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	50%			
Practical		**20**		
Attendance and Lab Participation	10%			

Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

Books

5. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2011). *Database system concepts.* McGraw Hill

References

1. Majumdar, A. K., & Bhattacharyya, P. (1996). *Database Management Systems*. McGraw-Hill.
2. Elmasri, R., & Navathe, S. B. (1994). Fundamentals of Database Systems Benjamin Cummings. *Redwood City, CA*.
3. Everest, G. C. (1986), *Database management*. McGraw-Hill, Inc.

Pokhara University Faculty of Science and Technology	
Course Code.: CMP 242 (3 Credits)	Full Marks: 100
Course Title: Computer Graphics and Multimedia Technology (3-0-2)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: VI	Program: Bachelor of Computer Application

1. Course Description:

This course is designed to encompass the fundamental concepts of Computer Graphics. It covers Graphics hardware, various drawing algorithms, animation and multimedia at the industry level. It includes areas such as 2D and 3D modeling, rendering, animation, image processing, virtual reality, and augmented reality. Multimedia, on the other hand, refers to the integration of different forms of media, such as text, graphics, audio, video, and animation, to create interactive and immersive experiences

2. General Objectives:

The general objectives of this course are as follows:

- To introduce the hardware and concept of object rendering.
- To introduce the concept of 2D and 3D Graphics Algorithm
- To familiarize students with the knowledge of multimedia and animation.

3. Methods of Instructions:

- Lecture
- Tutorial
- Practical and case studies

4. Course Contents:

Specific Objectives	Contents
Unit 1: Introduction to Graphics	**6 hours**
• Familiarize with the basic Graphics hardware and interfacing and software	1.1 Application of computer graphics 1.2 CRT monitor working 1.3 Display technology (Raster and Random) 1.4 Color generation technique in CRT 1.5 Frame buffer organization and video controller 1.6 Recent trend virtual reality and augmented reality
Unit 2: Scan Conversion	**6 hours**
• Implement Scan conversion of Graphics Primitive line and Circle.	2.1 Importance of scan conversion 2.2 Line Drawing Method: DDA and BLA 2.3 Mid-point circle generation algorithm
Unit 3: Two Dimensional Transformation and Windows Viewing	**8 hours**
• Implement the function of Basic and Advance transformation of solid body	3.1 2D Geometric Transformation (Translation, Rotation, Scaling) 3.2 Homogeneous Coordinate Representation 3.3 Successive Transformation 3.4 Window to Viewport Transformation 3.5 Cohen Sutherland Line Clipping Algorithm
Unit 4: Three Dimensional Graphics and Hidden Surface Removal	**6 hours**
• Describe the concept of 3D Graphics	4.1 3D Geometric Transformation

and 3D object representation projection and hidden surface removal	4.2 3D Object Representation- Polygon Surface and Polygon Mesh 4.3 Parallel and Perspective Projection 4.4 Hidden Surface Removal 4.5 Back Face Detection, Z-Buffer Method, Scan Line Method
Unit 5: Illumination Model and Shading	**6 hours**
• Explain the Effect of Light in a computer Scene and Shading techniques	5.1 Basic illumination model 5.2 Specular Reflection and Phong Model 5.3 Polygon Rendering Methods (Flat, Goraud and Phong shading)
Unit 6: Multimedia System and Representation	**9 hours**
• Introduction to Multimedia and media representation and various different kind of authoring tools and Animation	6.1 Introduction to Multimedia 6.2 Different kind of Image Format 6.3 Sound and Audio 6.3.1 Digital Audio Formats 6.3.2 MIDI hardware, software and Messages 6.4 Multimedia Authoring Tools
Unit 7: Multimedia Compression Techniques	**7 hours**
Explain video, compression technology and animation	7.1 Video 7.1.1 Digital video formats (AVI and MOV) 7.2 Compression techniques for audio and video (spatial and temporal) 7.3 Animation and its types

5. Practical Works:

Laboratory work should cover the basic principles of Computer Graphics and Multimedia Systems. It also insists the students to design graphical system with the reference of open source guideline. Students should complete the following tasks in laboratory:

1. To draw a straight line using Digital Differential Analyzer and Bresenham's Line Drawing Algorithm
2. Digitize a circle using mid-point circle Drawing Algorithm
3. Translation, Scaling and Rotation in 2D
4. 2D Screen Viewing Transformation
5. To draw a graphical output primitives using Open GL
6. To implement and test the Image Compression Algorithm (JPEG Compression)

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6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**		
Attendance /Class Participation	10%			
Assignments	20%			
Project Work/Presentations	20%			
Term Exam	50%			**50**

Practical		**20**	**Semester End examination**	
Attendance and Lab Participation	20%			
Lab Report	30%			
Lab Examination	30%			
Viva Examination	20%			
Total Internal Marks		**50**		
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

Text Books

1. Hearn D & Baker M.P (1997) Computer Graphics, C Version. Pearson Education India
2. Buford, J.F.K(2002), Multimedia Systems, Pearson, India

References

1. Malay K. Pakhira Computer Graphics Multimedia and Animation ,Second Edition PHI Publication

Pokhara University Faculty of Science and Technology	
Course Code.: MTH 320 (3 Credits)	Full Marks: 100
Course Title: Probability and Statistics (3-0-0)	Pass Marks: 45
Nature of the Course: Theory	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: IV	Program: Bachelor of Computer Application

1. Course Description:

This course aims to provide Bachelor of Computer Application (BCA) students with a solid understanding of applied statistics, as well as practical skills. It covers statistical concepts including data presentation, measures of central tendency, dispersion, correlation and regression analysis, probability, probability distributions, sampling, estimation, and hypothesis testing. The course focuses on hands-on learning, with laboratory sessions where students can use statistical analysis software tools. Having a mastery of these statistical tools is essential for creating effective dashboards in computer applications, software, web development, and research projects.

2. General Objectives:

The general objectives of this course are as follows:

- To provide students with a fundamental understanding of Statistics, Probability, and Sample Survey techniques, particularly in relation to Computer Applications.
- To equip students with the ability to compute various statistical measures of central tendency and dispersion for given data sets.
- To develop students' skills in utilizing statistical tools and techniques that are relevant to computer applications and research work.
- To familiarize students with statistical software used for data analysis, such as Microsoft Excel, SPSS, or R Studio.
- To make students competent in understanding and applying the concepts of hypothesis testing and significance tests.

3. Methods of Instructions:

- Lecture and discussions
- Demonstration
- Practical
- Case study

4. Course Contents:

Specific Objectives	Contents
Unit 1: Introduction	**3 hours**
• Acquaint fundamental concepts of statistics, its broad applications in various fields. • Familiarize different types of variables and data. • Equip with the understanding of key elements of statistical analysis.	1.1 Introduction to Statistics, 1.2 History and its scope of statistics 1.3 Variable and data, types of variables (quantitative and categorical), types of data (quantitative and qualitative), measurement and scaling, types of scale (nominal, ordinal, interval and ratio), fundamental elements of statistical analysis.

Unit 2: Data Collection	5 hours
• Enable in distinguishing between primary and secondary data sources and understand their uses in research. • Acquaint with the fundamental principles and approaches of experimental and survey research methods.	2.1 Sources of Data (Primary and Secondary) 2.2 Primary data collection methods (Field survey, Observation, Experiment) and tools (Questionnaire, Observation Checklist. 2.3 Introduction to Experimental research and survey research 2.4 Data preparation - editing, coding, and transcribing
Unit 3: Presentation of Data	**4 hours**
• Enhance proficiency in organizing and presenting data through the utilization of diverse tabular formats using computer software. • Equip with the skills to effectively visualize data using different graphical methods both manually and using computer software.	3.1 **Tabular Presentation:** Steam-and-leaf display, frequency distribution, relative frequency distribution, cumulative frequency distribution, bivariate frequency distribution. 3.2 **Data Visualization**: Bar diagram (simple, multiple, sub-divided, percentage), pie chart, histogram, frequency polygon, frequency curve, ogive and time-plots.
Unit 4: Summary Measures	**5 hours**
• Enable to compute and interpret various measures of central tendency using computer software. • Develop ability to compute and interpret measures of variation both manually and computer software. • Familiarize and calculate with the concepts of distribution shape	4.1 **Measure of central tendency**: mean (Arithmetic, Geometric and Harmonic), median, mode and mid-hinge 4.2 **Partition Values:** Quartiles, Deciles, Percentiles 4.3 **Measures of variation:** range, interquartile range, semi-inter quartile range standard deviation, Variance and coefficient of variations. 4.4 **Shape of the distribution:** five-number summary, box-and-whisker plot, concept of skewness and kurtosis.
Unit 5: Probability	**5 hours**
• Familiarize with fundamental probability rules and apply them in different scenarios of problem solving. • Make knowledgeable about the concept of conditional probability and Bayes' Theorem, and to enable to apply these concepts in problem-solving.	5.1 **Introduction of Probability:** Basic concepts, Counting rules, Objective and subjective probability, Marginal and joint probability 5.2 **Basic Probability:** Addition rule, Multiplication rules, 5.3 **Conditional Probability:** Concept of Conditional Probability, Bays' Theorem
Unit 6: Discrete Probability Distribution	**10 hours**
• Distinguish between discrete and continuous random variables. • Familiarize with probability distributions. • Explain Binomial, Poisson, and Normal distributions, including the normal approximation of binomial and Poisson distributions.	6.1 **Random variables**: Introduction to random Variable, types of random variable (discrete and continuous), Mean and standard deviation of discrete random variables, Mathematical expectation of discrete random variable 6.2 **Probability distribution:** Concept of probability distribution, types of probability distribution (discrete and continuous), Probability mass function, Probability density function, probability distribution function, Binomial distribution, Poisson distribution and Normal distribution, Normal approximation of binomial and Poisson distribution
Unit 7: Sampling and Estimation of Population Parameters	**5 hours**

• Acquaint with the concepts of population and sample, and sampling methods in statistical analysis. • Calculate and interpret confidence intervals for means and proportions, • Gain hands-on experience using computer software to analyze and interpret data.	7.1 **Sampling:** Population and sample, different types of sampling methods, 7.2 **Estimation:** Law of large numbers, Central limit theorem, point and interval estimation, Statistical confidence, Confidence intervals, Confidence interval estimation for mean and proportion.
Unit 8: Hypothesis Testing	**6 hours**
• Provide with an understanding of hypothesis testing, including steps involved, and the connection between confidence intervals and hypothesis testing. • Equip with skills single mean and double means using z-tests and t-tests both manually and using computer software. • Acquaint with the Chi-square test for independence and gain practical experience with computer software.	8.1 **Introduction to Testing of hypothesis: Statistical hypothesis, steps in hypothesis testing,** P-value approach to hypothesis testing, Connection between confidence intervals and hypothesis testing, 8.2 **Test of significance of mean:** Test of single mean, comparing two means (z and t-test) 8.3 **Test of proportion:** Test of single proportion, Comparing two proportions 8.4 **Test of independence using Chi-square test**
Unit 9: Correlation and Regression Analysis	**5 hours**
• Analyze correlations using scatter diagrams, Pearson's correlation coefficient, and Spearman's rank correlation. • Performed simple linear regression analysis using the least squares method to fit models. • Calculate and interpret the Coefficient of Determination and Standard Error, and analyze regression output with the help of statistical software.	9.1 **Correlation Analysis**: Scatter diagram, Correlation Coefficient and its Properties, Spearman's rank correlation, Pearson's correlation coefficient. 9.2 **Simple Linear Regression Analysis:** Model fit using least square method, Coefficient of Determination and Standard Error, interpretation of output of regression analysis using statistical software

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		50	Semester End examination	50
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:
Text Book 1. **Levin, Richard I. And David S. Rubin: Statistics for Management, Prentice-Hall of India** **Reference Books** 1. Fundamentals of Mathematical Statistics – 1 st Edition S. C. Gupta, V.K. Kapoor and S Chand. 2. Introduction to Probability & Statistics – 4 th Edition J. Susan Milton, Jesse C. Arnold Tata McGraw Hill. 3. Fundamentals of Statistics (7th edition), S C Gupta, Himalaya Publishing house 4. Probability and Statistical Inference (9th Edition), Robert V. Hogg & Elliot Tanis & Dale Zimmerman, Pearson.

Pokhara University Faculty of Science and Technology	
Course Code.: CMP 380 (3 Credits)	Full Marks: 100
Course Title: Web Technology II (3-0-3)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: IV	Program: Bachelor of Computer Application

1. Course Description:

This course is designed to give students hands on knowledge of server side development. After completing this course students will not only have theoretical knowledge but also practical, hands-on experience in building, securing, and deploying web applications, making them well-prepared for the challenges of professional web development. The course covers essential topics such as server architecture, web frameworks, database integration, web services, security best practices, and deployment strategies. Students will gain hands-on experience in building, securing, and deploying web applications.

2. General Objectives:

The general objectives of this course are as follows:

- To provide knowledge about the fundamentals of server-side developmentusing appropriate programming language.
- To enable students to develop web applications using the framework and manage databases in web applications.
- To introduce students to the basics of Content Management Systems (CMS).
- To develop skills in creating and consuming web services and APIs.
- To equip students with the knowledge to implement security best practices in web applications.
- To prepare students for the deployment of web applications to production environments.

3. Methods of Instructions:

This course can be delivered through a combination of lectures, hands-on coding exercises, and practical projects. Students will engage in interactive sessions that involve building real-world web applications and practicing new skills through lecturer and assignments. Each unit will include both theoretical explanations and practical tasks to reinforce learning and ensure students can apply their knowledge effectively.

4. Course Contents:

Specific Objectives	**Contents**
Unit 1: Introduction to Server-Side Development	**4 hours**
• Describe web technologies role of client-side verses server-side. • Explain HTTP request and response message format and different web servers. • Explore server architectures and backend programming languages. • Implement MVC design in web development.	1.1. Overview of Web-I 1.2. HTTP request message, HTTP response Message 1.3. Differences between client-side and server-side technologies 1.4. Web Clients 1.5. Web Server 1.5.1. Role of the server in a web application 1.5.2. Overview of web servers - Apache - Nginx - IIS 1.6. Server-side architecture 1.6.1. Monolithic 1.6.2. Micro-services 1.7. Introduction to backend programming languages (Python, PHP, Java)

	1.8. Understanding MVC (Model-View-Controller) architecture
Unit 2: Server-Side Programming with Python	**8 hours**
• Explain about Python data types, functions, and control structures. • Explore about lists, dictionaries, files, and understand OOP concepts. • Implement CSV and JSON files	2.1 Setting up the Python environment 2.2 Introduction to Python syntax and basic operations 2.3 Data Types and Variables 2.3.1. int, float, str, bool, list, dict, tuple, set 2.3.2. Variable assignment and naming conventions 2.3.3. Type conversion 2.4 Control Structures 2.4.1 Conditional statements: if, elif, else 2.4.2 Loops: for and while 2.4.3 List comprehensions 2.5 Introduction to functions 2.5.1 Defining functions, parameters, return values 2.5.2 Scope and global variables 2.6 Working with Python Data Structures (1 hour) 2.6.1 Lists 2.6.2 Creating, accessing, and modifying lists List methods: append(), remove(), pop(), etc. 2.6.3 Dictionaries: Key-value pairs, adding and accessing items Dictionary methods: get(), keys(), values() 2.6.4 Tuples and Sets: Creating and using tuples and sets 2.6.5 Differences between lists, tuples, and sets 2.7 File Handling in Python 2.7.1 Introduction to File Operations 2.7.2 Opening and closing files 2.7.3 File modes 2.7.4 Reading from Files Reading entire files, line-by-line reading 2.7.5 Using with statement for file operations 2.7.6 Writing to Files: Writing data to files, Appending data to existing files 2.8 Working with CSV and JSON Files 2.8.1 Reading and writing CSV files using csv module 2.8.2 Handling JSON files using json module 2.9 OOP Concepts
Unit 3: Working with Web Framework	**8 hours**
• Describe the importance of framework in web development • Implement HTTP requests, form submissions, cookies, and sessions • Explore Flask features like file uploads and sending emails	3.1 Framework overview: 3.2 Introduction to Flask Framework 3.3 Setting up a Flask environment 3.4 Flask basics: Routes, Views, and Templates 3.5 Handling HTTP requests and responses 3.6 Building Basic Web Functionality 3.6.1 Implementing form submission and input validation 3.6.2 Understanding and managing cookies and sessions

	3.7 Flask redirect, Message flashing , File upload 3.8 Sending Email
Unit 4: Database Integration	**8 hours**
• Implement web application with database • Implement basic CRUD operations and manage database transections.	4.1 Introduction to Databases 4.2 Overview of relational (SQL) vs. NoSQL databases 4.3 Database Connectivity with Flask 4.3.1 Connecting Flask applications to databases using SQL Alchemy 4.4 Performing CRUD operations using HTML template 4.5 Handling database connections and transactions
Unit 5: Introduction to Content Management Systems	**3 hours**
• Explain the knowledge about CMS and its importance in web. • Describe the knowledge of the popular frameworks.	5.7 Overview of popular CMS platforms (WordPress, Drupal, Joomla) 5.8 Advantages of using CMS in web development 5.9 Use cases and scenarios where CMS is beneficial 5.10 Brief about Web-programming frameworks: php, java, python frameworks.
Unit 6: Web Services and APIs	**5 hours**
• Describe SOA,SOAP, API , JSON and XML • Implement REST API.	6.1 Introduction to Web Services 6.2 Brief about service-oriented architecture 6.3 SOAP 6.4 RESTful web services 6.6 RESTful APIs vs. SOAP 6.6 JSON and XML data formats 6.7 Building and Consuming APIs 6.7.1 Creating simple RESTful APIs using Flask
Unit 7: Security in Web Applications	**8 hours**
• Analyse the security basics and common vulnerabilities. • Explore security principles and best practices. • Implement protection measures for your web apps.	7.1 Web application security fundamentals 7.1.1 Foundations of security 7.1.2 Vulnerabilities 7.1.3 Threats 7.1.4 Attacks, 7.2 Security Principles: 7.2.1 Least Privilege 7.2.2 Defense in Depth 7.2.3 Fail Securely 7.3 Common web vulnerabilities 7.3.1 SQL Injection, 7.3.2 XSS 7.3.3 CSRF 7.3.4 DoS 7.3.5 Live examples and demonstrations of vulnerabilities 7.4 Security best practices for web development 7.4.1 Input Validation and Output Encoding 7.4.2 Secure Authentication and Authorization Multi-Factor Authentication (MFA),Role-Based Access Control (RBAC), Secure Session Handling, Session Fixation and Hijacking Prevention, Error Handling and Logging, Secure File Uploads 7.4.3 SSL/TLS and HTTPS 7.5 Firewalls, proxies, and VPNs 7.6 Host and Network Security threats and countermeasures

	7.7 Application Security threats and countermeasures 7.8 Design guidelines for secure web applications
Unit 8: Web Application Deployment	**4 hours**
• Explore the deployment and deployment platforms • Implement SSL certificates to web app or website	8.1 Introduction to Deployment 8.2 Deployment platforms 8.3 Basic differences between development and production environments. 8.4 Setting up environment variables and basic production settings. 8.5 Introduction to using Gunicorn to serve Flask apps. 8.6 Setting up basic HTTPS using a free certificate (e.g., Let's Encrypt). 8.7 Brief introduction to monitoring and logging. 8.8 Importance of security in deployment.
5. Practical Works:	

-HTTP Request and Response Simulation
- Setting up web server
- Install and configure a local web server (Apache or Nginx). Create a simple HTML. page and serve it via the web server.
- Install Vscode or any IDE
- Setup Python Environment
-Install Python and create a simple script that performs basic arithmetic operations.
- Practice using data types like lists, dictionaries, and sets. Write a Python program to create a list of integers, append values, sort the list, and remove duplicates.
- Write a Python program that stores student names as keys and their marks as values in a dictionary. Print the average marks and the name of the top scorer.
- Perform operations like slicing a string and list manipulation.
- Practice control structures using if-else conditions.
- Write a Python function that takes two numbers as input and returns their sum, difference, product, and quotient.
- write program using recursion technique.
- Perform basic read and write operations.
- Write a Python program to read data from a CSV file and display the content. Then, modify the program to write the output to a JSON file.
- Write a Python class called Student with attributes like name, roll number, and marks. Create methods to display student details and calculate the average marks.
- Create a base class Vehicle with attributes like make and model. Derive a class Car from Vehicle with additional attributes like mileage and capacity. Create objects of both classes and display their details.

1. Install flask and understand directory structure.
2. Create a basic Flask application with a route that returns “Hello, World!” when accessed via the browser.
3. Develop a web form using HTML and Flask to capture user input (e.g., name, email) and display the input on another page.
4. Installation and setting MySQL database server.
5. Write a Flask application that connects to a MySQL database using MySQL-connector or SQL Alchemy. Create a table called students with columns id, name, age, and grade. Insert records and fetch data from the database to display on a webpage.
6. Implement HTML Dropdown to display dynamic options, Dynamic HTML tables, product list and product details using the flask.
7. Create a simple MySQL database and connect it to a Flask application. Develop a form that inserts user data into the database.
8. Create a Flask app that allows users to add, update, and delete records from a database using a web interface.

9. Use Bootstrap with Flask to style the CRUD application developed in above task. Create a form with proper layout and design. Use Bootstrap classes for buttons, tables, and forms.
10. Implement input validation for the form created in the previous exercise.
11. Template integration
12. Sending an email using flask.
13. Develop a login page using Flask that includes session management and secure password storage using hashing
14. Develop a REST API using Flask that performs CRUD operations on a student's table in the MySQL database. Implement API endpoints for creating, retrieving, updating, and deleting student records.
15. Use Python's requests module to fetch data from an open API (e.g., weather or currency exchange API) and display the results.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50				

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:

1. Grinberg, M. (2018). Flask Web Development: Developing Web Applications with Python (2nd ed.). O'Reilly Media.
2. Sweigart, A. (2020). The Big Book of Small Python Projects: 81 Easy Practice Programs. No Starch Press.
3. Beazley, D. (2020). *Python distilled*. Addison-Wesley.

Pokhara University Faculty of Science and Technology	
Course Code: CMP 323 (3 Credits)	Full Marks: 100
Course Title: Project II	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: II / Semester: IV	Program: Bachelor of Computer Application

1. Course Description:
Project-II is an immersive, hands-on course designed for BCA fourth-semester students to translate their classroom knowledge into a tangible, real-world application. The project should focus on the development of a web or desktop application. This course allows students to demonstrate their skills in areas such as user interface design, backend development, and database management. Building a project in web platform or desktop sharpens students' abilities in advanced software development and effective project management.

2. General Objectives:

1) To empower students to apply the knowledge and skills gained in courses like Java, Web Technolgies, DBMS to create a practical software application.
2) To help students develop essential project management, teamwork, and communication skills that are crucial for their future careers.
3) To encourage students to think critically and solve problems creatively as they work on their projects.
4) To enable students to produce a functional, user-friendly, and well-documented software application that meets specific needs.

3. Procedure:
The Project II is focused to implement the object oriented system concept in programming using Java, Web Technology, DBMS. A group of maximum four students can be formed to work in their project II. Students are themselves responsible to form the group and select project on any domain of common interest. Project supervisor(s) shall be assigned for guidance, tutoring and mentoring of the project activities. The project shall be evaluated by evaluation panel that includes internal evaluator(s) and/or supervisor and external examiner(s) appointed by the respective department. Appointment of External Examiner is mandatory for final defense of the project. Each group is required to complete all project work phases and develop software product that is SMART (Specific, Measurable, Achievable, Relevant, and Time-bound.

4. Project Phases:

The entire project work shall be divided into three phases and evaluation shall be done accordingly:

4.1 Phase I – Proposal Submission and Defense
In Phase I, students are required to form a team comprising of maximum 4 team members and draft a conceptual framework for their project work in the form of a Proposal. The project team must submit and present the proposal in presence of Examiner/Supervisor in a formal presentation and must be approved.

4.2 Phase II – Mid Term Defense:
The project team needs to present the work progress to Examiner(s)/Supervisor and demonstrate as required. Till the mid-term defense, students must have finished the design phase including the overall system/architectural design and validation scheme. The Mid-Term defense can be scheduled after 4 weeks of project proposal defense.

4.3 Phase III – Final Submission and Defense:
Before the semester ends, students must submit and defend the project final work (oral presentation and demonstration) in presence of the evaluation panel inclusive of external examiner. The project work should be presented in complete format that include project requirement analysis, design, coding and testing. Students are

required to submit the final report 1 week prior to the final defense. Students must submit the hardcopy/softcopy of the project in the standard format prescribed by respective department of the college.

6. Documentation Structure:

The documentation structure defines the components that must be included in the proposal and final report of the Project II. It includes the following components:

Proposal Format:

1. Introduction
2. Problem statement
3. Objectives
4. System requirement
   - 4.1 Functional
   - 4.2 Non functional
   - 4.3 Requirement prioritization table
5. Design
6. Development Cost estimation
7. Project Schedule
8. Conclusion
9. References

Report format:

Title Page
Student Declaration
Supervisor Acceptance
Approval Certificate
Acknowledgement
Abstract
List of Abbreviations
Table of Figures
List of Tables
Table of Contents

CHAPTER 1: INTRODUCTION

1.1 Background
1.2 Objectives
1.3 Purpose, Scope, and Applicability
   1.3.1 Purpose
   1.3.2 Scope and Limitation
   1.3.3 Applicability
1.4 Achievements
1.5 Organization of Report

CHAPTER 2: SURVEY OF TECHNOLOGIES

Review of the similar/relevant projects

CHAPTER 3: REQUIREMENTS AND ANALYSIS

3.1 Problem definition
3.2 Requirements specification
3.3 Planning and scheduling
3.4 System requirements
3.5 Preliminary product description

CHAPTER 4: DESIGN

4.1 Introduction
4.2 System Design
4.3 Database design
4.4 Interface Design

CHAPTER 5: IMPLEMENTATION AND TESTING
5.1 Implementation Approaches
5.2 Coding Details and Code Efficiency
5.2.1 Code Efficiency
5.3 Testing Approach
5.3.1 Unit Testing
5.3.2 Integrated Testing
5.3.3 Beta Testing
5.4 Modifications and Improvements
5.5 Test Cases

CHAPTER 6: RESULTS AND DISCUSSION
6.1 Test Reports
6.2 User Documentation

CHAPTER 7: CONCLUSIONS
7.1 Conclusion
7.1.1 Significance of the System
7.3 Recommendation
REFERENCES: IEEE references style

Pokhara University Faculty of Science and Technology	
Course Code.: MTH230 (3 Credit)	Full Marks: 100
Course Title: Numerical Methods (3-0-1)	Pass Marks: 45
Nature of the Course: Theory	Total Lectures: 48 hours
Level: Bachelor | Year: III / Semester: V	Program: Bachelor of Computer Application

1. Course Description:

This course covers how a computer can be used to solve the problems that may not be solvable by calculus and algebra. It develops the problem solving techniques required to study other courses and mathematical relationship that can be used to simulate real –world situation.

2. General Objectives:

The general objectives of this course are:

1. To solve elementary matrix, arithmetic problems analytically and numerically.
2. To introduce numerical methods for interpolation, regressions and finding of roots.
3. To find the solution of ordinary and partial differential equations.
4. To provide knowledge of relevant high level programming language for computing, implementing, solving and testing of algorithms.

3. Methods of Instructions:

Lecture
Tutorial/Discussion/Readings
Practical works and Project works

4. Course Contents:

Specific Objectives	Contents
Unit 1: Solution of Non-liner equation	**10 hours**
• Solve none linear equations by different methods of various fields and comparison of their convergence.	1.1. Introduction, Importance of Numerical Methods 1.2. Approximation and Errors in computation. 1.3. Bisection Method. 1.4. Secant method. 1.5. Newton Raphson method (for one variable) 1.6. Fixed point iterative method. 1.7. Rate of convergence and comparisons of these Methods.
Unit 2: Interpolation and Approximation	**8 hours**
• Develop mathematical relationship between observations to simulate real world situation.	2.1. Lagrange interpolation 2.2. Finite differences (forward, backward, and divided difference) 2.3. Newton's Interpolation (forward, backward). 2.4. Least square method of fitting linear and nonlinear curve for discrete data and continuous function. 2.5. Spline 2.5.1 Interpolation (Cubic Spline)
Unit 3: Numerical Differentiation and Integration	**6 hours**
• Solve the not solvable problem of calculus.	3.1. Numerical Differentiation formulae 3.2. Maxima and minima 3.3. Newton-Cote general quadrature formula 3.4. Trapezoidal, Simpson's 1/3, 3/8 rule

	3.5. Romberg integration 3.6. Gaussian integration (2 point and 3 point formula)
Unit 4: Solution of system of linear algebraic equations	**10 hours**
• Solve linear equations by different techniques.	4.1. Gauss elimination method. 4.2. Gauss-Jordan method 4.3. The inverse of a matrix. 4.4. Ill-conditioned system of linear equations. 4. 5. LU Factorization method (Dolittle's, Crout's, Cholesky's) 4.6. Iterative methods (Jacobi method, Gauss-Seidel method) 4.7. Eigen value and Eigen vector using Power method
Unit 5: Solution of ordinary differential equations	**6 hours**
• Solve the ordinary differential equations which may exist in real world but not solvable by calculus.	5.1. Over view of solution of differential equations 5.2. Taylor's series method. (for first and second order) 5.3. Euler's method (for first and second order) 5.4. Heun's method. (for first and second order) 5.5. Runge-Kutta methods (for first and second order) 5.6. Solution of boundary value problem by finite difference method and shooting method.
Unit 6: Numerical solution of Partial differential Equation	**5 hours**
• Solve the partial differential equations which exist in real world but not solvable by calculus.	6.1. Classification of partial differential equation(Elliptic, parabolic, and Hyperbolic) 6.2. Solution of Laplace equation (standard five point formula with iterative method) 6.3. Solution of Poisson equation (finite difference approximation) 6.4. Solution of one dimensional Heat equation by Schmidt method.

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End**	**50**
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		**20**		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

6.2 Students' Responsibilities:
To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References:
Text Books: **1.** C.F. Gerald and P.O. Wheatley ***Applied Numerical Analysis (7th edition),*** New York. **2.** Dr. B.S. Grewal, ***Numerical Methods in Engineering and Science,*** Khanna Publication, (10th edition)
References: **1.** E. Balagurusamy ***Numerical methods*** New Delhi; Tata McGraw Hill, 2010 **2.** Richard L. Burden, J. Douglas Faires, "Numerical Analysis 7th edition" , Thomson / Brooks/Cole **3.** S.S. Sastry ***Introductory Methods of Numerical Analysis (4th edition),*** Prentice-Hall of India, New Delhi, 2008. ***4.*** Dr. V. N.Vedamurthy & Dr. N. Ch. S. N. Iyengar ***Numerical Methods,*** Noida, Vikash Publication House 2009. ***5.*** Rudra Pratap ***Getting Started with MATLAB,*** Oxford University Press 2010

Pokhara University **Faculty of Science and Technology**	
Course Code.: CMP 317 (3 Credits)	Full Marks: 100
Course Title: .NET Technology (3-0-3)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: III / Semester: V	Program: Bachelor of Computer Application

1. Course Description:

This course provides a comprehensive introduction to the .NET ecosystem, focusing on essential programming and application development skills. Its major intent is to equip students with a solid understanding of .NET Core and C# programming fundamentals, enabling them to build, manage, and deploy modern .NET applications. The course covers key areas including .NET Core architecture, C# language basics, console applications, dependency management with NuGet, and data access using Entity Framework Core. It includes hands-on experiences with ASP.NET Core for web API development, control flow in C#, exception handling, and LINQ for data manipulation. Delivery approaches include a mix of self-learning modules, interactive and experiential exercises to enhance understanding and practical application. Students will engage in project-based assignments and use Visual Studio IDE to apply their knowledge, ensuring a practical and immersive learning experience.

2. General Objectives:

The general objectives of this course are:

- To familiarize students with the evolution, core principles, and differences between .NET Framework and .NET Core.
- To enable students to understand .NET Core's cross-platform, Common Language Runtime (CLR) and core .NET Core libraries.
- To guide students in using Visual Studio IDE for project creation, management, and version control.
- To equip students with skills in implementing control flow statements, object-oriented, exception handling, handling input/output and command-line arguments.
- To familiarize students with dependency management using NuGet, including adding packages and managing versions.
- To provide skills in modeling data, defining entities, and performing CRUD operations using Entity Framework Core.
- To introduce students to building dynamic Web APIs, including RESTful design principles, HTTP methods, and versioning strategies.

3. Methods of Instructions:

1. Lectures and discussion,
2. Group Work,
3. Experiential Exercises,
4. Self-Learning Modules.

4. Course Contents:

Specific Objectives	**Contents**
Unit 1: Introduction to .NET	**4 hours**
• Describing the evolution and core principles, key features, cross-platform capabilities and application structure of .NET Core. • Explore the role of the Common Language Runtime (CLR) in .NET execution, including memory management and JIT compilation. • Get acquainted with Visual Studio IDE, including project creation, management.	1.1. Overview of .NET and its evolution 1.1.1. Core principles of the .NET Framework 1.1.2. Relationship between .NET Framework and .NET Core 1.1.3. Key features and benefits of the .NET platform 1.1.4. Role of .NET in modern application development 1.1.5. Brief history of .NET 1.2. Introduction to .NET Core 1.2.1 .NET Core 1.2.2 Key differences between .NET Framework and .NET Core

	1.2.3 Cross-platform capabilities of .NET Core 1.2.4 .NET Core application structure 1.3. Common Language Runtime (CLR) 1.3.1 Role of CLR in .NET Execution 1.3.2 Memory management and garbage collection 1.3.3 JIT compilation, CTS and its importance 1.4. .NET Core libraries 1.4.1 Core libraries included in .NET Core 1.4.2 Namespace structure 1.4.3 Fundamental classes and their functionalities
Unit 2: C# Programming Fundamentals	**8 hours**
• Explain structure of a C# program and perform I/O operations with comments and documentation. • Implement control flow using selection, iteration, and jump statements. • Understand object-oriented programming (OOP) concepts, including classes, objects, inheritance, polymorphism, encapsulation, and abstraction. • Work with methods, properties, and indexers in C#. • Handle exceptions using try-catch-finally blocks, throw and create custom exceptions, and understand exception hierarchies.	2.1. C# language basics 2.1.1. C# Language History & Evolution 2.1.2. C# Syntax and Keywords 2.1.3. Basic Program Structure 2.1.4. I/O Operations, Comments and Documentation 2.2. Data types, variables, operators 2.2.1. Value Types & Reference Types 2.2.2. Variable Declaration and Initialization 2.2.3. Operators & its types 2.2.4. Type Casting and Conversation 2.3. Control flow statements 2.3.1. Selection, Iteration and Jump Statements 2.4. Object-oriented programming concepts (OOPs): 2.4.1. Introduction to OOPs, Classes and Objects 2.4.2. Methods, Properties & Indexers 2.5. Exception handling 2.5.1. Introduction to exception 2.5.2. Try-catch-finally blocks 2.5.3. Throwing exceptions and custom exceptions 2.5.4. Exception Hierarchies
Unit 3: Basic .NET Core Applications	**8 hours**
• Perform input/output operations in a console application. • Implement command-line arguments in a .NET Core console application. • Implement error handling and validate user input in console applications. • Understand the structure and role of the .NET Core project file (csproj). • Build, run, debug, and publish .NET Core applications for different platforms.	3.1. Console applications: Input/output, command line arguments 3.1.1. Creating a console application project 3.1.2. Basic console application structure 3.1.3. I/O operating in Console 3.1.4. Using Command-line arguments 3.1.5. Error handling and user input validation 3.2. Building and running .NET Core application 3.2.1. Understanding the .NET Core project file(csproj) 3.2.2. Building the application for different target platforms 3.2.3. Running the application from command line 3.2.4. Debugging and Publishing .NET Core application
Unit 4: Advanced .NET Core Concepts	**4 hours**
• Describe the role of NuGet packages in a project.. • Explain the concept of assemblies and their role in .NET. • Implement best practices for exception handling in .NET Core. • Develop and use custom exception types in applications.	4.1. Dependency management with NuGet 4.1.1. Introduction to NuGet packages 4.1.2. Adding NuGet packages to a project 4.1.3. Managing package version 4.2. Class libraries and namespaces 4.2.1. Creating class library projects 4.2.2. Organizing code using namespaces 4.2.3. Assembly concept and its role in .NET

	4.2.4. Referencing class libraries in other projects 4.3. Implementing Exception handling 4.3.1. Best practices for exception handling 4.3.2. Custom Exception Types
Unit 5: Data Access with Entity Framework Core	**8 hours**
• Explore the benefits of using EF Core in .NET Core applications. • Comapre code-first and database-first development approaches. • Implement CRUD operations (Create, Read, Update, Delete) using EF Core. • Develop databases and manage data using EF Core. • Master LINQ syntax for data manipulation. • Implement complex queries, including joins, projections, filtering, and grouping using LINQ to Entities.	5.1. Introduction to Entity Framework Core (EF Core) 5.1.1. Overview of ORM 5.1.2. Basic concept and architecture of EF Core 5.1.3. Benefits of using EF Core 5.1.4. Relationship with .NET Core 5.2. Data modeling concepts: Entities, relationships 5.2.1. Entity Framework Core Data Model 5.2.2. Defining entities and properties 5.2.3. Primary Keys and Foreign Keys 5.2.4. Relationships: 5.2.5. Navigation Properties 5.3. Working with databases: CRUD operations 5.3.1. Connecting to a database using EF Core 5.3.2. Creating new records 5.3.3. Retrieving data using LINQ 5.3.4. Updating and Deleting records 5.4. LINQ queries for data manipulation 5.4.1. Introduction to Language Integrated Query 5.4.2. Basic LINQ Syntax 5.4.3. Quering data using LINQ to Entities 5.4.3. Projecting, filtering, ordering, grouping 5.4.4. Joining data from multiple tables
Unit 6: ASP .NET Core Web Development	**8 hours**
• Describe the ASP.NET Core MVC framework. • Explore Razor syntax and page structure. • Implement model binding and data access in Razor Pages. • Implement Create and manage controllers to handle user requests. • Develop views using Razor templates and data binding techniques. • Create and validate HTML forms in ASP.NET Core. • Apply Dependency Injection (DI) principles for loose coupling.	6.1. Introduction to ASP.NET Core MVC 6.1.1. ASP.NET Core MVC framework 6.1.2. MVC architecture 6.2. Razor Pages for simple web applications 6.2.1. Razor syntax and page structure 6.2.2. Model binding and data access in Razor 6.2.3. Creating dynamic and interactive pages 6.3. Routing and controllers 6.3.1. Routing mechanisms in ASP.NET core MVC 6.3.2. Routes and URL patterns 6.3.3. Creating controllers to handle user requests 6.3.4. Controller actions and action methods 6.3.5. Returning different types of responses 6.4. Views and data binding 6.4.1. Roles of views in presenting data to the user 6.4.2. Using Razor templates to build views 6.4.3. Data binding as controllers to views 6.4.4. Model formatting and dynamic content 6.5. Forms and validation 6.5.1. Creating HTML forms for user input 6.5.2. Handling form submissions in controllers 6.5.3. Implementing data validation techniques 6.5.4. Error handling and validation messages 6.6. Overview of Dependency Injection (DI) concepts 6.6.1. Dependency Injection (DI) principles 6.6.2. Benefits of using DI for loose coupling

	6.6.3. Built-in DI container in ASP.NET Core 6.6.4. Registering and injecting dependencies 6.6.5. Implementing custom services, repositories
Unit 7: Building Modern Web APIs	**8 hours**
• Implement Web APIs APIs. • Explain best practices for clean and maintainable API design. • Explor RESTful API principles and design. • Implement HTTP methods (GET, POST, PUT, DELETE) in RESTful APIs. • Develop APIs using the ASP.NET Core framework.	7.1. Introduction to Web APIs 7.1.1. Web APIs and its purposes 7.1.2. Advantages of using Web APIs 7.1.3. Common use cases for Web APIs 7.1.4. Design considerations for APIs 7.2. Designing RESTful APIs 7.2.1. REST principles 7.2.2. Resources identifiers in RESTful APIs 7.2.3. HTTP methods *(GET, POST, PUT, DELETE)* 7.2.4. Designing URLs for API endpoints 7.2.5. Versioning strategies for Web APIs 7.3. ASP.NET Core Web API framework 7.3.1. Introduction to ASP.NET Core Web API 7.3.2. Building web API project in ASP.NET Core 7.3.3. Action methods for API requests. 7.3.4. Returning data from controllers JSON, XML 7.3.5. Handling input parameters and data binding 7.4. Data serialization (JSON, XML) 7.4.1. Importance of data serialization in Web APIs 7.4.2. JSON structure and serialization techniques 7.4.3. Introduction to XML 7.4.4. XML structure and serialization techniques

5. Practicals Works:

1. Develop console application using C# with .NET Core [2-4 hours]
2. Create .NET Core class libraries [2-3 hours]
3. Build database-driven applications using Entity Framework Core and LINQ [4-6 hours]
4. Develop web applications with ASP.NET Core MVC or Razor Pages [6-8 hours]
5. Create Simple Web APIs using ASP.NET Core [4-6 hours]

6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		

Full marks=50+50
6.2 Students' Responsibilities:
To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.
7. Prescribed Books and References:
1. Mueller, J. (2021). *C# for Dummies*. Wiley. 2. MacDonald, M. (2018). *Beginning ASP.NET in C#*. Wrox. 3. Nagel, C., & Glynn, J. (2021). *Professional C# and .NET Core*. Wrox.

Pokhara University Faculty of Science and Technology	
Course Code.: CMP 336 (3 Credits)	Full Marks: 100
Course Title: Data Communication and Computer Networks (3-0-2)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: III / Semester: V	Program: Bachelor of Computer Application

1. Course Description:

This course is designed to provide students with the fundamental concepts of data communication and the architecture, protocols, and technologies that underpin modern computer networks. It covers the principles of digital communication, network models, transmission media, and network devices, focusing on the OSI and TCP/IP models. Students will explore key topics such as LANs, WANs, IP addressing, routing, switching, and network security. Through practical exercises and simulations, the course aims to equip students with the skills to design, implement, and manage robust and efficient computer networks, preparing them for careers in network administration and IT infrastructure management.

2. General Objectives:

The general objectives of this course are as follows:

- To equip students with in-depth insights into the basic principles of data communication and networking, including architecture, protocols, and services.
- To build a strong grasp of essential concepts focusing on the OSI and TCP/IP models.
- To develop hands-on abilities in configuring and managing network and intermediate devices such as routers and switches.
- To equip students with knowledge of IP addressing schemes, subnetting, and the implementation of routing protocols

3. Methods of Instructions:

- Lecture, Discussion, Readings, Practical works, Case Studies and Project works

4. Course Contents:

Specific Objectives	Contents
Unit 1: Data Communications	**8 hours**
• Familiarize students with the basic concept of Data Communications, data transmission techniques and impairments, with performance parameters and related theorems.	1.1 Definition, Components of Data Communication 1.2 Data Transmission Modes: Simplex, Half-Duplex, Full-Duplex 1.3 Data Representation: Analog and Digital 1.4 Analog and Digital Transmission 1.5 Serial and Parallel Communication 1.6 Asynchronous and Synchronous Transmission 1.7 Transmission Impairments 1.8 Bandwidth, Throughput, Baud Rate, Jitter 1.9 Nyquist and Shannon Capacity Theorem 1.10 Digital Transmission Techniques
Unit 2: Multiplexing and Switching	**4 Hours**
• Explore the various types of multiplexing and switching techniques, and their usage in data communication.	2.1 Multiplexing and its types- FDM, TDM, WDM 2.2 Switching and its types- Circuit, Packet and Message switching
Unit 3: Reference Model	**6 Hours**

• Elaborate the concept of Protocol, Protocol architecture along with OSI, TCP/IP and Internet Protocol Stack.	3.1 Protocols and Layered Architecture 3.2 OSI Reference Model and functions 3.3 TCP/IP Protocol 3.4 Cross layers 3.5 Devices in different layers 3.6 Software defined networks
Unit 4: Physical Layer Design Issues	**4 Hours**
• Analyze different transmission media, including both guided and unguided methods, and explore the standards for twisted pair cabling, labeling, and testing.	4.1 Network Topologies 4.2 Transmission Medium 4.2.1 Guided Transmission Media (Twisted Pair, Coaxial, Optical Fiber) 4.2.2 Unguided Transmission Media (Bluetooth, Wi-Fi, Cellular, Radiowave, Microwave, Satellite) 4.3 Twisted Pair Cabling, Labeling and Testing Standards
Unit 5: Data Link Layer Design Issues	**6 Hours**
• Explore basics of error detection and correction techniques with data link layer protocols.	5.1 Framing 5.2 Error Detection (Parity, Checksum and CRC) 5.3 Error Correction (FEC, ARQ) 5.4 Flow Control Protocols (Stop and Wait, Go-Back-N, Selective Repeat) 5.5 ATM, HDLC, PPP
Unit 6: Network Layer Design Issues	**8 Hours**
• Gain understanding of network layer design, focusing on logical addressing, subnetting, routing protocols, and the transition from IPv4 to IPv6, including migration strategies and the role of Network Address Translation (NAT) in modern networks.	6.1 Logical Addressing (IPv4/IPv6) 6.2 Subnetting and Prefix Delegation 6.3 Routing Protocols (Static, RIP, EIGRP and OSPF) 6.4 IPv4 Frame Format 6.5 IPv6 Frame Format 6.6 IPv6 Migration Strategies 6.7 Network Address Translation
Unit 7: Transport and Application Layer	**6 Hours**
• Explore the transport and application layer Protocols and processes.	7.1 Process to Process Communication 7.2 TCP, UDP and Sockets 7.3 Three Way Handshake 7.4 Client Server Computing Model 7.5 Application Layer Protocols (DNS, DHCP, HTTP, SMTP, POP, IMAP) 7.6 Congestion Control Approaches
Unit 8: Network Management and Security	**6 Hours**
• Delve into cryptographic techniques, the functionality of VPNs and firewalls necessary for network management and security.	8.1 Network Management 8.2 Infrastructure of Network Management, FCAPS 8.3 Cryptography (Symmetric and Asymmetric) 8.4 Virtual Private Networks 8.5 Firewalls 8.6 Layered Approach of Network Security

5. List of Practical:

SN	Practical Works
1.	Network Cabling (Crossover and Straight Through)
2.	Configure IP addresses and Subnetworks
3.	Basic Network Commands (ipconfig, ping, netstat, arp, tracert, telnet, nslookup)
4.	Network Design and Visualization in Simulation Software
5.	Enterprise LAN Architecture, DNS and VLAN
6.	Deployment of Routing Protocols (Static, RIP, EIGRP and OSPF), NAT
7.	Deployment of Virtual Private Networks (VPN)
8.	Analyzing Network Traffic Using Wireshark

6. 6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50 =100				

6.2 Students' Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References:

Text Books:

1. B.A. Forouzan, *Data Communications and Networking*, 5th edition. McGraw-Hill Education, 2012.

References:

1. J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 7th ed. Pearson, 2017.
2. A.S. Tanenbaum and D. J. Wetherall, *Computer Networks*, 5th ed. Pearson, 2010.
3. W. R. Stevens, *TCP/IP Illustrated, Volume 1: The Protocols*, 1st ed. Addison-Wesley Professional, 1994.
4. T. Lammle, *CCNA Routing and Switching Study Guide*, 2nd ed. Sybex, 2016.
5. Lowe, *Networking All-in-One For Dummies*, 8th ed. Wiley, 2018.

Pokhara University Faculty of Science and Technology	
Course Code: ELE322 (3 Credits)	Full Marks: 100
Course Title: Research Methodology (3-0-1)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: III / Semester: V	Program: Bachelor of Computer Application

1. Course Description:

The course is a basic one and it should be treated at an introductory level of research methodology and method of doing practical knowledge. It introduces students to the fundamentals of research methodology and helps students to build a strong foundation in research methodology, enabling them to conduct independent research and contribute to their field of study by considering research ethics

2. General Objectives:

The general objectives of this course are:

- To provide the concept the basic concepts and principles of research, including the scientific method and ethical considerations
- To familiarize the students with different research design and gain skills in collecting, analyzing, and interpreting data using various tools and techniques
- To make the students familiar with the proposal and report writing along with publication process.

3. Methods of Instructions:

1. Lecture and discussion
2. Paper reviews
3. Presentation
4. Paper development

4. Course Contents:

Specific Objectives	Contents
Unit 1: Introduction	**7 hours**
• Explain the research process, including the sequential steps involved in conducting research from start to finish	1.1. Definition, objectives, characteristics, purpose and significance of research 1.2. Importance of research: emphasizing its role in innovation, problem-solving, and advancing technology 1.3. Types of research: fundamental research, applied research, experimental research, descriptive research, analytical research, empirical research, qualitative and quantitative approaches 1.4. Overview of the research process and its steps of research process: highlighting its key stages, such as problem identification, literature review, data collection, analysis, and reporting
Unit 2: Literature Review and Problem Statement	**7 hours**
• Describe the technique and source conducting literature review. • Explain the concept of research gaps and develop the ability to analyze existing research to spot areas that are underexplored	2.1. Conducting a literature review: techniques and sources 2.2. Identifying research gaps. 2.3. Formulating problem statement, research questions, and hypotheses in scientific settings

	2.4. Research objectives: align with the problem statement/ research questions and hypotheses, ensuring clarity and feasibility.
Unit 3: Research Design **Hours**	**12**
• Provide the clear concept of different variable and measurement scale • Develop the ability to select and implement appropriate research designs based on the research objectives.	3.1. Meaning, purpose, principles and importance of research design 3.2. Element of research design 3.3. Types of research design and their uses in research 3.4. Sources of error in research design 3.5. Features of a good research design 3.6. Types of research design: exploratory, descriptive experimental, observational, and survey research: strengths and weaknesses of each 3.7. Qualitative method, quantitative method, and mixed method
Unit 4: Data Collection and Analysis **Hours**	**10**
• Explain to perform collection of data using appropriate techniques. • Interpret data, draw logical conclusions, and relate findings to the research objectives.	4.1. Meaning, nature, and types of data collection; secondary and primary data-definition, sources, characteristics, advantages and disadvantage 4.2. Primary data collection methods: surveys, questionnaires, interviews, observation, and experiments 4.3. Secondary data collection: academic journals, books, online databases, government reports, and industry publications; evaluating secondary data 4.4. Concept of sampling techniques and types 4.5. Introduction to data organization, processing, and analysis
Unit 5: Research Ethics, Proposal, and Report Writing **Hours**	**9**
• Understand and apply key ethical principles in research, including integrity, objectivity, confidentiality, and avoiding plagiarism. • Enable students to structure and write comprehensive research proposals, reports and presentations, covering all essential sections and ensuring accurate referencing.	5.1. Ethical principles in research: Integrity, objectivity, confidentiality, and informed consent. 5.2. Plagiarism and academic integrity: plagiarism, consequences, avoidance. 5.3. Ethical issues in data collection and reporting 5.4. Structure of a research proposal: Introduction, objectives, rationale of study, review of literature, research gap, research design and methods, expected outcomes, and timeline, discussion. 5.5. Report writing: abstract, introduction, literature review, methodology, result and discussion, conclusion and future work. 5.6. Referencing and citation: Understanding citation styles (APA, MLA, and IEEE) and the importance of accurate referencing. 5.7. Preparing research presentations: effective visual and oral presentations of research findings.

	5.8. Publication process: journal selection, manuscript preparation, and peer review.

5.: Practical Works

5.1	**Laboratory work - Presentation**
	1. Development of questionnaire
	2. Data entry in any computer application
	3. Apply data filtration and validation rules
	4. Apply any statistical application tool for analysis and interpretation
5.2	5. Conducting a literature review
	6. Identifying research gaps, and formulating problem statement
	7. Compare different data organization, processing, and analysis technique

6. 6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50 =100				

6.2 Students' Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References:

Text Books

1. Kothari, C.R. (2004) Research Methodology: Methods and Techniques (2nd Edition), New Age International Publishers, New Delhi.
2. Creswell, J. W. and Creswell, J.D (2017). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches (5th ed) Sage.

References

1. Leedy, P. D. and Ormrod, J. E. (2019). Practical Research: Planning and Design (12th Edition) Pearson.
- Kumar, R (2023) Research Methodology: A Step by Step Guide for Beginners (4th Edition), SAGE Publications Pvt. Ltd; Fourth edition (13 August 2023)

Pokhara University Faculty of Science and Technology	
Course Code.: MTH 330 (3 Credit)	Full Marks: 100
Course Title: Mathematical Foundation and Computer Application (3-0-0)	Pass Marks: 45
Nature of the Course: Theory	Total Lectures: 48 hours
Level: Bachelor | Year: III / Semester: V	Program: Bachelor of Computer Application

1. Course Description:

This course is designed to provide a framework for solving complex problems mathematically. It supports for exploring abstract concept as the foundation of countless scientific and technological advancements. At its core, it deals with countable, distinct elements rather than continuous quantities. It incites mathematical reasoning, algorithms thinking, combinatorial analysis, discrete structures, application and modeling through mathematical logic, proof and induction; networks and graph theory; advanced counting techniques; and modeling computation. This course also introduces the emerging new trended concept and essence for explore its diverse applications across different domains such as encryption, networks and gaming theory, optimization, circuit analysis, structural analysis, quantum mechanics, etc. After completion of this course, students can select, implement and apply the best mathematical tools for solving real-world problems during ICT processes

2. General Objectives:

The general objectives of this course are:

1. To acquaint the students thinking solutions mathematically.
2. To develop the skills in students to select and measures the optimal techniques for real-life problem solution.
3. To acquaint the students with mathematical foundations for many computer science courses including data structures, algorithms, database theory, automata theory, formal languages, compiler theory, computer security, and operating systems, etc.
4. To acquaint the students success in developing the problem-solving skills needed in subsequent courses and professional work.

3. Methods of Instructions:

Lecture
Tutorial/Discussion/Readings
Practical works and Project works

4. Course Contents:

Specific Objectives	Contents
Unit 1: The Foundations: Logic and Induction	**6 hours**
• Familiarize with basic concepts of discrete mathematics. • Explain use of logic and conclude the different statements towards a goal. • Apply different proof techniques and verify formulas using induction.	1.1 Propositional Logic and its Applications 1.2 Propositional Equivalences 1.3 Predicates and Quantifiers 1.4 Nested Quantifiers 1.5 Expressing statement in the language
Unit 2: Mathematical Reasoning	**7 hours**
• Learning the different relation available with their types and properties	2.1 Mathematical Induction 2.2 Strong Induction and Well-Ordering 2.3 Recursive Definitions and Structural Induction

• Find the closure of relation.	2.4 Recursive Algorithms 2.5 Rules of Inference and proofs 2.6 Direct proof and Indirect proof 2.7 Formal and Informal proof
Unit 3: Recurrence Relations	**7 hours**
• Conceptualize the role and working procedure for sequence • Apply the concept of recursion used in both linear and nonlinear recurrence relation.	3.1 Recursive Definition of Sequences. 3.2 Differencing and Summation, 3.3 Solution of Linear Recursive Relation, 3.4 Solution of Non-linear Recurrence Relation. 3.5 Methods of Solving Recurrence Relations- *Substitution, Recurrence Tree and Master Method*
Unit 4: Graph Theory and Its Application	**14 hours**
• Understand the role of graph theory • Familiarize with basic terminologies and properties of graph, and how to simulate the real world problem through graph	4.1 Graph Terminology– definition, direct and indirect graphs, walk, path, circuits, connected components. 4.2 Graph Models and Special Types of Graphs 4.3 Representing Graphs and Graph Isomorphism 4.4 Connectivity 4.5 Euler and Hamilton Paths 4.6 Shortest-Path Problems 4.7 Planar Graphs 4.8 Graph Coloring
Unit 5: Modeling Computation	**14 hours**
• Visualize the computing model using language, grammars • Explain the actual computation of the computer and how language and expression are perceived by the machine.	5.1 Languages and Grammars- *Introduction, Phrase-structure grammars, Types of phrase-structure grammar, Derivation trees* 5.2 Finite-State Machines with output and with no output- *Definition, State diagram* 5.4 Language Recognition- *Regular sets, regular grammars, pushdown automata, Moore and Mealy machines* 5.5 Turing Machines- *Introduction, definition, representation and language of a Turing Machine*

6. 5. Evaluation System and Students' Responsibilities:

5.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End**	**50**
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Total Internal		**50**		

Full Marks: 50 + 50 = 100	

6.2 Students' Responsibilities:

To be eligible for the Semester End Examinations, each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

6. Prescribed Books and References:

Test Book

1. Rosen Kenneth H., *Discrete Mathematics and Its Applications,* Seventh revised edition, 2015, Tata McGraw-Hill Publishing Company Limited, ISBN 0–07–338309–0

References

1. Sarkar, Swapan Kumar, *A textbook of Discrete Mathematics*, Eighth Revised Edition, 2013, S. Chand & Company Ltd., ISBN 81-219-2232-1
2. Richard, Johnsonbaugh, *Discrete Mathematics*, Fifth Edition, Addison Wesley, Pearson Education Asia (LPE), ISBN: 81-780-82799, 2000
3. Trus, J., *Discrete Mathematics for Computer Scientists*, Second Edition, Addision Wesley ISBN: 0-201-36061,1999

Pokhara University Faculty of Science and Technology	
Course Code: MGT 322	Full Marks: 100
Course Title: Data Science and Analytics (3-0-2)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor Year: III / Semester: VI	Program: Bachelor of Computer Application

1. Course Description

This course is designed to introduce students to the core principles and practical applications of Data Science and Analytics. It aims to equip students with the essential skills required to transform raw data into meaningful insights, enabling data-driven decision-making in various industries. The course covers major content areas including data collection, preprocessing, exploratory data analysis (EDA), and the basics of machine learning. It also includes machine learning model validation, alongside data visualization and reporting.

The course is structured into two parts:

- **Theoretical Foundations** - focusing on understanding key concepts and methodologies in data science.
- **Practical Application** - emphasizing hands-on experience through laboratory work, where students will apply what they have learned using real-world datasets.

The delivery approach combines traditional lectures with self-learning modules, experiential exercises. This approach is designed to foster independent learning and critical thinking skills, preparing students for future challenges in the field of data science.

2. General Objectives

The course is designed with the following general objectives:

1. To familiarize students with key concepts in data science and analytics, including data types and roles.
2. To enable students to collect, clean, and preprocess data from various sources for accurate analysis.
3. To equip students with the skills to implement and evaluate basic machine learning models.
4. To develop students' skills in data visualization and reporting, emphasizing effective communication of insights.
5. To make students cognizant of ethical challenges in data science, including privacy, security, and bias.
6. To provide students with hands-on experience through practical laboratory work on real-world datasets.

3. Contents in Detail

This section contains the details to be taught under the course. Normally, the contents can be divided into 5 to 8 units, and each unit can be assigned 5 – 8 teaching hours.

Specific Objectives	Contents

• Explain the basic concepts and scope of data science. • Differentiate between data science, analytics, and business intelligence.	**Unit I: Introduction to Data Science and Analytics (4 hours)** 1.1 Definition and Scope of Data Science 1.2 Key Concepts in Data Science 1.3 Data Science, Data Analytics, Business Intelligence 1.4 Applications of Data Science
• Collect and preprocess data from various sources. • Clean data and handle missing values. • Transform data for analysis.	**Unit II: Data Collection and Preprocessing (6 hours)** 2.1 Types of Data: Structured, Unstructured, Semi-structured 2.2 Data Collection Methods 2.3 Data Cleaning and Handling Missing Data 2.4 Data Normalization and Standardization
• Perform exploratory data analysis using descriptive statistics. • Create visualizations to identify patterns and trends in data.	**Unit III: Exploratory Data Analysis (EDA) (6 hours)** 3.1 Descriptive Statistics 3.2 Data Visualization Techniques (Histograms, Box Plots, Scatter Plots) 3.3 Identifying Patterns and Anomalies 3.4 Correlation and Covariance Analysis 3.5 Introduction to Hypothesis Testing 3.6 Effective Data Interpretation
• Analyze real-world cases using data science techniques.	**Unit IV: Feature Engineering (6 hours)** 4.1 Time-series, Image, Text and Categorical features 4.2 Features selection techniques- Filter, Wrapper and Embedded 4.3 Data Reduction Techniques – Principal Component Analysis, Linear Discriminant Analysis
• Implement basic machine learning algorithms. • Understand the differences between different machine learning algorithms: supervised, unsupervised and reinforcement learning	**Unit V: Introduction to Machine Learning (10 hours)** 5.1 Basics of Machine Learning: Supervised Unsupervised, Reinforcement Learning 5.2 Linear Regression, Logistic Regression 5.3 Decision Trees and Random Forests 5.4 k-Nearest Neighbors (k-NN) 5.5 Clustering Techniques (k-Means, Hierarchical Clustering) 5.6 Frequent Pattern Mining and Association Rule Mining 5.7 Anomaly Detection Techniques

● Evaluate models like regression and clustering. ● Deploy the models and monitor them	**Unit VI: Model Evaluation and Deployment (7 hours)** 6.1 Basics of Model Deployment and Monitoring 6.2 Model Evaluation Metrics for different Machine Learning Methods (Accuracy, Precision, Recall, F1-Score, MSE, R^2.)
● Identify ethical issues in data science. ● Address challenges related to data privacy, security, and bias.	**Unit VII: Data Ethics and Governance (5 hours)** 7.1 Ethical Issues in Data Science 7.2 Data Privacy and Security 7.3 Data Governance and Compliance

4. Methods of Instruction

The course will employ a combination of lectures, and lab sessions to facilitate learning. Lectures will provide foundational knowledge. Lab sessions will offer hands-on experience, allowing students to apply theoretical concepts to real-world scenarios. This blend of instructional methods ensures a comprehensive learning experience, helping both theoretical understanding and practical skills essential for success in data science and analytics.

5. Practical Works

SN	Practical works
1	Data Collection and Preprocessing: Collect, clean, and preprocess a dataset from various sources. Students will gather data from online repositories or APIs, clean the data by handling missing values, and transform it for analysis.
2	Exploratory Data Analysis (EDA) - Descriptive Statistics: Perform exploratory data analysis using descriptive statistics. Students will calculate mean, median, mode, variance, standard deviation etc. to understand the distribution of the dataset.
3	Exploratory Data Analysis (EDA) - Data Visualization: Create and interpret data visualizations to identify patterns and trends. Students will generate various plots, including line charts, histograms, scatter plots, box plots, and heatmaps. Students will also customize visualizations with colors, labels, and titles to effectively explore and communicate data insights.
4	Introduction to Machine Learning - Regression Models: Implement and evaluate a simple linear regression model. Students will build a linear regression model using a given dataset, train the model, and evaluate its performance using metrics like MSE, RMSE.
5	Introduction to Machine Learning - Classification Models: Implement and evaluate basic classical models such as decision tree, random forest. Students will build basic classical models using a given dataset, train the model, and evaluate its performance using metrics like recall, precision, F1_score.
6	Introduction to Machine Learning - Clustering Models: Implement and evaluate clustering models such as k-Means, Hierarchical Clustering. Students will build basic classical models using a given dataset, train the model, and evaluate its performance in terms of recall, precision (for ground truth dataset) and in terms of compactness, separation, or silhouette if the ground truth dataset is not available.
7	Data Visualization and Reporting - Dashboard Design: Design a dashboard to present data insights.

7. Evaluation system and Students' Responsibilities

Evaluation System

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

External Evaluation	Marks	Internal Evaluation	Marks
Semester-End Examination	50	Class attendance and participation	5
		Field visit and field report	5+5
		Quizzes/assignments and presentations	10
		Internal Term Exam	25
Total External	50	Total Internal	50
Full Marks 50+5 100			

Students' Responsibilities:
Each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

8. Prescribed Books and References

a. Text Book

i. Grus, J. (2019). Data science from scratch: first principles with python. O'Reilly Media.
ii. McKinney, W. (2022). Python for data analysis. O'Reilly Media, Inc.
iii. C Müller, A., and Guido S. (2017). Introduction to machine learning with python.

b. Reference Books

i. McKinney, W. (2022). Python for data analysis. O'Reilly Media, Inc.
ii. Hastie, T. Tibshirani R. and Friedman J. (2009). The elements of statistical learning: data mining, inference, and prediction.

Papers

Agrawal, R., & Srikant, R. (1994, September). Fast algorithms for mining association rules. In Proc. 20th int. conf. very large data bases, VLDB (Vol. 1215, pp. 487-499).

Pokhara University Faculty of Science and Technology	
Course Code: CMP 314	Full Marks: 100
Course Title: Management Information Systems (3-0-1)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: III / Semester: VI	Program: Bachelor of Computer Application

1. Course Description

This course provides an in-depth understanding of Information Systems (IS) and their role in business and technology environments. It covers key concepts of IS, the use of information technologies to support business operations, and strategic decision-making. The course integrates theoretical knowledge with practical applications, enabling students to develop skills to design, implement, and manage information systems effectively. The practical component will involve hands-on projects using contemporary IS tools and software, preparing students for real-world applications.

2. General Objectives

- To provide students with a solid understanding of the fundamental concepts and components of Information Systems.
- To explore the strategic role of Information Systems in organizations and how they contribute to achieving business objectives.
- To develop the skills needed to analyze, design, and implement Information Systems that support organizational processes.
- To introduce students to current trends and emerging technologies in the field of Information Systems.
- To equip students with practical experience in using software and tools for IS design and management.
- To understand the ethical and societal implications of information systems.

3. Contents in Detail

Specific Objectives	Contents
– Understand the definition, types, and significance of Information Systems. – Describe the components of IS and how they interact. – Analyze the role of IS in supporting business operations and decision-making.	**Unit 1: Introduction to Information Systems (IS) (6 Hours)** o Definition and components of IS: Hardware, software, data, procedures, and people. o Types of Information Systems: Transaction Processing Systems (TPS), Management Information Systems (MIS), Decision Support Systems (DSS), and Executive Information Systems (EIS). o The role of IS in organizations. o Case studies on successful IS implementation

– Analyze how IS can provide a competitive advantage. – Understand the relationship between business strategy and IS strategy. – Explore frameworks like Porter's Five Forces and Value Chain Analysis in the context of IS.	**Information Systems and Business Strategy (6 Hours)** o The strategic role of IS in organizations. o Aligning IS strategy with business strategy. o Porter's Competitive Forces Model and Value Chain Analysis. o Case studies on strategic IS use in businesses.
– - Manage IT at an enterprise and global level. – - Understand outsourcing, offshoring, and IT governance.	**Unit 3: Enterprise and Global Management of IT (8 hrs)** - Managing IT functions: Organizing, staffing, and planning. - Business/IT planning. - IT architecture and managing IS operations. - Outsourcing, offshoring trends. - IT governance, failures, and risks. - The role of CIO and IT executives. - Managing user services, technology management, application development. - IT governance and management involvement
– Understand the ethical implications of information systems. – Analyze the societal impact of information systems. – Develop ethical frameworks for designing and using information systems.	**Security, Ethical, and Societal Challenges of IT (4 hrs)** - Ethics in IT and technology. - Data privacy and security. - Bias in AI and machine learning. - Social responsibility of IS professionals. - Ethical decision-making in IS. - Business ethics. -IT policy review - AI policy and review
– Understand the importance of cybersecurity in protecting information systems. – Learn about common threats, vulnerabilities, and attack vectors. – Develop practical skills in implementing security measures and best practices to protect IS.	**Unit 4: Cybersecurity, Information Assurance, and Security Management (10 hrs)** 4.1 Introduction to cybersecurity: Key concepts, importance, challenges. 4.2 Types of threats: Malware, phishing, denial of service, social engineering. 4.3 Security measures: Firewalls, IDS, encryption, MFA. Tools for security management: Firewalls, IDS, encryption, VPNs. 5 Information assurance: Data integrity, confidentiality, availability. 6 Privacy issues, computer crime, cyber law, and the current state of cyber law. 7 Computer libel and censorship. 8 Cyber and Digital transactions laws of Nepal
– Understand the concept of e-commerce and its role in modern business. – Analyze different e-commerce models and technologies.	**Unit 6: E-commerce and Digital Business (8 Hours)** 6.1 Introduction to e-commerce 6.2 E-commerce business models (B2B, B2C, C2C) 6.3 Online marketplaces and platforms 6.4 Digital marketing and e-commerce strategies 6.5 Payment gateways and online transactions

– Develop practical skills for implementing basic e-commerce solutions.	6.6 E-commerce logistics and supply chain management. 6.7 E-commerce web design essentials (design, UI, integration with business needs and customer type)
– Identify and understand emerging trends and technologies in IS. – Analyze the impact of these trends on businesses and society. – Explore the role of IS in digital transformation and innovation.	**Unit 7: Emerging Trends in Information Systems (6 Hours)** 7.1 Big Data and Analytics in IS 7.2 Artificial Intelligence and Machine Learning in IS. 7.3 Social Network and IS 7.4 Generative AI and IS

4. Methods of Instruction

- **Lectures:**
  - Lectures will cover core theoretical concepts and provide foundational knowledge on Information Systems. They will be interactive, using multimedia presentations, real-world examples, and case studies to illustrate key points. Lectures will encourage questions and discussions to deepen understanding.
- **Case Studies and Real-World Examples:**
  - Case studies will be used to demonstrate how businesses and organizations use Information Systems to solve problems and achieve strategic goals. Students will analyze these cases, identify challenges, and propose solutions, bridging the gap between theory and practice..
- **Hands-on Labs and Practical Sessions:**
  - Practical lab sessions will allow students to apply theoretical knowledge using various tools and technologies. For instance, students will set up virtual machines, implement basic cybersecurity measures, and develop a simple e-commerce website. These labs will provide hands-on experience with software like VMware, VirtualBox, MySQL, and web development tools, facilitating experiential learning.
- **Group Projects and Collaborative Learning:**
  - Group projects will encourage collaboration and teamwork, simulating real-world scenarios where students design, implement, or evaluate Information Systems. These projects will foster communication skills and the ability to work effectively in a team, which are crucial in the IT industry.
- **Guest Lectures and Industry Insights:**
  - Guest lectures from industry professionals and experts will provide students with insights into current industry practices, emerging technologies, and career opportunities in Information Systems. These sessions will connect academic learning with industry realities.

6. . **Practical Works**

SN	Practical works
1	• Hands-on labs: Wireshark, Snort, basic cryptographic techniques.
2	• Analyze how IS helped Netflix transform its business strategy using Porter's Five Forces Model Apply Porter's Value Chain Analysis to a manufacturing company like Toyota and identify IS applications in each value chain activity.
3	• Analyze the role of a Chief Information Officer (CIO) in a successful organization such as Apple or Google.

4	• Use a simulated environment to identify and classify threats (malware, phishing) using open-source cybersecurity tools (e.g., Wireshark).
5	• Set up a basic firewall or VPN on a virtual machine and test its functionality.
6	• Study and present the e-commerce strategy of a company like Amazon or Alibaba.
7	• Analyze a social network dataset using tools like Gephi to visualize relationships.
8	**Case Study on Emerging IS Trends** o Select a company leveraging emerging IS trends (e.g., Google with AI). o Analyze how these trends are implemented and their impact on the business.
9	**Develop a Proposal for an Emerging IS Solution** o Identify a business problem that can be solved using an emerging IS trend (e.g., Big Data analytics for customer segmentation). o Develop a proposal outlining the solution, required technologies, and expected benefits.
10	**Trend Analysis Report** o Research and report on the latest trends in Information Systems. o Discuss how these trends are shaping the future of businesses and IS management.

7. **Evaluation system and Students' Responsibilities**

Evaluation System

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Full Marks 50+50=100			

Students' Responsibilities:

Each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such a score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

8. **Prescribed Books and References**

Text Book

- Behl, R., O'Brien, J. A., & Marakas, G. M. (2019). Management information systems. McGraw Hill Education (India) Private Limited. ISBN 9353164664, 9789353164669.

Reference Books

- "Management Information Systems" by Kenneth C. Laudon and Jane P. Laudon, 15th Edition. This book provides comprehensive coverage of information systems' role in business, focusing on the real-world application of technology.
- "Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl, Ricardo Puttini, and Zaigham Mahmood. A key resource for understanding cloud computing fundamentals, models, and architectures.
- "Cybersecurity and Cyberwar: What Everyone Needs to Know" by P.W. Singer and Allan Friedman. This book offers insights into the importance of cybersecurity and various threats to information systems.
- "E-commerce 2024: Business, Technology, Society" by Kenneth C. Laudon and Carol Guercio Traver, 16th Edition. This book provides insights into the rapidly evolving world of e-commerce.
- "Ethics for the Information Age" by Michael J. Quinn, 8th Edition. A comprehensive guide on ethical issues in information technology, privacy, and digital security.

Pokhara University Faculty of Science and Technology	
Course Code: MTH 250	Full Marks: 100
Course Title: Simulation and Modeling	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: III / Semester: VI	Program: Bachelor of Computer Application

2. Course Description

This course is designed to solve real-world problems safely and efficiently. It supports for exploring valuable solutions by giving clear insights into complex systems. At its core, it deals with understanding the problem from real-life, developing the solution model using discrete and continuous system properties, nature of random number, simulation of software development process, verification and validation of model. This course also introduces the emerging new trended concept and essence for explore multi-agent based modeling and simulation of software development process. After completion of this course, students can create a representative system model including their properties and study the performance by time and space for solving real-world problems.

It analyses, designs and evaluates the complex systems by replicating a real or proposed system using computer software when changes to the actual system are difficult to implement, involve high costs, or are impractical.

3.General Objective

- To acquaint the students thinking solutions of any system safer and cheaper than the real world.
- To develop the skills in students to test a product or system works before building it.
- To acquaint the students with surfacing implicit issues, testing ability to detect and respond, etc.
- To acquaint the students success in developing system model and evaluate the process of decision making in a crisis setting.
- To provide the students with an opportunity for learners to absorb knowledge and practice skills in a realistic but simulated environment.

4.Methods of Instruction

- Lecture
- Practical
- Case study
- Assignment and discussion

5. Contents in Detail

Specific Objectives	Contents

• Familiarize with basic concepts of system, model and simulation • Explain the type of system by input and output • Apply the steps for model development life cycle	**Unit I: Introduction** [6 hours] 1.1 System, model and simulation 1.2 Discrete and continuous systems 1.3 Model of a system 1.4 Types of models 1.5 Computer simulation 1.6 Model development life cycle 1.7 Advantage and disadvantage; area of applications 1.8 Steps in simulation and modeling- *problem formulation, model construction, data collection, model programming, validation, design of experiment, simulation rum and analysis, documentation and implementation.*
• Learning the different techniques for simulation • Find the role of Monte Carlo method for various application	**Unit II: System Simulations** [7 hours] 2.1 The Techniques of Simulation 2.2 The Monte Carlo Method 2.3 Comparison of simulation and analytical methods 2.4 Experimental nature of simulation 2.5 Types of system simulation 2.6 Computation techniques for continuous models 2.7 Cobweb models
• Conceptualize the role and working procedure continuous system simulation • Apply the concept for real-life problem solving	**Unit- III: Continuous System Simulation** [7 hours] 3.1 Continuous system models 3.2 Differential and partial differential equations 3.3 Analog computer 3.4 Analog methods 3.5 Hybrid simulation 3.6 Digital-analog simulators 3.7 Example- *A serial chase problem* 3.8 Feedback systems
• Understand the role of random number • Familiarize with basic approaches and application of random number, and testing randomness	**Unit IV : Random Number** [6 hours] 4.1 Random number properties and tables 4.2 Pseudo-random numbers 4.3 Random number generation methods- *Mid-square, residue* 4.4 Testing numbers for randomness- *Uniformity test and Chi-square test* 4.5 Testing for autocorrelation- *Uniformity test and Chi-square test*
• Visualize the queuing model in service based computation • Explain the actual model, associated attributes and	**Unit V : Queuing System and Performance** [7 hours] 5.1 Elements of queuing system 5.2 Characteristics of queuing systems 5.3 Model of queuing system 5.4 Queuing notation

application of queuing system • Visualize the role of Markov chains in software services.	5.6 Measurement of system performance 5.7 Network of queues 5.8 Application of queuing system 5.9 Markov chains with example
• Visualize the model for software development processes • Test and explain the efficiency of software development process • Visualize the role of agent based simulation for decision making • Explain the system dynamics characteristics	**Unit VI : Software Process Simulation Modeling** [9 hours] 6.1 Software development as a process • The Water fall • Incremental models 6.2 Modeling the software development process • Effort Performance Model (EPM) • Knowledge Model (HKM) • Artefact Quality Model (AQM) • Integrating the models • Developer interaction model 6.3 Multi Agent-based Simulation Model • Agents and their environment-*Environment, Characters, Autonomy, Intelligence* • Implementation- *Agents, Decision making* 6.4 System Dynamics Simulator • Model prerequisites • A system dynamic model- *feedback, levels, flows, auxiliary variables, difficulties*
• Visualize the role of validation and verification in simulation and modeling • Explain the role of different testing and interpretation of output. • Learning the computed output using simple reporting statistical tools	**Unit VII : Design and Output Analysis** [6 hours] 7.1 Calibration and validation of models • Face validation • Validation for assumptions • Output data validation Verification and validation of simulator models 7.2 Input validation using Turing test 7.3 Design of simulation experiment with elimination of initial bias 7.4 Regression analysis 7.5 Analysis of variance (ANOVA)

6. List of Practical

The various practical activities covering all the essence of the course students must engage more actively in the laboratory presence of the instructor. Students should submit practical reports to the instructor for final evaluation. The following laboratory activities should cover the weightage of 1 credit hour using lab environment as per the convenient of student, faculty and institution.

Using any graphics program or open source mapping tools or any high level programming language students can perform following laboratory activities and submit the lab report.

1. Generation of random numbers using computer.
   - *Use algorithms or function related to randomness independently, described as pseudorandom and pseudo independent; interpret the output with its application.*
2. Simulation, testing the correctness of hypothesis using One-sample Kolmogorov-Smirnov.
   - *It should include the types of hypothesis that data in vector x comes from a standard normal distribution, against the alternative.*
3. Monte Carlo simulation
   - *It explores the sensitivity of a complex system by varying parameters within statistical constraints. These systems can include financial, physical, and mathematical models that are simulated in a loop, with statistical uncertainty between simulations. The results from the simulation are analyzed to determine the characteristics of the system.*
4. Create a complete conceptual map of the traffic model.
   - *It should include all of the factors that would contribute either to the increase or decrease in traffic speed as occur in a real situation.*
5. Simulation of Single Server Queuing System
   - *It should include all of the factors that would contribute either to the increase or decrease in traffic speed as occur in a real situation.*
6. Simulation of Software Development Process
   - *It should include the dynamics of software development models and processes with Why, What and How simulation in software engineering. It must include timely review and update the current state-of-art to portray a full overview of Software Process Simulation Modeling (SPSM).*

7. Evaluation System and Students' Responsibilities

Evaluation System

Student academic performance is assessed through internal and external methods. Internal evaluation is carried out by faculty members, monitoring day-to-day student progress. On the other hand, external evaluation is conducted by the university, involving semester-end examinations worth 100 marks. This evaluation system operates as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			

Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Text Books and References Books

Test Books

2. Gordon Geoffrey, *System Simulation*, Second edition, PHI Learning

References

4. Hira D.S., *System Simulation,* Second edition-2009 by S. Chand & Company Ltd, ISBN 81-219-259-0.
5. Law Averill M., Simulation Modeling and Analysis FIFTH EDITION, Averill M. Law & Associates, Inc. Tucson, Arizona, USA www.averill-law.com
6. Singh V.P., System Modeling and Simulation, New Age International (P) Limited, Publishers
7. Jerry Banks, John S. Carson II, Barry L. Nelson, Devid M. Nicol, P. Shahabudeen: Discrete-Event system simulation

Pokhara University Faculty of Science and Technology	
Course Code: MGT 322	Full Marks: 100
Course Title: Organization Management	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: III / Semester: VI	Program: Bachelor of Computer Application

15. Course Description

This course on Organization Management provides a comprehensive exploration of key management principles, organizational structures, and human resource practices. It covers essential topics such as motivation, leadership, and industrial relations, emphasizing their impact on organizational success. Students will engage with strategic planning and decision-making processes, analyze technological innovations and their implications for management, and address contemporary issues like ethics and globalization. Through case studies and practical applications, the course aims to equip students with the knowledge and skills needed to effectively manage and lead organizations in a dynamic business environment.

16. Course Objectives

The general course objectives are to enhance the ability of students with the conceptual of fundamental knowledge about Organization Management. After completing this course, the students will be able:

1. To understand key management concepts, functions, and principles.
2. To explore organizational structures, types, and the relationship with management.
3. To analyze motivation theories, leadership styles, and their impact on organizational success.
4. To examine industrial relations, HRM practices, and compliance with labor laws.
5. To apply strategic planning processes and decision-making techniques.
6. To understand technological innovations, their impact on management, and change management.
7. To explore contemporary management issues, including ethics, globalization, and future trends.

17. Methods of Instruction

The course is taught in English, with faculty members employing a range of teaching methodologies tailored to student needs. These include traditional methods such as lectures and discussions, as well as innovative approaches like Project-Based Learning (PBL), Flipped Classroom, Active Learning, and Gamification. The aim is to engage and motivate students through various instructional strategies. Assessment methods include theoretical and practical evaluations, VIVA, tests, assignments, project work, and terminal examinations, allowing faculty to gauge student understanding comprehensively.

18. Learning Outcomes

Upon completing this course, students will be equipped to effectively analyze and apply management principles, develop and implement strategic plans, and utilize decision-making techniques in organizational contexts. They will understand and address motivation and leadership dynamics, navigate industrial relations and HRM practices, and manage technological innovations and organizational change. Students will also be prepared to tackle contemporary issues such as ethics, globalization, and future management trends. Through practical case studies, they will gain the skills necessary to enhance organizational performance and drive successful management outcomes.

19. Course Contents in Details

Unit-1 Topic: Introduction to Management	**8 hours**
Specific Unit wise Objective	**Course Contents**
• Describe core concepts, characteristics, advantages, limitations, scope, and functions.	1.1 Management 1.1.1. Definition 1.1.2 Characteristics

• Compare various approaches and hierarchical levels within organizations.	1.1.3 Advantages and Limitation 1.1.3 Scope of Management 1.1.4 Function of Management 1.1.5 Principles of Management 1.1.6 Types of Management 1.1.4 Level of Management 1.1.5 Theories of Management 1.2 Simple Case Study
Unit-2 Topic: Organizational Structure and Design	**7 hours**
• Describe organization and its principles. • Explore organizational culture and its impact. • Compare formal and informal organizations.	2.1 Organization 2.1.1 Definition 2.1.2 Characteristics 2.1.3 Principles of Organization 2.1.4 Culture of Organization 2.1.5 Types of Organization (Formal and Informal Organization with advantages and disadvantages) 2.1.6 Types of Formal Organization (advantages and disadvantages) 2.1.7 Difference between Organization and Organizational Chart 2.2 Organizational Structure Design 2.2.1 Definition 2.2.2 Organizational Framework 2.2.3 Key Components of an Organizational Framework 2.3 Relation Between Management and Organization 2.4 Simple Case study
Unit-3 Topic: Motivation and Leadership	**8 hours**
• Describe the importance and, motivation and demotivation in the workplace. • Explore various motivation types, strategies, and theories. • Comprehend the importance and characteristics of effective leadership. • Distinguish between leadership and management functions. • Analyze leadership styles, motivational leadership, and emerging trends.	3.1 Motivation .1.1 Introduction and Importance of Motivation .1.2 Indicators of Motivation and demotivation in an organization .1.3 Process of Motivation .1.4 Types of Motivation .1.5 Motivation Strategies in the Workplaces .1.6 Theories of Motivation .1.7 Role in Organization Success 3.2 Leadership 3.2.1 Introduction and Importance of Leadership 3.2.2 Characteristics of Good (Effective) Leadership 3.2.3 Function or Need of Leadership 3.2.4 Theories of Leadership 3.2.5 Leadership Styles 3.3 Leadership Vs. Management 3.4 Motivational Leadership 3.5 Challenges in Leadership and Motivation 3.6 Emerging Trends in Leadership and Motivation 3.7 Developing Future Leaders 3.8 Simple Case Study
Unit-4 Topic: Industrial Relationship and Human Resource Management	**14 hours**
• Describe the key definitions and concepts of Industrial Relations and HRM.	4.1 Introduction to Industrial Relations and Human Resource Management 4.1.1 Definitions and Concepts

• Explore the importance of effective employer-employee relationships. • Analyze the role of trade unions in organizations. • Describe the process of collective bargaining and negotiation. • Develop strategies for conflict resolution and handling grievances. • Implement effective recruitment, selection, and diversity practices.	4.1.2 Importance in Organizational Success 4.2 Key Concepts in Industrial Relations 4.2.1 Employer-Employee Relationships 4.2.2 Role of Trade Unions 4.2.3 Collective Bargaining and Negotiation 4.3 Labor Laws and Legal Framework 4.3.1 Overview of Key Labor Laws 4.3.2 Workers' Rights and Employer Obligations 4.3.3 Compliance with Industrial Regulations 4.4 Conflict Resolution and Grievance Handling 4.4.1 Causes of Industrial Conflicts 4.4.2 Mediation, Arbitration, and Negotiation 4.4.3 Handling Grievances and Disputes 4.5 Introduction to Human Resource Management (HRM) 4.5.1 Role and Functions of HRM 4.5.2 Strategic Importance of HR in Organizations 4.5.3 HRM vs Personnel Management 4.6 Recruitment and Selection 4.6.1 Recruitment Strategies and Processes 4.6.2 Selection Techniques and Best Practices 4.6.3 Importance of Diversity and Inclusion in Hiring 4.7 Training and Development 4.7.1 Employee Skill Development 4.7.2 Career Planning and Growth 4.7.3 Importance of Continuous Learning in Organizations 4.8 Performance Management and Appraisal 4.8.1 Methods of Performance Appraisal 4.8.2 Setting Goals and KPIs 4.8.3 Feedback and Coaching for Improvement 4.9 Compensation and Benefits 4.9.1 Designing Compensation Structures 4.9.2 Types of Employee Benefits 4.9.3 Pay Equity and Reward Systems 4.10 Employee Engagement and Retention 4.10.1 Strategies for Employee Motivation 4.10.2 Importance of Job Satisfaction 4.11 Simple Case Study
Unit-5 Topic: Strategic Planning and Decision Making	**5 hours**
• Explain the introduction, objectives, and importance of strategic planning. • Explore the objectives and importance of decision making in organizations. • Explore SWOT analysis for evaluating organizational strengths and weaknesses.	.1 Introduction, Objective, Importance of Strategy Planning .2 Introduction, Objective, Importance of Decision Making .3 Strategic Planning Process .4 SWOT Analysis .5 Decision-Making Techniques .6 Simple Case Study
Unit-6 Topic: Technology and Innovation in Management	**3 hours**
• Describe the fundamentals of technological innovation and its significance. • To analyze the impact of technology on management practices and decision-making. • Explore innovation management techniques and strategies.	6.1 Introduction to Technological Innovation 6.2 Technology's Impact on Management 6.3 Innovation Management 6.4 Managing Change

1.	
Unit-7 Topic: Contemporary Issues in Management	**3 hours**
• Explain contemporary issues and their relevance in modern management. • Explore future trends and their potential impact on management practices.	Introduction to Contemporary Issues in Management 4.1. Ethics and Social Responsibility 4.2. Globalization Challenges 4.3. Future Trends in Management

6. Evaluation System

Student academic performance is assessed through internal and external methods. Internal evaluation is carried out by faculty members, monitoring day-to-day student progress. On the other hand, external evaluation is conducted by the university, involving semester-end examinations worth 100 marks. This evaluation system operates as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Total Internal Marks		**50**		
Full marks=50+50 =100				

7. Students' Responsibilities

To be eligible for the Semester End Examinations, students must achieve a minimum of 45% marks in internal evaluation including both theoretical and practical sections. Additionally, they must maintain an attendance of at least 80% in their classes. Failure to meet these criteria will result in a "NOT QUALIFIED" (NQ) status, rendering the student ineligible for the Semester End Examinations. Students are strongly advised to attend all classes, formal exams, tests, and fulfill all requirements within the stipulated time frame. It is essential for students to fulfill all course requirements to successfully complete the course.

Pokhara University **Faculty of Science and Technology**	
Course Code: PRJ 351 (3 Credits)	Full Marks: 100
Course Title: Project III	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: III / Semester: VI	Program: Bachelor of Computer Application

1. Course Description:
Project-III is a hands-on, practical course designed for BCA sixth-semester students. It focuses on the development of web application or data analysis tool. The project should be original and reflective of the students' skillset in software development or data analysis.The course encourages students to apply their knowledge in creating a functional web application or performing data analysis to solve a real-world problem. Students may work independently or in small teams to develop a web tool, data analysis system, or any other software project of interest. Guided by a faculty supervisor, the course helps students explore and manage various stages of the software or data lifecycle. Project III offers students exposure to the practical side of development, testing, and documentation.

2. General Objectives:

- To allow students to apply theoretical knowledge from previous courses to create a functional web-based application or a robust data analysis project.
- To encourage students to think critically and solve problems creatively as they work on their projects.
- To help students develop essential project management, teamwork, and communication skills that are crucial for their future careers.
- To enable students to produce a functional, user-friendly, and well-documented software application that meets specific needs.
- To provide students knowledge about data analysis projects, where students will learn to gather, clean, process, and visualize data meaningfully.

3. Procedure:

The Project III is focused to develop web application for event management; data analysis project that identifies trends from social media data; an interactive tool for financial planning or budgeting; dashboard for visualizing data and providing analytics.

- Use modern frameworks such as Django, Flask, or any prefered framework for web projects.
- For data analysis, Python (with libraries like Pandas, Matplotlib) is recommended.
- Teams of 2-3 students ensure effective collaboration and project management.
- Projects must be original and free from plagiarism.
- Follow coding standards with properly documented, modular code, adhering to naming conventions.
- Use Object-Oriented Programming (OOP) and ensure code reusability.

4. Project Phases:

The entire project work shall be divided into three phases and evaluation shall be done accordingly:

I. Team formation (Week 1)

- Students will form a team consisting of a maximum of 3 members based on their interests and skills.
- Teams should be balanced to ensure diverse skill sets and effective collaboration.

II. Project Proposal Submission: (Week 2-4)

- Students will start by defining the project's scope, objectives, and deliverables.
- A detailed project proposal must be submitted, outlining the project idea, timeline, and required resources.

Proposal Format:

1. *Introduction*
2. *Aim and Scope*
3. *Objectives*
4. *Study of the Previous Systems*
5. *Current Problem and Proposed Solution*
6. *High level requirements*
   - *5.1 Functional*
   - *5.2 Non functional*
   - *5.3 Requirement prioritization table*
7. *Design*
   - *6.1 ER Diagram(for database systems)*
   - *6.2 Use case Diagram*
   - *6.3 DFD*
   - *6.4 UML Class Diagram*
8. *Detail Budget*
   - *7.1 Detail Budget*
   - *7.2 Budget Breakdown*

9. *Detail Timeline*
10. *Conclusion*
11. *References*

III. UI/UX Design (Week 5-6)

- Students should begin by gathering user stories to understand the needs, goals, and pain points of potential users. These user stories will guide the design and development of the user interface and user experience, ensuring the final product meets the end users' expectations.
- Collect atleast three user story for each page.
- User Interface designed with any UI/UX Design tool.
- Present and defend the UI/UX design.

IV. First Milestone (Week 7-8)

- Present 60% of the project work. Demonstrate functionality, including data flows, UI/UX, and initial versions of features.

V. Second Milestone (Week 9)

- Present **100% of the completed project**. Include both front-end and back-end functionalities, or full data analysis reports with insights.

VI. Final Defense (Week 10)

- The final demonstration should focus on both the **functionalities** and **documentation**..
- Final defense includes project demo, addressing both technical aspects and usability.

8 Guidelines for Project using Framework

- Use modern frameworks such as Django, Flask, or any preferred framework for web projects.
- For data analysis, Python (with libraries like Pandas, Matplotlib) is recommended.
- Teams of 2-3 students ensure effective collaboration and project management.
- Projects must be original and free from plagiarism.
- Follow coding standards with properly documented, modular code, adhering to naming conventions.
- Use Object-Oriented Programming (OOP) and ensure code reusability.

9 Report format:

Title Page
Original Copy of the Approval
Certificate of Authenticated work
Role and Responsibility Form
Abstract
Acknowledgement
Table of Contents
Table of Figures
List of Abbreviations
List of Tables
CHAPTER 1: INTRODUCTION
1.1 Background
1.2 Objectives
1.3 Purpose, Scope, and Applicability
1.3.1 Purpose
1.3.2 Scope
1.3.3 Applicability

1.4 Achievements
1.5 Organization of Report

CHAPTER 2: SURVEY OF TECHNOLOGIES
Review of the similar/relevant projects
CHAPTER 3: REQUIREMENTS AND ANALYSIS
3.1 Problem Definition
3.2 Requirements Specification
3.3 Planning and Scheduling
3.4 Software and Hardware Requirements
3.5 Preliminary Product Description
3.6 Conceptual Models

CHAPTER 4: DESIGN
4.1 Introduction
4.2 System Design
4.3 Database design
4.4 Interface Design
4.5 Summary

CHAPTER 5: IMPLEMENTATION AND TESTING
5.1 Implementation Approaches
5.2 Coding Details and Code Efficiency
5.2.1 Code Efficiency
5.3 Testing Approach
5.3.1 Unit Testing
5.3.2 Integrated Testing
5.3.3 Beta Testing
5.4 Modifications and Improvements
5.5 Test Cases

CHAPTER 6: RESULTS AND DISCUSSION
6.1 Test Reports
6.2 User Documentation

CHAPTER 7: CONCLUSIONS
7.1 Conclusion
7.1.1 Significance of the System
7.2 Limitations of the System
7.3 Future Scope of the Project
REFERANCES

Pokhara University **Faculty of Science and Technology**		
Course Code.: CMP 401 (3 Credits)		Full Marks: 100
Course Title: Cyber Law and Professional Ethics (3-0-1)		Pass Marks: 45
Nature of the Course: Theory/Practical		Total Lectures: 48 hours
Level: Bachelor	Year: IV / Semester: VIII	Program: Bachelor of Computer Application

1. Course Description

This course provides an in-depth understanding of cyber law, cybersecurity, privacy protection, intellectual property rights, and professional ethics for IT professionals. It aims to equip students with the knowledge and skills necessary to navigate the legal and ethical challenges in the digital world.

2. General Objectives

The general objectives of the course are: -

- To understand the fundamentals of cyber law and its application.
- To explore cybersecurity principles and practices.
- To examine privacy laws and issues related to freedom of expression.
- To understand intellectual property rights and their implications.
- To develop ethical decision-making skills in the context of IT.
-

3. Methods of Instructions:

- Lecture and discussion
- Practical
- Project and presentation

4. Contents in Detail

Specific Objectives	Contents
Unit 1: Ethics and Professional Practice	**9 hours**
To explore the ethical behaviour and professional practice.	1.1 Ethics and ethical analysis 1.2 Ethical theories - Ethical relativism, Deontological theories, Utilitarianism 1.3 Profession and attributes of a profession 1.4 Professional roles and responsibilities, 1.5 Conflict of interest 1.6 Whistleblowing 1.7 Corporate social responsibility 1.8 Code of conduct- ACM and IEEE
Unit 2: Cybersecurity and Computer Crime	**9 hours**
To have an overview of cyber security and cybercrime.	2.1 Attack, Threat, Vulnerability and their types 2.2 Security, CIA Triad, Security Service, Security Mechanism 2.3 DMZ, Firewall and their limitations, Need of encryption 2.4 Network Security Model, Network Security Access Model 2.5 Secure Socket Layer, Transport Layer Security 2.6 IPSec – transport and tunnel 2.7 Computer crime: terms and definitions 2.8 Taxonomy / types of Computer crime
Unit 3: Cyberspace and Freedom of Expression	**10 hours**

To find the relation between the Internet and the expression posted and the freedom of speech and its social implications.	3.1 Historical context and legal foundations 3.2 Key principles of free speech 3.3 First Amendment Rights and Freedom of Expression 3.4 The Internetours platform for free expression 3.5 Online and offline speech 3.6 Ethical Issues in Social Networking
Unit 4: Privacy Issues and Protection	**10 hours**
To explore the privacy violations issues and its ways of protections through the technology.	4.1 Privacy then and now 4.2 Information Privacy and Privacy Laws 4.3 Key Privacy and Anonymity Issues 4.4 Consumer Profiling, Workplace Monitoring 4.5 Data Surveillance 4.6 Privacy Protection 4.6.1 Individual responsibility, 4.6.2 Role of IT professionals, 4.6.3 Legislative provisioning
Unit 5: Cyber Law and Legal Issues	**10 hours**
To make students understand legal issues and exhibit professional practices for IT professionals	5.1 Intellectual Properties 5.1.1 Copyright, patent, design, trademark, and trade-secrets 5.1.2 IP Laws 5.4 IT related policies and laws in Nepal 5.4.1 ICT Policy 5.4.2 Electronics Transaction Act 5.4.3 Communication Act 5.4.4 Provisions and Electronic Transaction Act

5. Practical Works (Not all are mandatory, selective ones may be done. Practical work may also be replaced with case study, where students shall analyze the existing implementation of the assigned practical work.)

S.N.	Tutorials / Case studies (C)
1	Unit I: Case studies on ethical discussions and dialectic processes
2	Unit II: Case studies on various computer attacks
3	Unit III: Case studies on internet censorship
4	Unit IV: Case studies on Digital surveillance and privacy violations
5	Unit V: Case studies national and international cyber laws and their violations

6. **Evaluation System and Students' Responsibilities**

Evaluation System

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester End examination	50
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	50%			
Practical		20		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		50		
Full marks=50+50				

Students' Responsibilities:
Each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such a score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References

Text Book

1. Anirudh Rastogi, *Cyber Law: The Law of the Internet and Information Technology*, Second Edition, Lexis Nexis, ISBN-10: 9351432548 ISBN-13: 978-9351432548, 2014
2. George Reynolds, *Ethics in Information Technology*, Fourth Edition, Course Technology, Cengage Learning, 2015
3. Pavan Duggal, *Cybersecurity and Cyber Law*, Third Edition, Universal LexisNexis, 2023.
4. Johnson, D. G., *Computer Ethics*, Pearson Education Asia, Third Edition, 2001, ISBN 81-7808-306-X.

Online References:

1. Cyber Law and Professional Practice Workbook
2. National and International Policies and Laws
3. Office of Controller of Certification, https://occ.gov.np/
4. Nepal Law Commission, https://www.lawcommission.gov.np/
5. Online book and information on cyber law, ethics and professional practice

Pokhara University Faculty of Science and Technology	
Course Code.: CMP 404 (3 Credits)	Full Marks: 100
Course Title: Mobile Application Development Technology (3-0-3)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: IV / Semester: VII	Program: Bachelor of Computer Application

1. Course Description:

This course provides an in-depth exploration of Android application development using Kotlin and modern techniques, focusing on Jetpack Compose for UI design. The primary intent is to equip students with the knowledge and skills necessary to create, manage, and deploy Android applications. The course covers major content areas such as Kotlin fundamentals, Android architecture components, Jetpack Compose for building modern UIs, data storage, networking, and app monetization. The delivery approach includes hands-on projects, self-learning modules, and practical exercises to ensure a comprehensive learning experience

2. General Objectives:

The course is designed with the following general objectives:

- To familiarize students with Kotlin programming fundamentals and its application in Android development.
- To enable students to understand and implement Android architecture components effectively.
- To provide students with skills to design modern UIs using Jetpack Compose.
- To make students capable of integrating data storage solutions in Android applications.
- To equip students with the knowledge to handle network operations and APIs in Android.
- To develop students' understanding of app monetization strategies and publishing on app stores.

3. Methods of Instructions:

The course will employ a variety of instructional methods to provide a comprehensive learning experience:

- **Lectures**: To introduce and explain core concepts of Android development and Kotlin programming.
- **Discussion**: To facilitate deeper understanding through interactive conversations.
- **Group Work**: To foster collaboration and practical application through team-based projects.
- **Lab Work**: To offer hands-on experience with coding, debugging, and deploying Android applications.
- **Experiential Exercises**: To reinforce learning through real-world programming tasks and projects.
- **Self-Learning Modules**: To support independent study and accommodate different learning paces.

4. Course Contents:

Specific Objectives	Contents
Unit I: Kotlin Programming Fundamentals	**6 hours**
• Explore Kotlin's features, syntax, and structure to leverage its advantages in Android development.	1.1 Introduction to Kotlin 1.1.1 Overview of Kotlin's features 1.1.2 History of Kotlin Language 1.1.3 Why Kotlin over Java?

• Utilize key programming constructs such as conditional statements, loops, and exception handling to control program flow and manage errors. • Apply object-oriented and functional programming principles, including classes, functions, and collections to develop efficient Kotlin applications.	1.2 Basic Syntax and Structure 1.2.1 Variables and Data Types 1.2.2 Type Inference 1.2.3 Type Casting 1.2.4 Lists, Sets and Maps in Kotlin 1.3 Control Flow in Kotlin 1.3.1 Conditional Statements (if, when) 1.3.2 Loops (for, while and do-while) 1.4 Exception Handling in Kotlin 1.4.1 Try, Catch and Finally 1.4.2 Throwing exception using throw 1.5 Functions in Kotlin 1.5.1 Function Declaration and Parameters 1.5.2 Lambda Expressions 1.5.3 Higher-Order Functions 1.6 Object-Oriented Programming 1.6.1 Classes and Objects 1.6.2 Inheritance and Polymorphism 1.6.3 Interfaces and Abstract Classes
Unit II: Android OS & Basics	**6 hours**
• Understand the evolution of Android OS, its market impact, and the reasons for its dominance in the mobile OS landscape. • Learn the components of Android architecture, including the Linux Kernel and Android Runtime (ART), and their roles in performance and security. • Apply knowledge of the Android application lifecycle, Manifest File, Fragments, intents, and navigation to effectively manage and develop Android apps.	2.1 Introduction to Android OS 2.1.1 Overview of the history and evolution of Android OS 2.1.2 Android's Market share and impact in mobile development 2.2 Android Architecture Overview 2.2.1 Linux Kernel 2.2.2 Android Runtime 2.2.3 Android OS Architecture 2.3 Android Application Components 2.3.1 Overview of Activity 2.3.2 Lifecycle of Activity 2.3.3 Android Manifest File 2.3.4 Permission Mechanism in Android 2.3.5 Fragments and its uses 2.4 Components- button, dialog, checkbox, date picker, snackbar, list and grid 2.5 Intents and Navigation 2.5.1 Explicit and Implicit Intent 2.5.2 Passing data between activities
Unit III: Jetpack Compose Fundamentals	**8 hours**
• Compare Jetpack Compose with XML-based UI development, highlighting the benefits of Compose for modern Android app development. • Set up Jetpack Compose in an Android project, including necessary configurations and dependencies.	3.1 Introduction to Jetpack Compose 3.1.1 Jetpack Compose vs XML 3.1.2 Setting up Compose in Android Project 3.2 Composable Functions 3.2.1 Reusable UI Components 3.2.2 Recomposition 3.3 Layouts and Theming 3.3.1 Layout Components (Column, Row and Box)

• Create and utilize composable functions and layout components to build and manage reusable, flexible, and efficient UI components. • Apply Material Design principles, manage UI state effectively, and design responsive UIs to ensure consistency and adaptability across different devices and screen sizes.	3.3.2 Custom Layouts 3.3.3 Material Theme and Styles 3.4 State Management 3.4.1 remember and mutableStateOf() 3.4.2 State Hoisting 3.5 Responsive and Adaptive UIs 3.5.1 Supporting multiple screen sizes
Unit IV: Coroutines in Kotlin	**6 hours**
• Understand asynchronous programming basics and how Kotlin coroutines provide an efficient approach for handling background tasks. • Compare coroutines with traditional multithreading, focusing on their lightweight nature, structured concurrency, and lifecycle management. • Learn to create, manage, and optimize coroutines using functions like launch and async, handle exceptions, and utilize various dispatchers and context switching for performance.	4.1 Introduction to Coroutines 4.1.1 Asynchronous Programming 4.1.2 Coroutines and its advantages 4.1.3 Lifecycle of Coroutines 4.2 Creating and Managing Coroutines 4.2.1 launch and async Function 4.2.2 Context Switching 4.3 Coroutines Scopes and Dispatchers 4.3.1 CoroutineScope and GlobalScope 4.3.2 Structure Concurrency 4.3.3 Dispatchers and their types 4.4 Exception Handling And Suspend in Coroutines 4.4.1 Exception Handling Basics 1.4.3 Supervisor Scope 4.4.3 Suspend Function in Coroutines
Unit V: Data Storage and Management	**6 hours**
Implementation SQLite's purpose and core components, and write SQL commands for basic database operations. • Explore Room's role in simplifying database interactions, configure it in an Android project, and use Data Access Objects (DAOs) for CRUD operations.	5.1 Brief Introduction to SQLite 5.1.1 Overview of SQLite 5.1.2 Basic SQLite structure concepts 5.1.3 SQL Commands CREATE, INSERT, UPDATE, DELETE, SELECT 5.2 Room Database 5.2.1 Introduction to Room 5.2.2 Setting up Room 5.2.3 Creating Data Access Objects (DAOs) 5.2.4 Performing CRUD Operations with Room 5.3 Shared Preferences
Unit VI: Networking and APIs	**7 hours**
• Explain RESTful API principles and HTTP operations and implement HTTP requests • Set up Retrofit for HTTP requests, integrate Gson for JSON parsing, and handle errors effectively. • Apply coroutines with Retrofit for asynchronous network operations and manage notifications in Android using Notification Manager.	6.1 Understanding RESTful APIs 6.1.1 Overview of REST Principles: GET, POST, PUT, DELETE operations 6.1.2 Http URL Connection for HTTP request 6.2 Networking Libraries 6.2.1 Retrofit for HTTP request 6.2.2 Gson in Retrofit for JSON Parsing 6.2.3 Error Handling for failed Requests 6.2.4 Coroutines with Retrofit 6.3 Notifications and its types

Unit VII: Monetization and Publishing	6 hours
• Explain various app monetization strategies, including ads, in-app purchases, and subscriptions, and identify different types of ads used in mobile apps. • Introduce Google AdMob, detail its integration for ad monetization, and differentiate between APK and AAB files in app distribution. • Distinguish between demo and release APKs, manage app versioning and changelogs, and understand the process of uploading an app to the Google Play Store.	7.1 App Monetization strategies 7.1.1 Overview of Monetization Models (Ads, In App Purchases, Subscription Services) 7.1.2 Types of Ads 7.1.3 Introduction to Google AdMob 7.2 Publishing Applications 7.2.1 APK and AAB files 7.2.2 Demo vs Release apk 7.2.3 Versioning and Changelog 7.2.4 Developer account setup and app upload

5. List of Practical:

SN	Practical Works
1.	Develop Android applications using Kotlin [6-8 hours]
2.	Create data storage solutions with Room database [4-6 hours]
3.	Implement network operations using Retrofit [4-6 hours]
4.	Build and test user interfaces. [4-6 hours]
5.	Monetize and publish Android applications [4-6 hours]

6. 6. Evaluation System and Students' Responsibilities:

6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50 =100				

6.2 Students' Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References:
Reference Books 1. Zawinski, J. (2022). *Kotlin Programming: The Big Nerd Ranch Guide*. Big Nerd Ranch. 2. Swaine, M. (2021). *Android Programming with Kotlin for Beginners*. Packt Publishing. 3. Grobmeier, A., & McGowan, B. (2023). *Mastering Jetpack Compose*. Apress.

Pokhara University Faculty of Science and Technology	
Course Code.: ECO 311 (3 Credits)	Full Marks: 100
Course Title: Applied Economics (3-0-0)	Pass Marks: 50
Nature of the Course: Theory	Total Lectures: 48 hours
Level: Bachelor, Year: IV / Semester: VII	Program: Bachelor of Computer Application

1. Course Description:

This course introduces students to the fundamental principles of economics with a focus on applications relevant to the technology and computing sectors. It covers essential microeconomic and macroeconomic concepts such as market forces, demand and supply, theory of production, product pricing and factor pricing, national income, inflation and unemployment, business cycle, etc. Students will learn to apply these economic principles to decision-making in tech-driven environments, analyze the impact of economic policies on the IT industry, and understand the global economic factors influencing innovation and technological advancements.

2. General Objectives:

The general objectives of this course are as follows:

e) To equip students with a foundational understanding of microeconomics and macroeconomics concepts relevant to computing and technology industries.
f) To enable the students to apply economic theories and quantitative methods to analyze economic problems in IT and tech sectors.
g) To enhance the capacity of the students to evaluate market conditions, competitive dynamics, and the economic environment and formulate effective strategies.
h) To teach students how to optimize the use of resources within an organization under different market conditions.
i) To make the students knowledgeable about the impact of government policies, regulations, and global economic trends on operations and strategy.

3. Methods of Instructions:

5. Lecture and disctussion
6. Demostration
7. Presentation
8. Case study

It is expected that students are fully engaged with the teacher in subject matters and lessons to excel their interactive and presentation skills.

4. Course Contents:

Specific Objectives	**Contents**
Unit 1: Introduction to Economics	**4 hours**
• Explain a clear understanding of the nature and scope of fundamentals of economics, including its definition, purpose, relevance it to the IT and technology sector.	1.1 Nature and scope of economics 1.2 Fundamental of economic problems: scarcity, choice and opportunity cost 1.3 Microeconomics verses macroeconomics 1.4 Role of economics in the IT and technology Sectors
Unit 2: Demand, Supply and Consumer Behaviour	**6 hours**

• Apply quantitative measures (such as elasticity) and graphical tools (such as supply-demand diagrams and indifference curves) to analyze economic phenomena. • Devise decision-making skills how consumers optimize their choices given constraints, and how firms can respond to changes in market conditions and policies.	2.1 Market forces of demand and supply 2.2 Elasticity and its application 2.3 Consumer behaviour: utility, indifference curve, budget constraints 2.4 Consumer equilibrium, price, income and substitution effect
Unit 3: Theory of Production and Cost Analysis	**6 hours**
• Describe how firms make production and cost-related decisions in both the short-run and long-run, and how these decisions impact overall performance. • Apply strong analytical and graphical skills, enabling them to visualize and analyze production processes and cost structures. •	3.1 Law of variable proportion and law of returns to scale 3.2 Isoquant and isocost lines 3.3 Producer equilibrium, output maximization, cost minimization 3.1 Nature and types of cost curves (both traditional and modern), relationship among total, average and marginal cost curves. 3.2 Nature and types of revenue curves, relationship among total, average and marginal revenue curves
Unit 4: Market Structure and Pricing	**6 hours**
• Compare and contrast different market structures, understanding the implications of each for making strategy and public policy. • Calculate how firms in various market structures determine pricing, output, and competitive strategies.	4.1 Market structure and degree of competition 4.2 Perfect competition: Price and output determination 4.3 Monopoly: Sources of monopoly, price and output determination 4.4 Monopolistic competition: Price and output determination 4.5 Oligopoly: Concept and characteristics of oligopoly market
Unit 5: Factor Pricing	**4 hours**
• Define and explain the concepts and types of rent, wages, interest and profit. • Analyze how wage rate is determined under perfect competition market and imperfect competition market structures.	5.1 Concept and types of factor prices: Rent, wages, interest and profit 5.2 Wage rate determination under perfect competition market. 5.3 Wage rate determination under monopsony.
Unit 6: National Income	**6 hours**
• Explain the circular flow of national income under two, three and four sector models addressing the leakages and injections of the economy as well. • Define and explain the concepts of national income and learn the methods of calculating national income and application • Identify the challenges and importance of calculating national income.	6.1 Circular flow of national income in two, three and four sectors economy 6.2 Concept of national income: GDP, GNP, NNP, personal income, disposable income, and per capita income 6.3 Methods of calculation of national income: product, income and expenditure methods 6.4 Diffculties in calculation of national income 6.5 IT Industries and national income

Unit 7: AD-AS Model and IS-LM Model	8 hours
• Explain the macroeconomic equilibrium under two, three and four sector economy • Illustrate the concept of product market (IS) and money market (LM) equilibrium and explain the determination of interest rate and output level. • Identify the factors causing shift in IS and LM curves and evaluate the effects of fiscal and monetary policy.	7.1 **AD-AS Model** 7.1.1 Aggregate Demand and Aggregate Supply: Components of Aggregate Demand (AD): Consumption, Investment, Government, Spending, and Net Exports Aggregate Supply (AS): Short-run and Long-run as Curves. 7.1.2 Equilibrium in AD-AS Model: Short-run vs. Long-run Equilibrium, Output, Gaps (Recessionary and Inflationary Gaps) 7.2 IS-LM Model 7.3 IS Curve: Equilibrium in the Goods Market Deriving the IS Curve: Relationship between Interest Rates and Output Factors Affecting the IS Curve: Investment, Government Spending, and Taxes 7.4 LM Curve: Equilibrium in the Money Market Deriving the LM Curve: Relationship between Interest Rates and Money Supply 7.5 General Equilibrium in IS-LM Model Interaction of Goods and Money Markets, Determination of Interest Rates and Output 7.6 Fiscal and Monetary Policy Implications in IS-LM Framework
Unit 8: Macroeconomic Issues and Policies	**8 hours**
• Define and describe the business cycle its features, phases and explain the counter-cyclical policies for economic stabilization. • Define and explain inflation, its measure, types, causes, effects and control measures. • Explain the concept of Phillips curve. • Define fiscal policy and monetary policy with the major objectives and tools. • Describe the role of deficit financing in an economy. • Distinguish between the fiscal policy and monetary policy.	8.1 Business cycle: Definition, features and phases, counter-cyclical policies. 8.2 Inflation: Definition and measurement, Types, causes and effects, control measures 8.3 Fiscal policy: Objectives and tools, deficit financing 8.4 Monetary policy: Objectives and tools. 8.5 Assessment of the comparative effectiveness of fiscal and monetary policies
5. Methods of Instruction	
To effectively convey the concepts and their practical applications in the applied economics, here are some possible teaching methods: Lecture method applied to introduce and explain key concepts, theories and models. To make lectures more engaging, real-world examples are discussed. Besides, group discussion to foster collaborative learning; quantitative excersise to strengthen students' ability in quantitative tools in economic analysis; guest lectures and industry speakers to bring real-world	

insights into the classroom; and, assessment method to evaluate understanding and application of concepts.

6. Evaluation System and Students' Responsibilities

6.1 Evaluation System

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

External Evaluation	Marks	Internal Evaluation	Marks
Total Internal	50	Class attendance, quizzes, assignment	15
		Presentation	10
		Internal Term Exam	25
Total External	50	Total Internal	50
Full Marks 50+50 = 100			

6.2 Students' Responsibilities:

Each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the semester end examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the semester end examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References

7.1 Text Books

1. Mankiw, N. G. (2021). *Principles of Microeconomics (9th ed.).* Cengage Learning.
2. Mankiw, N. G. (2019). *Macroeconomics (10th ed.).* Worth Publishers.

7.2 Reference Books

3. Salvatore, D. (2020). *Microeconomics: Theory and Applications (6th edition.)* Oxford University Press.
4. Pindyck, R.S, & Rubinfeld, D.L. (2018). *Microeconomics (9th ed.).* Pearson.
5. Vaish, M.C. (2010). *Macroeconomic Theory (14th ed.),* Vikas Publishing House.

Note: Latest editions of the prescribed text books and references are recommended.

Pokhara University
Faculty of Science and Technology

Course Code.: INT461 (3 Credits)	Full Marks: 100
Course Title: Internship (3-0-0)	Pass Marks: 50
Nature of the Course: Lecture	Total Lectures: 48 hours
Level: Bachelor, Year: IV / Semester: VII	Program: Bachelor of Computer Application

20. Course Description

The Internship Program for BCA is designed to provide students with hands-on experience in the IT industry, bridging the gap between academic knowledge and practical application. Students will work on real-world projects, allowing them to apply their skills, learn new technologies, and understand the dynamics of working in a professional environment.

21. General Objectives

- To expose students to real-world applications of computer science and IT.
- To enhance students' problem-solving, technical, and communication skills.
- To develop knowledge about the workflow and project management processes within an organization, further making them understand professional ethics and practices.
- To provide an opportunity to work on project(s) that aligns with their career interests.

22. Internship Structure

3.1 Internship Orientation
3.2 Internship Placement
3.3 Duration
3.4 Documentation
3.5 Reporting
3.6 Monitoring and Evaluation

3.1 Internship Orientation

An internship orientation (3-5 hrs) should be conducted prior to internship start that shall include;

- Introduction to the internship objectives, processes, expectations, and deliverables.
- Workshop on professional behavior, communication, and workplace ethics.
- Industry professionals should be invited to conduct one session within the orientation to introduce the real-work environment and the requisites.

3.2 Internship Placement

3.2.1 Internship Selection

Selection of intern organization shall be done by students at a reputable organization in the IT sector prioritizing their interest. College shall also assist in placement with the partnering organization.

3.2.2 Placement Process

- Students should submit their resume to the college along with a letter introducing the organization of their interest. Respective department of the college shall then prepare recommendation for placement in the internee organization.
- Internship for any student is not allowed in a company owned by themselves, or if it is a family business and/or if they are the employee at the same organization.
- Respective students should submit the college recommendation including their resume to the intern organization.

3.3 Duration

Total duration for the Internship = Minimum 8 Weeks

3.4 Documentation

- Internship Agreement should be signed by the student, the organization, and the college prior to start of internship.
- Internship confirmation letter from the internee organization should be submitted to the college.
- After completion of the internship, students should submit internship completion certificate from the intern organization.

- Respective college department/academic supervisor should collect a sealed evaluation sheet from the intern organization.

3.5 Reporting

3.5.1 Weekly Progress Reports

Students must submit weekly reports (for minimum of 8 weeks) outlining their progress, challenges faced, and skills acquired each week.

3.5.2 Mid-Internship Review

A formal review session should be scheduled in 4-5 weeks of internship involving the student and academic supervisor to assess progress and address any issues. This shall be the mid-term presentation summarizing all the activities within the past weeks and planning for remaining internship period.

3.5.3 Final Internship Review

The Final internship review shall be conducted after completion of the internship duration, as scheduled by academic supervisor or the respective department of the college.

- The final internship review shall cover Reflection on the learning experience and how it has impacted the student's career goals.
- A comprehensive Final Report with details of the internship activities, including objectives, issues addressed, methodology, results, and conclusions should be submitted 3 days prior to internship final presenation.
- An external evaluator should be appointed for Final Internship Review.

All report and document formats shall be prepared by academic supervisor or the respective department of the college.

3.6 Monitoring and Evaluation

3.6.1 Monitoring

Visit to the intern organization should be made by the academic supervisor at least once so as to confirm that the intern is following procedures and guidelines.

3.6.2 Evaluation Criteria

The evaluation for the overall activies of the Internship should be covered in two stages Internal 60% and External 40%.

3.6.2.1 Internal Evaluation (Academic Supervisor) - 60%

- Weekly Reporting & Documentation = 20
- Mid-Internship Review = 20
- Internship Final Report = 10
- Overall Activities = 10

3.6.2.2 External Evaluation - 40%

- Performance and feedback from the Industry Mentor = 20
- Internship Final Report (External Evaluator) = 10
- Presentation and viva-voce (External Evaluator) = 10

23. Students' Responsibilities:

Each student must secure at least 45% marks in the internal evaluation. All activities mentioned should be completed and punctuality at intern organization should be maintained.Each intern must complete the minimum duration of internship. Final evaluation shall only be done based on internship completion recommendation made by the intern organization. Any confirmation not received from the intern organization or student failing to obtain minimum pass marks as stated

above shall be marked as NOT QUALIFIED (NQ). All activities mentioned in the Internship structure are mandatory.

Pokhara University **Faculty of Science and Technology**	
Course Code.: CMP 415 (3 Credits)	Full Marks: 100
Course Title: **Cloud Computing (3-0-2)**	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: IV / Semester: VIII	Program: Bachelor of Computer Application

1. Course Description:

This course provide the theoretical as well as practical knowledge of cloud computing including designing, implementing and managing the cloud computing

2. General Objectives:

The general objectives of this course are to provide theoretical as well as practical knowledge of cloud computing to make students capable of designing, implementing and managing the issues of cloud computing in their personal as well professional life.

The course is designed with the following general objectives:

- To familiarize the students with cloud computing introduction
- To make the students familiar with benefits and challenges of cloud computing
- To make the students knowledgeable about cloud deployment model
- To enhance the knowledge/skills of the students with virtualization
- To develop skills in/among the students cloud security
- To develop students' knowledge/skills/competencies cloud platform

To expose the students to cloud applications in business

3. Methods of Instructions:

- Lecture, Discussion, Group Work, Practical works, Case Studies.

4. Course Contents:

Specific Objectives	**Contents**
Unit 1: Introduction	**10 hours**
• Introduction to cloud computing , challenges and cloud and dynamic infrastructure	1.1 Introduction of cloud computing 1.1.1 Need for Cloud Computing 1.1.2 NIST definition 1.1.3 Characteristics and Benefits 1.1.4 Application 1.2 Cloud reference model 1.2.1 NIST Architecture 1.2.2 Design principles of cloud architecture 1.2.3 Infrastructural Constraints 1.3 Evolution of Cloud Computing 1.4 Cloud Service Model 1.4.1 Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service (IaaS) 1.5 Cloud Deployment Model 1.5.1 Public, Private, Hybrid and Community 1.6 Jericho Cloud Cube model 1.7 Challenges and Ethical Issues

Unit 2: Cloud Virtualization technology	**8 hours**
• To make the students familiar with cloud virtualization	2.1 Introduction 2.1.1 Virtualization Technologies: Virtual machines, Hypervisors (Type 1 & Type 2) 2.1.2 Overview of Containerization and Kubernetes 2.2 Virtual Machine (VM) 2.2.1 VM management and configuration 2.3 Virtualization Level 2.3.1 Hardware, Software, Operating System, Data and Networking
Unit 3: Storage in Cloud Computing	**6 hours**
• To understand the various cloud storage models, redundancy and data durability, accessibility and availability, data management service and data security • Design relational and non-relational database in cloud computing	3.1 Database as a Service 3.2 Storage model and file system 3.2.1 Parallel file system and Distributed file system 3.3 Cloud Storage Services 3.3.1 Object storage, file storage and block storage, Archive Storage 3.3.2 Elastic storage, Data redundancy 3.4 Databases 3.4.1 Relational: SQL, Non-relational: NoSQL and Graph Database
Unit 4: Cloud Programming Models	**12 hours**
• To make the students familiar with cloud programming model and efficiency	Thread programming, Task programming, Map-reduce programming, Parallel efficiency of Map-Reduce, Enterprise batch processing using Map-Reduce , Comparisons between Thread, Task and Map reduce
Unit 5: Cloud Security	**6 hour**
• To make the students familiar with Cloud Security issues, challenges and Risks, Software-as-a-Service Security	Cloud Security issues, challenges and Risks, Software-as-a-Service Security, Security Monitoring, Security Architecture Design, Data and application Security, Virtual Machine Security, Legal issues and Aspects, Multi-tenancy issues
Unit 7: Cloud Governance	**6 hours**
• To make the students familiar with Cloud applications tools	7.1. Introduction to cloud governance 7.2. Principles 7.3. frameworks 7.4. Control through cloud governance 7.5. Impact and challenges

5. List of Practical:

The practical work consists of all features of cloud computing.

6. 6. Evaluation System and Students' Responsibilities:
6.1 Evaluation System:
In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50 =100				

6.2 Students' Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References:

Text Books

1. Raj Kumar Buyya, Christian Vecchiola, S. Thamarai Selvi, Mastering Cloud Computing

Reference Books

1. David S. Linthicum, Cloud Computing and SOA Convergence in your enterprise
2. Barrie Sosinsky, Cloud Computing Bible
3. Saurabh, K. (2011). Cloud Computing – Insights into New -Era Infrastructure, Wiley India.

Pokhara University
Faculty of Science and Technology

Course Code.: CMP 336 (3 Credits)	Full Marks: 100
Course Title: Digital Economy (3-0-2)	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 48 hours
Level: Bachelor | Year: IV / Semester: VIII	Program: Bachelor of Computer Application

1. Course Description:

This course is designed to help students build cognitive foundation for fully understanding the linkages among digital technologies, economy and society at large in the context of emerging knowledge economy and knowledge societies. At the completion of the course, the students will be able to articulate overarching fundamentals of digital economy and will develop sufficient conceptual grounding for further studying and analyzing key elements of digital economy and the ebusiness perspectives. The course delivery method would entail class room lectures based on both primary and secondary sources of information and knowledge backed by deliberations on case studies and real world examples. Pedagogical method to be employed would encourage classroom discussions and references to materials from varied sources.

2. General Objectives:

- To develop a solid understanding of the connections between digital technologies, the economy, and society within the context of the emerging knowledge economy.
- To articulate the fundamental concepts of the eBusiness contributing to digital economy and build a conceptual foundation for analyzing its key elements.
- To engage students in classroom discussions, case studies, and real-world examples to enhance learning through a variety of sources and pedagogical methods.

3. Methods of Instructions:

- **Lectures:**
  Lectures will cover core theoretical concepts and provide foundational knowledge on Digital Economy, Payment Gateway,E-commerce and M-commerce. They will be interactive, using multimedia presentations, real-world examples, and case studies to illustrate key points. Lectures will encourage questions and discussions to deepen understanding.
- **Case Studies and Real-World Examples:**
  Case studies will be used to demonstrate how Digital payment software companies use/develop cloud based systems to solve problems and achieve strategic goals. Students will analyze these cases, identify challenges, and propose solutions, bridging the gap between theory and practice.

4. Course Contents:

Specific Objectives	Contents
Unit I: Fundamentals of Knowledge Dimensions of Economy	**(3 hours)**
• Understand how information and knowledge contribute to economic value. • Analyze the role of knowledge management in enhancing organizational efficiency. • Evaluate contemporary practices and their impact on organizational success.	1.1 Role of Information and Knowledge in society and Economy. 1.2 Value of Information: 1.3 Information as a Factor of Production 1.4 Knowledge Management Concepts 1.5 Contemporary Practices in Knowledge Management in creating Organizational Values.
Unit II: Development of Knowledge-Based Societies	**(5 hours)**
• Explore how ICTs drive societal and economic development. • Assess the challenges and policy considerations for digital service delivery. • Understand emerging technological trends and their implications for policy.	2.1 ICTs and Development(e-Government-business) 2.2 Digital Services Delivery: 2.3 Policy and Regulatory Framework:(Infrastructure issue and challenges) 2.4 Emerging Technologies: Media convergence, broadband, IP services, telecom regulation, and radio-spectrum management.
Unit III:Role of ICT in Economic Development	**(4 hours)**
• Learn how to measure and improve e-readiness. • Apply assessment techniques to evaluate digital capabilities. • Analyze the impact of ICTs on economic and social development. • Understand IT-enabled business models and their global implications.	3.1 e-Readiness Concepts: Definitions and frameworks for assessing digital readiness. 3.2 Assessment Techniques: Methods for evaluating e-readiness at individual and organizational level 3.3 Economic and Social Dimensions of ICTS 3.4 IT Enabled Business Models: Business process outsourcing (BPO), business process management (BPM), and IT-enabled services (ITeS) landscape and Fundamentals 3.5 Global Sourcing: Positioning Nepal in the global IT sourcing landscape.

Unit IV: ICTs and Enterprise Level Value Creation	**(4 hours)**
• Explore how ERP and CRM systems contribute to enterprise value creation. • Assess the benefits and challenges of implementing these systems.	4.1ERP (Enterprise Resource Planning): Key concepts and benefits. 4.2CRM (Customer Relationship Management): Importance and implementation in enterprises.
Unit V: Fundamentals of Knowledge Economy	(6 hours)
• Understand the components and benefits of the internet economy and e-business. • Analyze integrated e-business models and their impact on organizational performance.	5.1 Internet Economy: Concepts and topologies of e-business. 5.2 Leveraging ICTs: For brand awareness and product visibility. 5.3 Integrated E-business Models: Components and examples. 5.4 Value Chain Management: Concepts of value proposition and integrated value chains. 5.5 ICTs in Supply Chain Management: Electronic sourcing, procurement, and forecasting 5.6 Integrated Forecasting, Quality System Management,Electronic Commerce,Customer Service, Interactive marketing,Demand forcasting, Order management,Developing e-business model
Unit VI: Issues on Developing E-Commerce Based Business Models	**(6 hours)**
• Learn about different e-commerce models and their applications. • Examine challenges and opportunities in the Nepalese e-commerce sector. • Develop and analyze e-commerce business models. • Understand the impact of ICT implementations on traditional business mode	6.1E-commerce Topologies: B2C, B2B, and B2G models. 6.2 Prospects and Challenges: Payment mechanisms, gateways, and order fulfillment. Payment Gateway, Nepal e-commerce Landscape issues and Challenges 6.3Nepal's E-Commerce Landscape: Specific issues and challenges. 6.4Analyzing Value Propositions: Understanding and developing value propositions. 6.5Techno-graphic Profiling: Techniques for profiling technology and market trends. 6.6Case Studies: Real-world examples in travel, entertainment, and hospitality Industries.
Unit VII: Financing	**(3 hours)**
• Explore the financing mechanisms available for knowledge industries. • Understand the roles of venture capitalists and angel investors.	7.1 Knowledge Industry Growth Ecosystem: Key elements and components. 7.2 Venture Capital and Angel Fund: Concepts and roles in financing.
Unit VIII: Information Economy: Requisite Infrastructure	**(3 hours)**
• Analyze the infrastructure required for a successful information economy.	8.1 Regulatory Regime, Frameworks for managing the information economy.

• Understand regulatory and governance challenges.	8.2 Trust and Governance, Issues related to accountability and operational issues.
Unit IX: Emerging Technology Landscape and their Implications on Digital Economy (8 hours)	
• Examine the impact of emerging technologies on the digital economy. • Understand the implications of business intelligence, analytics, and cybersecurity. • Analyze the current state of the information economy in Nepal. • Identify opportunities and constraints in Nepal's ICT and e-business landscape.	9.1Business Intelligence and Analytics: Concepts of competitive intelligence and business analytics. 9.2Technologies: Social media, mobile and cloud computing, data mining, and big data analytics. 9.3 Cyber Security: Importance of security and Public Key Infrastructure (PKI). 9.4 IT policy and Strategy. 9.5 Application of ICTS in public and Private sector. 9.6Level of e- business related activities, Prevailing regulatory instruments, Opportunities/Constraints/Limitation
X: Internet Access and Infrastructure Related Issue (3 hours)	
• Gain insights into how telecom policies influence the development of infrastructure and the provision of internet services. • Identify key components of telecom infrastructure and their roles in facilitating • Learn about the functions and market dynamics of ISPs, including challenges they face in providing widespread internet access. • Explore the challenges and barriers to ICT penetration and propose solutions for improving internet access and bridging the digital divide.	10.1 Telecom Policy and Telecom Infrastructure Telecom Policy Overview: National and international telecom policies, regulatory frameworks, and their impact on infrastructure development. 10.2 Internet Service Providers (ISPs) and ICT Penetration Issues and Digital Divide

5. List of Practical:

SN	Practical Works
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6. 6. Evaluation System and Students' Responsibilities:
6.1 Evaluation System:

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		**30**	**Semester End examination**	**50**
Attendance / Class Participation	10%			
Assignments	20%			
Project Work/Presentations	10%			
Term Exam	60%			
Practical		**20**		
Attendance and Lab Participation	10%			
Lab Report	20%			
Lab Examination	40%			
Viva Examination	30%			
Total Internal Marks		**50**		
Full marks=50+50 =100				

6.2 Students' Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References:

Text Books:

References:

3.Tissen, R., Andriessen, D., & Deprez, F. L. *The Knowledge Dividend, Financial Times*.New York: Prentice Hall.
4.Timmers, P. *Electronic Commerce*. New York: John wily & Sons, Ltd.

Pokhara University
Faculty of Science and Technology

Course Code: PRJ 451 (5 Credits)	Full Marks: 100
Course Title: Project IV	Pass Marks: 45
Nature of the Course: Theory/Practical	Total Lectures: 80 hours
Level: Bachelor | Year: IV / Semester: VIII	Program: Bachelor of Computer Application

1. Course Description:

Project-III is a hands-on, practical course designed for BCA sixth-semester students. It focuses on the development of web application or data analysis tool. The project should be original and reflective of the students' skillset in software development or data analysis.The course encourages students to apply their knowledge in creating a functional web application or performing data analysis to solve a real-world problem. Students may work independently or in small teams to develop a web tool, data analysis system, or any other software project of interest. Guided by a faculty supervisor, the course helps students explore and manage various stages of the

software or data lifecycle. Project III offers students exposure to the practical side of development, testing, and documentation.

2. General Objectives:

- To empower students to apply the knowledge and skills gained throughout the BCA program to create a practical software application.
- To encourage students to think critically and solve problems creatively as they work on their projects.
- To help students develop essential project management, teamwork, and communication skills that are crucial for their future careers.
- To enable students to produce a functional, user-friendly, and well-documented software application that meets specific needs.
- To provide students with hands-on experience in every phase of project development, from planning and design to implementation and testing.
- To provide students knowledge about the API and how it is implemented on mobile or web projects.

3. Procedure:

The Project IV is focused to develop web or mobile application for event management; data analysis project that identifies trends from social media data; an interactive tool for financial planning or budgeting; dashboard for visualizing data and providing analytics.

- It is suggested to develop an application using emerging frameworks for web platform and use Android platform for mobile application.
- It is recommended that the team should be of 2-3 students. Teams cannot exceed 3 members to ensure effective collaboration and manageability.
- The project should be free from plagiarism of any kind. Students are expected to produce their own work and properly attribute any external resources or inspiration.
- Coding standards should be followed meticulously. At the minimum, the code should be self documented, modular, and should use the meaningful naming convention.
- It is advisable that object-oriented methodology is used with reusability of classes and code, etc.
- Mentor/ Internal guides (i.e. the faculty members) must devote time, allocated as per the time table to guide the students for the project. The time allocation will be in accordance with the teaching scheme for 4th semester project.

4. Project Phases:

The entire project work shall be divided into three phases and evaluation shall be done accordingly:

24. Project Milestone

The schedule can be given to the students according to the following phases.

VII. Team formation (Week 1)

- Students will form a team consisting of a maximum of 3 members based on their interests and skills.
- Teams should be balanced to ensure diverse skill sets and effective collaboration.

VIII. Project Proposal Submission: **(Week 2-4)**

- Students will start by defining the project's scope, objectives, and deliverables.
- A detailed project proposal must be submitted, outlining the project idea, timeline, and required resources.

Proposal Format:

12. Introduction
13. Aim and Scope

14. Objectives
15. Study of the Previous Systems
16. Current Problem and Proposed Solution
17. High level requirements
    - 5.4 Functional
    - 5.5 Non functional
    - 5.6 Requirement prioritization table
18. Design
    - 6.5 ER Diagram
    - 6.6 Use case Diagram
    - 6.7 DFD
    - 6.8 UML Class Diagram
19. Detail Budget
    - 9.1 Detail Budget
    - 9.2 Budget Breakdown
20. Detail Timeline
21. Conclusion
22. References

IX. UI/UX Design (Week 5-6)

- Students should begin by gathering user stories to understand the needs, goals, and pain points of potential users. These user stories will guide the design and development of the user interface and user experience, ensuring the final product meets the end users' expectations.
- Collect atleast three user story for each page.
- User Interface designed with any UI/UX Design tool.
- Present and defend the UI/UX design.

X. First Milestone (Week 7-8)

- Students should present their project progress. At this stage 60% of the work should be completed.

XI. Second Milestone (Week 9)

- Students should present their project progress. At this stage 100% of the work should be completed.

XII. Final Defense (Week 10)

- Final Project demonstration/presentation with documentation.
- The final defense includes a demonstration of the project, focusing on both functionality and documentation quality.

10 Guidelines for Project using Framework

11 Report format:

Title Page
Student Declaration
Supervisor Acceptance
Approval Certificate
Acknowledgement
Abstract
List of Abbreviations
Table of Figures
List of Tables
Table of Contents

CHAPTER 1: INTRODUCTION
- 1.1 Background
- 1.2 Objectives
- 1.3 Purpose, Scope, and Applicability
  - 1.3.1 Purpose
  - 1.3.2 Scope
  - 1.3.3 Applicability
- 1.4 Achievements
- 1.5 Organization of Report

CHAPTER 2: SURVEY OF TECHNOLOGIES
- Review of the similar/relevant projects

CHAPTER 3: REQUIREMENTS AND ANALYSIS
- 3.1 Problem Definition
- 3.2 Requirements Specification
- 3.3 Planning and Scheduling
- 3.4 Software and Hardware Requirements
- 3.5 Preliminary Product Description
- 3.6 Conceptual Models

CHAPTER 4: DESIGN
- 4.1 Introduction
- 4.2 System Design
- 4.3 Database design
- 4.4 Interface Design
- 4.5 Summary

CHAPTER 5: IMPLEMENTATION AND TESTING
- 5.1 Implementation Approaches
- 5.2 Coding Details and Code Efficiency
  - 5.2.1 Code Efficiency
- 5.3 Testing Approach
  - 5.3.1 Unit Testing
  - 5.3.2 Integrated Testing
  - 5.3.3 Beta Testing
- 5.4 Modifications and Improvements
- 5.5 Test Cases

CHAPTER 6: RESULTS AND DISCUSSION
- 6.1 Test Reports
- 6.2 User Documentation

CHAPTER 7: CONCLUSIONS
- 7.1 Conclusion
  - 7.1.1 Significance of the System
- 7.2 Limitations of the System
- 7.3 Future Scope of the Project

REFERANCES

Note: *3 copies of the hard bind report should be submitted to the department of computer application with proper binding. The report should be approved and signed by an external, internal examiner and head of department.*